

1.1 المقدمة (INTRODUCTION)

1.1.1 (THIS DOCUMENT – SCOPE OF WORK) نطاق العمل

1.2 المشروع (THE OSS911 PROJECT)

1.2.1 موجز المتطلبات (SYNOPSIS OF REQUIREMENTS)

- 1.2.1.1 The BIDDER shall note that this RFP is non-prescriptive regarding vendor and vendor technologies, and is vendor agnostic.
- 1.2.1.2 The BIDDER shall propose their solutions that help them to deliver all the requirements of the RFP while complying to industry best practices.
- 1.2.1.3 The BIDDER shall note that the design and construction of any of the OSS911 facilities is not part of this RFP.
- 1.2.1.4 The BIDDER shall note that each OSS911 Center will be linked to all other OSS911 Centers via an-MOI provided wide area network backbone dark fiber (MiFON).
- 1.2.1.5 The BIDDER shall calculate the bandwidth of the network connectivity required with other OSS911 centers in order to meet this RFP requirements for its concept of operations, functionality, performance and availability, and shall include those in the network edge devices that they will be using for connectivity to the MiFON backbone for various OSS911 solution needs.
- 1.2.1.6 The BIDDER shall note that the MOI have the right to involve their technical specialists during the project contract duration to work closely with the contractor and the subcontractor technical teams for continual knowhow and knowledge absorption.
- 1.2.1.7 The BIDDER proposal shall follow the format and bid structure and content instructions that have been provided in the Arabic section of this RFP.
- 1.2.1.8 The BIDDER shall provide a complete and integrated high-level design of their OSS911 system solution in their proposal along with a high-level narrative describing each functional elements of the design along with example use-cases of the function and the workflows.
- 1.2.1.9 The BIDDER shall ratify any and all of their assumptions as part of the bid Q&A process.
- 1.2.1.10 The BIDDER shall note that when BIDDER confirms compliance to any and all clauses within this RFP, it implies that the BIDDER is fully compliant with the MOI requirements without any conditions, assumptions, constraints, exclusions and exceptions – stated and/or implied.
- 1.2.1.11 The BIDDER shall note that further to the confirmation of their compliance to any clause, any notes within in their proposal that imply an assumption or condition or constraint or exclusion or exception implicitly or explicitly, shall not be considered as part of the BIDDER proposal. Further, such conditional content in the proposal will negatively impact the evaluation of the proposal compliance with the MOI requirements.
- 1.2.1.12 The BIDDER response shall include the planning, design, supply, installation, configuration, testing, training, commissioning, management and assurance of all components.
- 1.2.1.13 The BIDDER shall provide documentation including the proposed technical architecture and systems design for the solutions showing where and how the necessary interfaces and integration between the various systems will take place.

- 1.2.1.14 The BIDDER shall show in their high-level systems design, any phasing or incremental implementation of the various components, and a comprehensive structured description of how the various systems are to be deployed and integrated.
- 1.2.1.15 The BIDDER shall undertake the required site surveys as per the instruction outlined in the bidding documents and validate the various datacenters, technical and operational rooms for sufficiency for their solutions to be deployed and operated.
- 1.2.1.16 The BIDDER shall submit a project implementation schedule based on the project delivery outline that is detailed to level 3 (task level) and which delivers the implementation schedule requirements outlined in this RFP.
- 1.2.1.17 In their proposal, the BIDDER shall confirm the delivery of the project as per the timeline stated in this RFP.
- 1.2.1.18 The BIDDER shall confirm that the BIDDER and all of their proposed subcontractors have resolved any and all mutual interdependencies in their proposed solution
- 1.2.1.19 The BIDDER shall confirm that their proposed solution is holistic without any implicit or explicit interdependencies.
- 1.2.1.20 The BIDDER shall note that MOI does not allow remote deployment of any solutions.
- 1.2.1.21 The BIDDER shall finalize the teaming agreement(s) / partnership(s) / subcontracting relationship(s) prior to bid submission and shall provide associated proofs as part of their proposals with the certified affirmation from each of the vendors / subcontractors / partners / distributors / suppliers that each one of them shall be fully operational for the contract (having covered off all the necessary administrative, legal, contractual, finance parts with between them and the CONTRACTOR) as part of a single team with the CONTRACTOR immediately after the completion of the mobilization period as indicated in this document and its appendices. The BIDDER shall note that any noncompliance to this requirement shall impact the bid evaluation.
- 1.2.1.22 The CONTRACTOR shall own the single, turnkey responsibility of the OSS911 services at the center in scope and as per the requirements of this RFP.
- 1.2.1.23 Based on the turnkey responsibility for the contract, the CONTRACTOR shall be responsible to supply any and all components and works that are required to ensure that all the OSS911 requirements are delivered as acceptable to the MOI according to the MOI requirements within this document.
- 1.2.1.24 As part of the CONTRACTOR's turnkey responsibility, the CONTRACTOR shall be fully and absolutely responsible for any stated /unstated interdependency between the BIDDER and any of their subcontractors, and between the subcontractors.
- 1.2.1.25 From the completion of the allowed mobilization period and for the duration of the contract, the CONTRACTOR shall deliver all the resources mobilized as per the work plan indicated in the Arabic section of this RFP.
- 1.2.1.26 The CONTRACTOR shall note that any delay in resource and team mobilization as per the requirements indicated in this document and its appendices shall result in associated penalties as per the contractual general terms and conditions.
- 1.2.1.27 The CONTRACTOR shall deliver the analysis, design, development, integration, testing, implementation, training, governance, operations, management and assurance of the OSS911 functional, human capital, processes, interfaces, procedures, services, operations, systems and technical infrastructure for the OSS911.

- 1.2.1.28 The CONTRACTOR shall design and deploy OSS911 solutions such that the OSS911 Center is working 24 hours a day and 7 days a week for all the weeks in a calendar year.
- 1.2.1.29 The CONTRACTOR shall implement all requirements in this contract with the understanding that:
- 1.2.1.29.1 Unless otherwise explicitly stated, the phrase "OSS911 center" refers to OSS911 centers in each of the provinces that are included in the scope of this contract.
 - 1.2.1.29.2 Unless otherwise explicitly stated, the phrases "OSS911 solution" or "OSS911 systems" or "OSS911 subsystems" refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.
 - 1.2.1.29.3 Unless otherwise explicitly stated, the word "data" in this document includes all types of data including voice, text, video and SMS.
- 1.2.1.30 Further to the contract, the CONTRACTOR shall submit a detailed transition plan per each province that will be evaluated by the MOI for applicability to the different agencies' transition to the OSS911 center of a province and shall be implemented by the CONTRACTOR.
- 1.2.1.31 When the transition of each of the existing Makkah 911 center systems to the OSS911 center systems delivered by the CONTRACTOR occurs, the CONTRACTOR shall ensure that at the OSS911 Center, each of the agencies is able to perform all of their operations that they were performing at the existing Makkah 911 center.
- 1.2.1.32 For the duration of the contract, in the event of technology advances may occur which make certain components obsolete and in the event of component changes or unavailability, the CONTRACTOR shall be responsible for ensuring interoperability is maintained by upgrading all such obsolete components with their contemporary models / contemporary equivalents, all at no additional costs to the MOI.
- 1.2.1.33 The CONTRACTOR shall be responsible for all activities pertaining to gathering information, conducting stakeholder analysis, conducting stakeholder interviews, surveys and any other data and information gathering activities, any and all data extraction, transformation and loading activities that the CONTRACTOR deems necessary for them to deliver the project.
- 1.2.1.34 For the duration of the contract, and as required from the existing Makkah 911 center, the CONTRACTOR shall extract, transform and load into the OSS911 systems all the operational and support data that is required by the OSS911 systems.
- 1.2.1.35 The CONTRACTOR shall provide the Project Management Office, the complete raw data that they receive from all the sources (MOI and non-MOI agencies) that will contribute to making the operational data ready for the OSS911 systems.
- 1.2.1.36 The CONTRACTOR shall provide the on-time delivery of full project data and documentation including consolidated data, detailed layout diagrams, line diagrams, schedules, interfaces and interconnections drawings as required and described in this RFP.
- 1.2.1.37 The CONTRACTOR shall prepare and deliver detailed test and acceptance plans, forms, and procedures for all OSS911 project components within the CONTRACTOR scope of supply.
- 1.2.1.38 The CONTRACTOR shall prepare and deliver all project management requirements including project implementation plan and project master schedule as detailed in the Project Management section of this RFP document.
- 1.2.1.39 The CONTRACTOR shall be available to provide training, maintenance, spare parts procurement and support for the MOI for the useful system life.

1.2.1.40 The CONTRACTOR shall deliver solutions such that multiagency co-location and consolidation within the OSS911 Center for the complete OSS911 center province is the desired end state, as per a transition scheme agreed with the MOI.

1.2.2 موجز معلومات تسليم المشروع (OSS911 PROJECT DELIVERY OUTLINE)

1.2.2.1 The duration of the project is thirty-six months during which the OSS911 solutions implementation will be done for the first six months which will include compliance to the service level agreement for all OSS911 solutions from the first day of live operations of the any agency the OSS911 center.

1.2.2.2 The CONTRACTOR shall clearly identify in their proposed schedules the periods within the project duration that shall be used for systems development, training, testing, fit-out, user transition and live operations.

1.3 مفهوم العمليات (CONCEPT OF OPERATIONS (CONOPS))

1.3.1 لمحة عامة (OVERVIEW)

1.3.1.1 The CONTRACTOR shall deliver the OSS911 solutions to comply with the following:

1.3.1.1.1 For all the provinces in scope of the project, the OSS911 solutions, systems, applications stack, technologies, products, technical infrastructure and databases shall be identical for each of the OSS911 solution deployments within the scope of the project.

1.3.1.1.2 Unless permitted by the MOI, all the OSS911 solutions shall utilize a central single database for all their data using a single database technology and product, and shall not have their own individual databases.

1.3.1.1.3 The CONTRACTOR shall ensure that the systems they are delivering as part of their contract are compatible with natively without the use of any middleware or SDKs.

1.3.1.1.4 The CONTRACTOR shall implement the OSS911 systems and solutions such that the operations in the Makkah 911 center have the capability to failover to another MOI 911 center for delivering the following two use-cases:

- a. For sharing of operational load, as determined by the MOI.
- b. For disaster recover while ensuring continuity of operations should the Makkah 911 center be affected partially/fully by faults that impact the operations.

1.3.1.1.5 The CONTRACTOR shall implement the OSS911 systems and solutions such that capability is delivered for any authorized operational personnel located in a paired MOI911 center to be able to login to the solutions in Makkah 911 center and handle operations for Makkah 911 center, and vice versa, without compromise in functionality, performance and availability while complying with all functional, performance and availability requirements within the scope of this project.

1.3.1.1.6 The CONTRACTOR shall deliver a Production Environment comprising OSS911 solutions and related technical infrastructure to be hosted the OSS911 center within the scope for the following solutions: Computer Aided Dispatch (CAD) solution as part of the IRMS solution, the IP Telephony and Analog telephony solution,

the common operational picture solution, the middleware solution including the enterprise service bus and related integrations and the data warehouse solution, the centralized directory structure and LDAP solution, the cybersecurity solutions, the online and offline backup solution and the management solution.

1.3.1.1.7 The CONTRACTOR shall deliver a centralized training environment solution and related technical infrastructure to be hosted the OSS911 center within the scope for all training requirements for the scope of the project.

1.3.1.1.8 The CONTRACTOR shall deliver a centralized test environment solution and related technical infrastructure to be hosted the OSS911 center.

1.3.1.1.9 Irrespective of the mode of operation, the CONTRACTOR shall dimension their systems as per the dimensioning principle mentioned in this RFP.

1.3.1.1.10 Irrespective of the mode of operations, the CONTRACTOR shall deliver the OSS911 solutions as per the performance, availability and reliability requirements mentioned in this RFP.

1.3.1.1.11 The BIDDER response shall include in their proposal the details for:

1.3.1.1.12 The different fault scenarios for all levels of the workflow – from call reception to incident closure, along with the associated built-in resilience, the associated fallback mechanisms including the proposed procedures for failing over to the fallback centers and the expected recovery times.

1.3.1.1.13 The above scenarios description shall include defined criteria that help to indicate when the OSS911 Center systems availability have degraded to the point it is necessary to move operations elsewhere

1.3.1.1.14 When services have been restored at the failed OSS911 Centre then an orderly return shall be made.

1.3.1.1.15 The BIDDER shall clearly explain the capabilities of their solution with respect to the above modes of OSS911 solutions operations and shall clearly document any associated dependencies for their solutions.

1.3.1.2 The CONTRACTOR delivered OSS911 center operating model shall include handling calls for emergency service and the resulting dispatch of resources and shall deliver multi-agency operations.

1.3.2 **حلول وخدمات المقاول لمراكز العمليات الأمنية الموحدة (CONTRACTOR SOLUTION AND SERVICES FOR THE OSS911 CENTER)**

1.3.2.1 The CONTRACTOR shall implement all requirements in this section with the understanding that:

1.3.2.1.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

1.3.2.1.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

1.3.2.1.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

1.3.2.2 The CONTRACTOR shall deliver a solution that complies with the CONOPS described for the OSS911 in this section.

1.3.2.3 The CONTRACTOR shall conduct due diligence, data and information gathering workshops for business and stakeholder analysis for all OSS911 subsystems.

1.3.2.4 The CONTRACTOR shall provide a dedicated business analyst covering each operational domain of the OSS911 subsystems for their respective workshops with the MOI identified agencies for all OSS911 subsystems.

1.3.2.5 Following each workshop, the CONTRACTOR shall deliver to MOI a report on the workshop outcome including detailed minutes of the workshop, attendance in the workshop, actions resulting from the workshop and an analysis report on each workshop.

1.3.2.6 The CONTRACTOR shall deliver the OSS911 systems such that the availability of Artificial Intelligence (AI) augmented capabilities and electronically stored data relating to call taking and incident response, coupled with documented processes and procedures, shall enable performance monitoring and improvement which in turn shall lead to improved time to answer and resource utilization.

1.3.2.7 The OSS911 solution delivered by the CONTRACTOR shall include ability to baseline the response to calls and incidents so that quantifiable measures can be defined to improve the service through better resource management or use of technology and AI capabilities.

1.3.2.8 The OSS911 solutions delivered by the CONTRACTOR shall include use of AI capabilities across all the solutions to enable the collection of a rich source of historical data, to provide immediate access to standard operating procedures and to prompt operators to provide a consistent high-quality response.

1.3.2.9 The OSS911 solutions delivered by the CONTRACTOR shall include AI-augmented geographic representation of data integrated with new and existing systems.

1.3.2.10 The OSS911 solutions delivered by the CONTRACTOR shall align with the concept of operations developed by the MOI, and with the international standards like NIMS, ISO 22320 as applicable.

1.3.2.11 The OSS911 solution delivered by the CONTRACTOR shall deliver the following capabilities:

1.3.2.11.1 Call Taking – AI-enabled handling emergency and non-emergency demands.

1.3.2.11.2 Dispatching – AI-assisted handling demands requiring dispatch to the field resources.

1.3.2.11.3 Supervision – AI-enabled supervising the handling and coordination of any demand.

1.3.2.11.4 Quality function – AI-assisted monitoring and management of the operational metrics, KPIs and reports.

1.3.2.11.5 Joint Operations and Crisis Management - providing a solution that enables dedicated command and control actions in response to planned or spontaneous large-scale incidents.

1.3.2.11.6 Integration of solutions delivering all the above capabilities.

1.3.2.12 The following operational activities shall be delivered by the CONTRACTOR that shall support the delivery of the above capabilities using AI capabilities:

1.3.2.12.1 Receiving, entering, and processing of all received calls.

1.3.2.12.2 Referring calls or incidents which are not managed by the center to the appropriate agency.

1.3.2.12.3 Transcription of voice calls to text that is displayed simultaneously in real-time to the operators while the voice conversation is ongoing.

1.3.2.12.4 Storing and retention of all voice calls and related transcriptions with the transcriptions being captured in the associated event records, and are searchable.

1.3.2.12.5 Identifying duplicate requests based on location, telephone number, or name of caller and to visual and audibly alert operator and supervisors of occurrence. Identification of the duplicate requests based on location, telephone number and semantics is required to be done automatically while the querying of the name is done through the interfaces to MOI-authorized databases. The system shall suggest to the operator confidence scores for the system judgement of duplicates or linked incidents.

1.3.2.12.6 Eliciting accurate information (location and telephone number) and to visual and audibly alert operator when accuracy of information is in question. Regarding the accuracy, if the caller provides a different location for himself as opposed to the location obtained from the different ALI interfaces, then this should be recorded in the system as a difference, and this should then be available for reporting so that the MOI can take the discrepancies to the concerned authorities and resolve these for accurate location reporting.

1.3.2.12.7 Recording, logging, disseminating and retrieval of all voice, video and data.

1.3.2.12.8 Providing self-tips and immediate help and advice when information is not in the proper format.

1.3.2.12.9 Assisting the operator in adhering to the standard operating procedures.

1.3.2.12.10 Retrieving information for review and for analysis.

1.3.2.12.11 Identifying and recommending the need for an appropriate response by gathering and programming the event types and sub-types for different agencies.

- 1.3.2.12.12 Arriving at deployment solutions by gathering and programming the response priorities based on event types and sub-types vis-à-vis the different agencies.
- 1.3.2.12.13 Aligning the response plans to the resource deployment plan by gathering and programming agencies' the plans received from the agencies.
- 1.3.2.12.14 Supporting operational staff by providing applications that aid and automate the processes to receive, dispatch and alerting resources.
- 1.3.2.12.15 Providing management information, reports, dashboards and performance data against specified indicators. Such information shall be made available through any form of access – either through browsers running on computers or through apps running on iOS and/or Android devices.
- 1.3.2.12.16 Providing an assistance capability that allows callers to add/modify/supplement information securely to their emergency calls and related events through a web page – such information to be added shall include validation of location, uploading of images and videos, and a chat capability with operator.
- 1.3.2.12.17 Providing assistance in dispatching function including assistance for field unit selection as per the response plan, and for alerting field units.
- 1.3.2.12.18 Protecting all information recorded and stored in archives, to provide access to critical forensic and video data evidence of response actions.

1.3.2.13 The CONTRACTOR solution shall ensure the capability to receive, record and process all urgent and non-urgent demands that will enter OSS911 Center using a common dedicated emergency telephone number, by existing agency-specific telephone numbers (998, 996, etc.) and by any other number authorized by the MOI.

1.3.3 تلقي البلاغات (CALL TAKING)

- 1.3.3.1 The CONTRACTOR shall take into account that service provided by the Call Takers is outward facing, handling demands and requests for service from the public and other stakeholders, through channels other than by radio.
- 1.3.3.2 The Call Taking solution delivered by the CONTRACTOR shall include an application to record information relating to all calls presented.
- 1.3.3.3 The CONTRACTOR solution shall insure that Incident records are retrievable, such that when once created, they cannot be deleted and that an access log is maintained for all users' access and for all actions.
- 1.3.3.4 The CONTRACTOR delivered IRMS integration with the Geographical Information System (GIS) shall, without intervention by the operator, utilize all information available from the telephony service supplier to locate the caller using all location sources and present all received information to the operator on the maps on the screen.
- 1.3.3.5 The CONTRACTOR delivered IRMS solution shall utilize AI capabilities be able to access Standard Operating Procedures (SOPs) and response plans, and shall alert the operator when

information obtained matches set criteria, providing the operator with additional visual and audible prompts related to the incident type.

- 1.3.3.6 The CONTRACTOR delivered IRMS solution shall intelligently analyze text entered to help reduce demands upon operators, for example by using AI capabilities to determine what further information is required and to pre-fill required fields where possible.
- 1.3.3.7 The CONTRACTOR delivered IRMS shall provide visual and audible indications of what actions an operator should follow and shall propose the most appropriate responses, whether that is telephone resolution or transfer to dispatcher(s) or to another agency.
- 1.3.3.8 The CONTRACTOR delivered IRMS shall include solutions such that complex or serious incidents that require multiple routing to more than one agency, supervisors, specialist units or areas within OSS911 are attended to through the CONTRACTOR delivered IRMS.
- 1.3.3.9 Call Routing in Unified Call Taking shall be delivered by the CONTRACTOR to include:
 - 1.3.3.9.1A call distribution system that will have the capability to automatically distribute incoming calls from emergency numbers based on specified business rules to designated Call Handling positions.
 - 1.3.3.9.2Call presentation to Call Handlers on the basis of business rules set by OSS911, irrespective of the agency he represents. These rules will be skill or profile based, and at the simplest level, by calls being presented to the 'logged in' Call Taker who is shown as available the longest period of time.
 - 1.3.3.9.3Prioritization of emergency demand queues over nonemergency demand queues.
 - 1.3.3.9.4Capability that in subsequent operational models developed as the OSS911 environment matures, emergency and non-emergency police-demand telephony and 'customer care' non-emergency telephony may be separated and presented to different Call Taker pools inside and outside the OSS911 Center.
 - 1.3.3.9.5Capability for the supervisors within the environment to monitor the telephony queues and operator performance.
 - 1.3.3.9.6All telephone calls into OSS911 center (incoming, outgoing and internal) shall be recorded in a single integrated system and immediately accessible by nominated operating personnel.
 - 1.3.3.9.7The call distribution technology shall collate and present telephony management information to assist in the management of telephony queues, to inform on performance, and in the determination of staffing patterns.
- 1.3.3.10 The CONTRACTOR delivered solution shall include integration with GIS to deliver a capability providing a single, integrated map at each operator position.
- 1.3.3.11 The CONTRACTOR delivered solution shall include access to MOI-authorized databases to be able to validate locations and caller profiles, and to provide location information based on address.
- 1.3.3.12 Based on the location the OSS911 solution shall represent any known history on the address, local places of interest, location of public buildings, and cameras within a configurable radius around

the location, along with the location details (acronyms, latitude, longitude, direction etc.), and warnings associated with the location or area as captured from MOI-authorized sources.

- 1.3.3.13 An enterprise or active directory capability integrated into the OSS911 systems for phone number lookups and user authentication to facilitate single logon to applications.

1.3.4 ترحيل البلاغات (DISPATCH)

- 1.3.4.1 The CONTRACTOR shall take into account that the service provided by Dispatch is essentially inward facing, capable of handling requests for service from operational resources via radio, some electronic messaging, and direct dial telephony from some field agents.

- 1.3.4.2 The CONTRACTOR delivered OSS911 solution shall insure that there is a dispatch capability within each OSS911 Center for each of the agencies and provinces under the scope of the contract.

- 1.3.4.3 The CONTRACTOR delivered OSS911 solution shall insure that agency dispatchers will be co-located, thus supporting collaborative business processes whilst reflecting their own particular needs.

- 1.3.4.4 The CONTRACTOR delivered OSS911 solution shall insure that demands that form the normal business of Dispatch include the following but be not limited to:

1.3.4.4.1 Demands from MOI's radio networks.

1.3.4.4.2 Events created from call taking.

1.3.4.4.3 Direct inward Dial (DID) telephone calls (from field operators)

- 1.3.4.5 The CONTRACTOR delivered OSS911 solution shall ensure that the dispatchers will be able to record information and access information from a number of sources to properly assign and support resources. These include but are not limited to:

1.3.4.5.1 A method of capturing assignment and situational information.

1.3.4.5.2 Communications that allow 'one to many' and 'one to one' secure conversation between Dispatch and field resources.

1.3.4.5.3 Resource availability information.

1.3.4.5.4 Mapping and Gazetteer information.

1.3.4.5.5 Reference material such as SOPs and response plans.

1.3.4.5.6 Information contained in MOI-authorized databases.

- 1.3.4.6 The CONTRACTOR delivered OSS911 solution for the operation of Dispatch shall ensure that:

1.3.4.6.1 Incidents that require the deployment of resources are forwarded to Dispatch for appropriate action.

1.3.4.6.2 When the created Incident passes to Dispatch, the location of the incident and the incident types identify the relevant dispatch queue – this may be a single dispatcher position, a group of positions or to a functional (non-geographically aligned) position.

1.3.4.6.3 Functional queues may exist for units that are not geographically aligned, for example region units such as helicopter assets, Special Weapons and Tactics (SWAT) etc.

- 1.3.4.6.4 Where more than one agency is shown the incident record goes to each agency queue; an indication that this has happened must be visible to each agency dispatcher on the system. As Dispatchers update the incident record with comments and actions, these should be visible to all dispatchers unless there is sensitive information on the message, where that particular data should be restricted and only visible to dispatchers in queues identified by the restricting authority.
- 1.3.4.6.5 Dispatch shall have the capability to assign the field units/resources to events and shall have the capability to provide operational support to those resources via bidirectional electronic data exchange and/or bidirectional voice communications.
- 1.3.4.6.6 The resource management function within the IRMS shall allow Dispatchers to both monitor and collate the status, and therefore the likely availability, of assignable resources once they are confirmed as assignable.
- 1.3.4.6.7 The IRMS will identify to the Dispatchers when there are no assignable resources available for deployment within their patrol area.
- 1.3.4.6.8 The IRMS will have readily available in database for Dispatchers the location, status and availability, skill/competency and equipment of resources when considering modification of system recommendation of which resources are appropriate to assign based on the differing types of demand.
- 1.3.4.6.9 The IRMS will support Dispatchers by recommending the most appropriate assignments with regard to skills, proximity and availability.
- 1.3.4.6.10 Dispatchers will assign units either by the use of radio, by the passing of electronic messaging (where mobile data systems exist) or a combination of both. Dispatchers and field supervisors will have the capability to request alternative resource assignment than the one recommended by IRMS.
- 1.3.4.6.11 IRMS system will have the capability to create an incident or perform MOI-authorized databases information queries by Dispatcher on the request of field personnel or if a mobile data system terminal is available to field personnel, they may perform their own queries or create incidents.
- 1.3.4.6.12 The IRMS shall have the capability to display, track, record and playback all mobile and handheld MOI radio terminals which have GPS location capability enabled and which have been integrated into the IRMS system.

1.3.5 إدارة سجلات الأحداث (INCIDENT RECORDS MANAGEMENT)

- 1.3.5.1 The CONTRACTOR delivered OSS911 solution shall ensure that an incident record can only be closed when acting on the recommendations of an Incident Records Management (IRM) function with appropriate permission.

1.3.5.2 The CONTRACTOR delivered OSS911 solution shall ensure capabilities for Incident Records Management (IRM) which will comprise one or more people whose role will be to track all open incident records, once any initial dispatch operations have taken place. Their capabilities will be to:

1.3.5.2.1 Retrieve incident record to determine if incident has been classified properly.

1.3.5.2.2 Review recorded information and actions performed.

1.3.5.2.3 See the entire chronological record from creation to closure.

1.3.5.2.4 Display the results of any actions, or progress updates, are recorded on incident records.

1.3.5.2.5 Inform authorized personnel when the incident has been completed and all actions are either closed or progressed to the point where they can be considered such.

1.3.6 **الدعم التشغيلي (OPERATIONAL SUPPORT LIAISONS)**

1.3.6.1 The CONTRACTOR shall supply IRMS integrated solutions to enable the operational support liaisons to take care of liaising operations with their or other agencies that are required to participate in related situations.

1.3.6.2 The CONTRACTOR IRMS solution shall ensure that each of the liaisons is able to view and provide updates to incidents that are related to his agency only unless mandated by the MOI authority to be able to have additional agency incidents' view and edit capabilities.

1.3.7 **العمليات المشتركة وإدارة الأزمات (JOINT OPERATIONS AND CRISIS MANAGEMENT)**

1.3.7.1 The CONTRACTOR shall supply IRMS integrated solutions for Disaster and Crisis management functionality that enables:

1.3.7.1.1 Access to active and archived IRMS data in operational areas including but not limited to the crisis management meeting room.

1.3.7.1.2 The capability to perform dispatch functionality of escalated incidents and crisis.

1.3.7.1.3 The capability to support the joint operations liaisons functionality of the escalated incidents and crisis (which does not require dispatch function).

1.3.8 **AI-ENABLED FUNCTIONS**

1.3.8.1 For purposes of this RFP, 'Artificial Intelligence (AI)' refers to systems and solutions using machine learning, deep learning, natural language processing, computer vision, or similar techniques to perform tasks requiring human-like intelligence. This includes but is not limited to predictive analytics, automated classification, speech recognition, image analysis, and decision support. Rule-based automation without learning algorithms is excluded.

1.3.8.2 The BIDDER shall refer to the AI-powered reference architecture in the appendix 11 for integrating the required AI-capabilities across the operational solutions.

1.3.8.3 The BIDDER shall note that the reference architecture included in this RFP is provided for guidance and illustrative purposes only to help bidders understand the target technical direction and integration expectations.

1.3.8.4 Based on the appendix 11, the BIDDER shall propose their architecture with their proposed solutions, systems and integrations mapped to it.

1.3.8.5 The BIDDER shall describe their proposed architecture and their proposed AI capabilities and shall include the following:

1.3.8.5.1 Human-in-the-loop controls

1.3.8.5.2 Explainability for AI decisions

1.3.8.5.3 Languages' dialects support

1.3.8.5.4 Integration capabilities

1.3.8.5.5 Monitoring and management capabilities

1.3.8.5.6 Similar deployments with metrics

1.3.8.6 Apart from implementing the data, security and monitoring and management controls included in this RFP, the CONTRACTOR shall also include the following governance controls in their implementation for the AI solution including but not limited to:

1.3.8.6.1 A Governance Board with defined roles and responsibilities providing strategy and oversight

1.3.8.6.2 Human-in-the-loop

1.3.8.6.3 Explainability for the AI solution to explain its actions and decisions

1.3.8.6.4 Centralized logging with complete audit trail

1.3.8.6.5 Compliance to PDPL and SDAIA policies

1.3.8.6.6 Continuous bias monitoring for bias reduction and fairness

1.3.8.6.7 Emergency Kill switch with the capability to immediately suspend or disable AI decision-making and switch to manual or fallback processes

1.3.8.7 The CONTRACTOR delivered OSS911 solutions shall include the following AI-enabled functional capabilities integrated into all the OSS911 systems and solutions delivered by the CONTRACTOR:

- 1.3.8.7.1 Live on-the-fly real-time transcription of the voice to text, from the voice in the following defined languages (all Arabic dialects of Saudi Arabia, all commonly spoken English dialects, Urdu, Bengali, Hindi, Punjabi, Pashto, etc. with the final list to be determined by the MOI) to Arabic and English on the authorized OSS911 personnel system displays, and to include the transcribed text as part of the recorded conversation, and in case of incident/event creation, as a part of the event chronology mapped to the different speakers in the conversation with timestamps.
- 1.3.8.7.2 Such a transcription should include the ability to identify and tag keywords as relevant to OSS911 defined incident/event types and subtypes and call types and subtypes.
- 1.3.8.7.3 Further, from the location information captured in transcription, the transcription shall identify situational awareness from information contained in MOI and OSS911 databases as relevant to locations captured and should then display this information in the relevant areas within the telephony caller information dialog box and in the incident/event card display.
- 1.3.8.7.4 Live on-the-fly real-time speech analysis and triage capabilities to identify the voice and the speakers therein, tag voice streams by speakers, identify emotions and sentiments within the voice and to present these to the authorized OSS911 operational personnel within the telephony caller information dialog box and in the incident/event card display.
- 1.3.8.7.5 On-Demand offline interactive support for the authorized OSS911 personnel in the form of SOPs, response plan guidance and systems knowhow.
- 1.3.8.7.6 Based on the voice and transcribed text, the AI capability shall provide silent listening and in-call real-time guidance and prompt to the authorized OSS911 personnel involved in voice and text conversations.
- 1.3.8.7.7 Further, the guidance and prompts themselves, along with the OSS911 personnel actions on them shall be stored as part of the recorded conversation and in case of incident/event creation, as part of the event chronology mapped to the individual actions and action-taker.
- 1.3.8.7.8 Live on-the-fly real-time quality monitoring and reporting capability for authorized OSS911 personnel including speech analysis as defined in this RFP, OSS911 operations' functional personnel performance assessment with real-time data for real-time insights, visualizations of KPIs, and ad-hoc reports generation.
- 1.3.8.7.9 Capability to handle non-emergent calls and demands till closure using the corresponding SOP and response plan, once these calls and demands are identified and tagged as such by authorized OSS911 operational personnel.

- 1.3.8.7.10 Capability to automatically generate request, for configurable number of attempts, for calling the caller automatically using an interactive voice response (IVR) capability for feedback based on predefined rules and using the language of choice (Arabic and English).
- 1.3.8.7.11 Capability in the IVR call back scenarios to have the option of scheduling a convenient time with the caller for making the call back for feedback.
- 1.3.8.7.12 Capability to capture and store the received feedback, to transcribe it (if needed), and to link it to the related event IDs, for reporting and analysis by authorized OSS911 personnel.
- 1.3.8.7.13 E-Assist feature to enable callers to upload their location relating to their call for help, to be able to chat with authorized OSS911 personnel and to upload supporting material including doc/voice/video/image files that add information to their call.
- 1.3.8.7.14 Capability to intelligently analyze the incoming call details (voice, text or both) and assist in smart incident classification and smart prioritization as assistance to OSS911 operational personnel.
- 1.3.8.7.15 Capability to intelligently analyze the OSS911 solutions (all OSS911 systems, all OSS911 databases, all OSS911 integrations, all OSS911 interfaces, all OSS911 network infrastructure, all OSS911 cybersecurity, and other elements) for any symptoms or faults and to automatically alert the authorized OSS911 personnel and the capability to take corrective actions automatically such that they do not cause any other impact.
- 1.3.8.7.16 Capability for authorized personnel to flag incoming calls on the predefined 9XX numbers as possible spam and/or malicious calls based on configurable rules and providing visual and audible alerts to the authorized personnel with the details including but not limited to the caller details including caller ID, called number, location, previous calls count, previous calls details.
- 1.3.8.7.17 Capability for adaptive learning within the various OSS911 systems based on previous calls and incidents, and based on previous outcomes, enabling these systems to become more accurate and effective over time.
- 1.3.8.7.18 Capability to generate warnings and alerts to authorized operational personnel based on predefined trigger mechanisms – for example: based on high-priority incidents/events, based on severity warnings from linked warning systems and/or data sources, etc.
- 1.3.8.7.19 Capability to expose all OSS911 systems and databases via secure REST APIs for integrations.
- 1.3.8.7.20 Capability to integrate all OSS911 systems and databases via secure REST APIs and Webhooks.

- 1.3.8.7.21 Capability for authorized external systems to be able to interact with the OSS911 solutions to create and update incidents/events and to create and update alarms within the respective OSS911 systems.
- 1.3.8.7.22 Capability for smart management and assurance to the OSS911 personnel of the OSS911 solutions, systems, applications, databases, integrations and technical infrastructure using a knowledgebase of frequently asked questions and their solutions.
- 1.3.8.7.23 Capability for the management solution to send smart alerts to the authorized personnel based on performance alerts for OSS911 solutions, systems, applications, databases, integrations and technical infrastructure.
- 1.3.8.7.24 Capability to decipher electronically and include in the OSS911 systems for operation use, the SOPs, plans and operational documents that may exist in paper format, and to analyze them and present them to the operators using AI-tools if they are relevant to the incident/event.
- 1.3.8.7.25 Capability of real-time data ingestion and processing for capturing, streaming and processing spatial and non-spatial data for analysis by demography, geography and for smarter decision making, rapid analysis and for predictive modeling and analysis.
- 1.3.8.7.26 Capability with Webhooks, Open APIs, plug-ins and built-in tools for machine learning within the OSS911 solutions to integrate for prediction and analysis with external AI frameworks, data science tools and ML platforms.
- 1.3.8.7.27 Capability for AI-powered natural language assistants to assist in common tasks with real-time structured and unstructured data including but not limited to tasks such as querying, analyzing and generating insights, visualizations and reports in real-time using natural language prompts (for all dialects of Arabic, and for English).
- 1.3.8.7.28 Capability for AI-powered real-time dispatch impact analysis providing simulation, and coverage analysis that is based on unit positions, potential assignment and demand patterns resulting in optimal coverage maps, impact analysis vis-à-vis the response times and possible gap alerts.
- 1.3.8.7.29 Capability for AI-powered real-time report generation based on calls and events data including reports for call handling efficiency, operator utilization, briefings, situation reports for authorized MOI persons – with the reports, their contents, the format and the distribution to be defined by the CONTRACTOR with the MOI during the design phase of the contract.
- 1.3.8.7.30 Capability for AI-powered real-time analysis of incident details, location history and the environment to present to the operators a risk assessment as supplemental information for dispatch decision support.

1.3.8.7.31 Capability for sharing event-related data using text, speech-to-text, summarization, entity extraction and report generation for the voice recordings, event-related notes and information including multimedia with MOI-authorized apps so as to be able to generate summarized reports via the app.

1.3.8.7.32 Capability for AI-powered demand and utilization patterns for units' optimization and positioning using historical events and units' data and patterns.

1.3.8.8 For each of the functional capabilities delivered by the CONTRACTOR through their AI solutions, the CONTRACTOR will agree a set of metrics and KPIs with the MOI during the design stage of the project, and shall implement them prior to go-live of these capabilities.

1.3.8.9 The CONTRACTOR shall continually provide reports relating to the accuracy, reliability and safety for the metrics from each of the capabilities including but not limited to data labeling accuracy, precision, false negatives and positives, word error rate, location extraction, confidence levels, uncertainty thresholds, overrides, bias deviations, latency etc.

1.4 إدارة المشروع (PROJECT MANAGEMENT)

1.4.1.1 The BIDDER shall provide a Project Management Plan that demonstrates to MOI their understanding of the requirements, and their ability to manage and deliver all the work as per the requirements.

1.4.1.2 The BIDDER shall propose a delivery Governance and Organization, headed by their Project Management Office, demonstrating how they will:

1.4.1.2.1 Establish the Governance and Organization following commencement of the engagement.

1.4.1.2.2 Manage the entire project successfully.

1.4.1.2.3 Implement a full range of effective project management techniques and tools, and associated reporting and communications.

1.4.1.2.4 Provide full, open and frequent communications and reporting on all aspects of the project

1.4.1.3 The BIDDER shall propose key governance processes including as a minimum: Project Resource Management, Issue Management, Risk Management, Contract Management, Project Change Management, Project Configuration Management, Quality Management, Reporting, Organizational Change Readiness Management, Communications, Project Financial Management, Project Training Management and Stakeholder Management.

1.4.1.4 The BIDDER shall include a concise practical description of their proposed project management methodology and framework within their proposal with examples.

1.4.1.5 The BIDDER shall address these requirements in their proposals in a practical manner, specific to the challenges of the OSS911 Program, and not in a high-level generic manner.

1.4.1.6 As part of the proposal, the BIDDER shall supply a Microsoft Project-based preliminary program schedule detailed up to activity level.

- 1.4.1.7 The BIDDER shall include in their proposals the full CV for all key personnel proposed by them for the project.
- 1.4.1.8 The CONTRACTOR shall be responsible for the data collection, analysis and delivery to the MOI of periodic Program Status Reports (PSR), preparation and delivery of program documentation such as executive and technical presentations, updates to program plans, deliverable reviews, budget/financial reports, delivery schedules, team performance and evaluation reviews, technical and functional papers, conformance certificates, acceptance certificates and policies and procedures.
- 1.4.1.9 The CONTRACTOR shall be responsible for the implementation and delivery of financial processes that provide proper financial controls and correct financial reporting of the OSS911 project.
- 1.4.1.10 The CONTRACTOR shall be responsible for the implementation and active management of the plan including but not limited to ensuring that all change introduced to the project environment is appropriately recorded, defined, evaluated and agreed to and approved with the MOI prior to implementation.
- 1.4.1.11 The CONTRACTOR shall implement and actively manage a Risk Management process that ensures that all project risks are identified, recorded, analyzed and assessed and, appropriate response/mitigation plans put in place and, continually managed and reported.
- 1.4.1.12 The CONTRACTOR shall implement and actively manage an Issue Management process which ensures that all issues are systematically identified, recorded, evaluated and actively managed, reported and resolutions communicated across the project.
- 1.4.1.13 The CONTRACTOR shall be responsible for the development and maintenance of the baselined OSS911 project master schedule.
- 1.4.1.14 The CONTRACTOR shall maintain baseline control of the master schedules, with changes to baseline values restricted to those formally approved by MOI.
- 1.4.1.15 The CONTRACTOR shall be responsible for developing a schedule that identifies milestones and key periods and dates for decisions, MOI input and actions, and external actions, and shall clearly communicate to MOI the time frames for response to these items such that MOI is fully informed of the requirements placed on their organization and external KSA organizations, agencies, ministries, and parties within the stakeholder community.
- 1.4.1.16 The CONTRACTOR shall be responsible for the delivery/distribution of information and communications to all the relevant appropriate stakeholders as per the agreed Communication Management plan with the MOI.
- 1.4.1.17 The CONTRACTOR shall be responsible for the development, implementation and active management of a Quality Management System which documents the policies, quality standards and requirements, review and acceptance processes, for all program/project planning, controlling, monitoring and reporting activities.
- 1.4.1.18 The CONTRACTOR shall implement, apply, maintain, audit and report on the Quality Assurance requirements as defined within the Project Management Plan (to be delivered by the CONTRACTOR) and the latest ISO 9001 and ISO 10006 standards.
- 1.4.1.19 For Contract Management, the CONTRACTOR shall deliver monitoring of contract compliance, terms and conditions, overseeing and reviewing overall performance of service level agreements, managing documentation of change requests and variations, delivery of contract deliverables, invoicing & payment procedures, and frequent concise and accurate management reporting.
- 1.4.1.20 The CONTRACTOR shall be responsible for timely review of deliverables for functional and technical accuracy/completeness and, compliance with applicable regulations.

- 1.4.1.21 Configuration Management by the CONTRACTOR shall include ensuring that the Configuration Items (CI's) are placed under proper configuration management (version control, etc.) control to ensure the latest version of a CI is being used/evaluated and that there is an audit trace of all changes.
- 1.4.1.22 The CONTRACTOR shall employ configuration and release management on each of the OSS911 system releases.
- 1.4.1.23 All system upgrades shall be tested, proven and accepted within the CONTRACTOR'S environment before roll out onto the operational systems to ensure maintenance of the systems operational availability to the MOI.
- 1.4.1.24 Weekly standard Program Status Report (PSR) formats shall be delivered by the CONTRACTOR to the MOI. The format shall include the following as a minimum:
- 1.4.1.24.1 Program Status Summary including brief Red Amber Green (RAG) summary of the project.
 - 1.4.1.24.2 A dashboard document for quick, high-level status view by MOI stakeholders and decision-makers on a project management reporting system portal.
 - 1.4.1.24.3 Progress during the reporting period
 - 1.4.1.24.4 Issues
 - 1.4.1.24.5 Risks
 - 1.4.1.24.6 Changes
 - 1.4.1.24.7 Forward look for progress in the next reporting period
- 1.4.1.25 The CONTRACTOR's Project Manager shall be an employee of the CONTRACTOR and must be authorized to make decisions on behalf of the CONTRACTOR and all subcontractors.
- 1.4.1.26 As part of their team and their subcontractor team, the CONTRACTOR shall assign sufficient qualified subject matter expert resources to the project to deliver the solutions required in this RFP for each of the RFP work streams / work packages.
- 1.4.1.27 Unless otherwise agreed with the MOI the CONTRACTOR shall provide Documentation using Microsoft Office components. Adobe Acrobat PDF format is permitted but all submitted documentation shall also be provided in its native format (i.e. .docx, .doc, .xlsx, .xls, .pptx, .ppt, .mpp etc.).
- 1.4.1.28 All documents (drafts or final versions) submitted by the CONTRACTOR shall have a document control section that includes the document versioning, unique reference, history of the document editing, document review by all subcontractors and document approval by the CONTRACTOR.
- 1.4.1.29 The OSS911 project schedule – covering all significant deliverables and milestones, for all sub-projects and work-streams, shall be maintained by the CONTRACTOR in MS Project format and reported periodically in electronic and hard copy.
- 1.4.1.30 The CONTRACTOR shall provide a detailed breakdown of all items in the bill of quantity which have been:

1.4.1.30.1 Grouped together and use the word “solution” in the item description.

1.4.1.30.2 Identified with item type as “LOT”.

1.5 TECHNICAL AND OPERATIONS SERVICE MANAGEMENT AND SUPPORT (TOSMS)

1.5.1.1 The CONTRACTOR shall deliver provide an Information Technology Infrastructure Library (ITIL) compliant technical and operations service management and support (TOSMS) for all 911 systems and technologies for MOI 911 centers under the scope of this RFP – for Makkah 911 the upgrade implementation by the CONTRACTOR will be the first six months of the contract followed by the operations and maintenance by CONTRACTOR for the next thirty months, and for Madinah 911, the operations and maintenance will be for thirty-six months of the contract.

1.5.1.2 The CONTRACTOR TOSMS shall be functional 24 hours a day, 365 days a year i.e., all around the year at all times including any and all holidays.

1.5.1.3 The 911 Centers shall have permanent on-site support personnel in a tiered-support model to be provided by the CONTRACTOR for the duration of the contract as per the BOQ

1.5.1.4 The CONTRACTOR provided TOSMS Solution shall include as a minimum:

- 1.5.1.4.1 A 24x7 centralized virtual helpdesk reachable by a single telephone number and by a web-based self-service portal.
- 1.5.1.4.2 An integrated framework and controls-based approach that is modelled on latest versions of internationally accepted standards including ISO 20000-1:2011 and ITIL.
- 1.5.1.4.3 A single point of responsibility for the management and delivery of the TOSMS.
- 1.5.1.4.4 Competent and qualified team for TOSMS.
- 1.5.1.4.5 Web based system which provides trouble ticketing, records alarms and fault reports, performs trend analysis as part of the TOSMS.
- 1.5.1.4.6 Compliance with the Service Level Agreements (SLA) requirements as defined in this document.
- 1.5.1.4.7 Compliance with the 911 systems availability requirements defined in this document and in accordance with the principle of Total Quality Management with ISO standards.

1.5.1.5 Within their proposal, the BIDDER shall provide the overall approach for TOSMS that includes:

- 1.5.1.5.1 How the CONTRACTOR shall support and maintain the 911 systems and technologies including listing and description of all hardware, software and equipment.
- 1.5.1.5.2 Human resources mobilization schedule along with their CVs.
- 1.5.1.5.3 The spares holding level for the spares identified as necessary and urgent by the BIDDER for maintaining the SLA and for meeting the performance and availability requirements.
- 1.5.1.5.4 Proposed team structure with roles and responsibilities, and governance.
- 1.5.1.5.5 Proposed maintenance planning including the scheduled and non-scheduled preventative maintenance plans and systems fault response and rectification requirements (including timings).
- 1.5.1.5.6 Proposed response times and priority levels for all systems and technologies of the 911 centers.
- 1.5.1.5.7 Proposal for system and component upgrades, updates and patches.

1.5.1.6 As part of the TOSMS implementation, the CONTRACTOR shall:

- 1.5.1.6.1 Design, document and implement all the required service processes, procedures and metrics for TOSMS including incident management, problem management, configuration management, change management, release management, availability management and service level management.
- 1.5.1.6.2 Design, document and implement a service catalogue and an integrated ITIL-compliant Configuration Management Database (CMDB) updated with a unique Asset Number for each item, including a bar code label or bar code tag label, which can be read electronically.
- 1.5.1.6.3 Deliver all the equipment necessary for reading the bar codes electronically, a bar code generator and printer, and for storing the bar codes in a database in the TOSMS solution.
- 1.5.1.6.4 Deliver service reports, trends and fault analysis reports, Mean Time to Repair (MTTR) reports, Mean Time between Failure (MTBF) reports, capacity utilization reports, availability reports and service level reports as agreed with the MOI.
- 1.5.1.6.5 Conduct 911 systems and technologies' knowledge transfer to the MOI.
- 1.5.1.6.6 Maintain the spares at the spares holding levels and shall submit the human resource plans for each month of live operations prior to the start of the month.
- 1.5.1.6.7 Design, document and implement preventative and corrective maintenance procedures.
- 1.5.1.6.8 Design and deploy a smartphone app that shall be capable of being accessed securely based on role-based-access-control, for the latest versions of iOS and Android systems.
- 1.5.1.6.9 Design, document and implement a dashboard that displays at multiple levels, a set of Key Performance Indicators (KPIs) for all the activities and actions, as agreed with the MOI.
- 1.5.1.6.10 Design, document and implement automatically generated alarms to authorized personnel when thresholds are exceeded for various metrics – availability levels, service levels, capacity levels, spare levels, faults, etc. for all 911 applications, systems, integrations, services, technologies and infrastructure.
- 1.5.1.6.11 Design, document and implement a database of its own, a knowledge base of known incident workarounds, errors and problem solutions, root cause analyses, and knowledge of similar incidents and resolutions.

1.5.1.7 Should the MOI so desire, the CONTRACTOR shall make available some or all of their TOSMS team members to be contracted by subsequent MOI contractors who will operate and maintain the 911 centers.

1.5.2 اتفاقيات مستوى الخدمة (SERVICE LEVEL AGREEMENTS (SLAs))

1.5.2.1 The CONTRACTOR shall deliver the 911 solutions including the CMA solution the Service Level specifications (including service availability, service SLAs and response times) set out in this section.

1.5.2.2 The BIDDER shall agree and confirm that the list of problems stated below under the different priority levels is not exhaustive and MOI shall have the full right to classify and categorize a problem or a request in a certain priority level according to operational criteria that may arise. Lower

priority faults can be escalated into higher priority fault if it is determined by the MOI that the severity increases to the extent that it falls within the higher priority fault definition as described below or should special circumstances arise which are recognizable as such in the contract.

1.5.2.3 All the CONTRACTOR solutions shall be bound by the following definitions:

1.5.2.3.1 Initial response time – The time within which the CONTRACTOR support personnel will provide initial response to any observed and/or reported problem, alarm and incident that occurred and that is logged into the Help Desk.

1.5.2.3.2 Workaround – A workaround is a temporary way to restore service failures to an MOI acceptable level and that includes the action(s) required to limit the effects of a problem.

1.5.2.3.3 Resolution Time – The time between the response time and the time the effected service is resolved. This means that the affected service is operational again to the required service level following a correction or permanent solution that was implemented by the CONTRACTOR.

1.5.2.3.4 Downtime – Inability to perform the relevant service or function by the user at the performance levels defined as per the contract, and which is resulting from factors within the direct responsibility and scope of the CONTRACTOR.

1.5.2.3.5 Interval of downtime – Time difference between the start and end of the downtime as defined above.

1.5.2.3.6 Number of Impacted Operator – Number of unique logged-in operators which are impacted by the downtime for the relevant service or function.

1.5.2.3.7 Total Number of Operators – Total number of unique logged-in operators within a 24-hour period for the relevant service or function.

1.5.2.4 If CONTRACTOR replaces a damaged configuration item (hardware / software) with a new one until the faulty one is repaired then the permitted time to repair the faulty one shall be within resolution time.

1.5.2.5 The CONTRACTOR shall implement any planned outages and/or downtime for the defined priority classes based on MOI approval after submitting a detailed change request for any planned change – however, any violation of the planned outage and/or downtime shall be considered as service disruption under CONTRACTOR responsibility.

1.5.2.6 The BIDDER shall agree and confirm that MOI has the right to define time periods in which no planned outages are allowed.

1.5.2.7 The CONTRACTOR CMA solution shall cover all trouble tickets and incidents type including critical, major, minor and service requests and shall specify response time, and mean time to repair along with recovery time objectives and recovery point objectives.

1.5.2.8 The CONTRACTOR shall provide monthly service level compliance reports to the MOI that also includes the availability, reliability and maintainability of the CONTRACTOR 911 solutions, technical infrastructure and integrations as required by the MOI.

1.5.2.9 The CONTRACTOR shall be bound by the Service Level and Availability Matrix of Table1 below, and the severity and response SLAs of Table 2 below from the start of the live operations of the first agency at a 911 center within the scope of the project.

Table 1: Service Level and Availability Matrix

Service level and availability matrix with total available period of any service being one day (24 hours x 60 minutes x 60 seconds)			
Service	Service Level	Availability Target	Formula
Answering calls	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds} \times \text{number of impacted call takers}}{\text{total number of call takers} \times 24 \times 60 \times 60} \right\} \%$
Recording of all calls on IP telephony	1	99.999%	Total Recording Duration/Total call duration %
IRMS functions and services including supervision	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds} \times \text{number of impacted operators}}{\text{total number of IRMS operators} \times 24 \times 60 \times 60} \right\} \%$
GIS functions, maps, data and services including supervision	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds} \times \text{number of impacted operators}}{\text{total number of operators} \times 24 \times 60 \times 60} \right\} \%$
Monitoring and management of all elements – applications, systems, integrations, services and operations	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds} \times \text{number of impacted elements}}{\text{total duration}} \right\} \%$
Security services	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds} \times \text{number of affected devices}}{\text{total number of devices} \times 24 \times 60 \times 60} \right\} \%$
Integrations within the CONTRACTOR delivered OSS911 solutions	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
Network and Domain services	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
Database solutions	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
TOSMS solution and services	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
Location services for callers and for field responders	1	99.999%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
Backups	1	99.999%	Total backup achieved /Planned backup
Dashboard operations	2	99.99%	$1 - \frac{\text{impacted seconds}}{24 \times 60 \times 60} \%$
Liaison events	2	99.99%	$1 - \left(\frac{\text{Number of calls for which liaison events should have been created but not created and/or dispatched}}{\text{Total number of calls for which liaison events should have been created and/or dispatched}} \right) \%$
Business Intelligence and Reporting	2	99.99%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$
Integrations with all external non-OSS911 applications/systems/services	2	99.99%	$1 - \left\{ \frac{\text{Downtime in seconds}}{24 \times 60 \times 60} \right\} \%$

All other OSS911 solution services and functions	2	99.99%	$1 - \left\{ \frac{\text{Downtime in seconds}}{(24 \times 60 \times 60)} \right\} \%$
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Table 2: Severity and Response SLAs

Severity and response SLAs			
Incident severity level	Incident Description	Initial response	Time to resolution
Severity Level One (Critical and/or Urgent)	a. Loss / prohibition / interruption of any of the P1 services affecting 5% of designated service capacity b. Loss and/or corruption of any data and/or records c. Issues without any workaround d. Service and/or program aborts e. Loss of redundancy f. Security breaches g. More than one servers / storage / network / security devices and/or components disrupted at the same time h. Loss and/or corruption of backups i. Loss of monitoring and management of any system and/or device and/or service j. Thresholds exceeded for critical parameters of P1 services	Centralized help desk available 24x7 – telephonic response available immediately, in-person response shall be within 10 minutes of notification	4 hours, regardless of time of the day the incident is reported, and for any day of the week
Severity Level Two (High Severity)	a. Primary purpose compromised with a loss / prohibition / interruption of any of the P1 services affecting less than 5% of designated service capacity b. Productive use significantly impacted and workaround generally not available c. Significant degradation of system alarms or trouble reporting d. Critical risks with system capacity / performance / redundancy parameters and thresholds e. Components failure without loss of service f. Databases corruption without loss of service g. Intermittent faults h. Queries for information in databases and systems i. Faults in reporting systems	Centralized help desk available 24x7 – telephonic response available immediately, in-person response with a workaround shall be within 30 minutes of notification	8 hours, regardless of time of the day the incident is reported, and for any day of the week

Severity Level Three (Medium Severity)	a. Primary purpose compromised with a loss / prohibition / interruption of any of the P1 services affecting less than 1% of designated service capacity b. Productive, but incomplete operation and workarounds generally available	Centralized help desk available 24x7 – telephonic response available immediately, in-person response with a workaround shall be within 60 minutes of notification	48 hours, regardless of time of the day the incident is reported, and for any day of the week
Severity Level Four (Low Severity)	a. Cosmetic problems b. Workarounds or configurable options generally available	Centralized help desk available 24x7 – telephonic response available immediately, in-person response shall be within 2 hours of notification if necessary	72 hours, regardless of time of the day the incident is reported, and for any day of the week

1.6 (SYSTEMS DEVELOPMENT, INTEGRATION AND TESTING) **تطوير الأنظمة وتكاملها واختبارات القبول**

1.6.1 (REQUIREMENTS FOR A FULLY INTEGRATED OSS911 SOLUTION) **متطلبات حل التكامل للمشروع**

1.6.1.1 The BIDDER shall propose all OSS911 solutions that are based on private cloud architecture delivering secure web-based access for all user and for all system management and administration functions.

1.6.1.2 The BIDDER shall not be constrained by architecture descriptions and features in the document.

1.6.1.3 The BIDDER shall describe their approach to each of the integration elements for OSS911 systems.

1.6.1.4 The CONTRACTOR shall implement all requirements in this contract with the understanding that:

1.6.1.4.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

1.6.1.4.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

1.6.1.4.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

1.6.1.5 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

1.6.1.6 The CONTRACTOR shall supply a microservices-based Service-oriented architecture (SOA)-based smart and AI-powered solution

enabling loose coupling, independent evolution of components and standardized integration.

1.6.1.7The CONTRACTOR solution for middleware shall allow creation of secure interfaces to other systems using service-oriented architecture such that the demand for data and databases is served on-the-fly through secure web services.

1.6.1.8The BIDDER shall clearly explain with how their proposed OSS911 solution supports SOA by providing a middleware that allows creation of interfaces to third party systems using microservices-based service-oriented architecture. This explanation shall include:

1.6.1.8.1New modules development and integration

1.6.1.8.2Native interoperability

1.6.1.8.3Architectures supported

1.6.1.8.4Application servers supported

1.6.1.8.5Extensible Markup Language (XML) support

1.6.1.8.6JavaScript Object Notation (JSON) support

1.6.1.8.7Representational State Transfer (REST) support

1.6.1.8.8Web Services standards support

1.6.1.8.9Integration with legacy systems

1.6.1.8.10Dependencies on any third-party product to deliver Web Services

1.6.1.8.11Modelling tools supported

1.6.1.9The BIDDER shall explain the security architecture and capabilities of the proposed SOA solution.

1.6.1.10The BIDDER shall indicate in their proposal the scalability and expandability in terms of the percentage of installed capacity.

1.6.1.11The BIDDER shall clearly identify the transaction volume limits of the OSS911 solution under normal operating conditions, and in the event of the different scenarios of fallback between centers.

1.6.1.12The BIDDER shall explain the components/web services or other building blocks at a high and fairly detailed level and show how they lead to a scalable solution.

1.6.1.13The BIDDER shall describe the loose coupling among various components and how a component can be replaced without affecting the other workable components so as to minimize the downtime.

1.6.1.14The BIDDER shall explain how the solution responds gracefully in case of an unexpected event.

1.6.1.15The BIDDER shall describe in the proposed solution how the entire solution will be configured in a way that is easy to change and does not

require major changes to the solution and will allow the administrators to make deployment decisions more robust.

1.6.1.16The BIDDER shall describe the approach to interfacing in their proposal. These interfaces shall have the capability to allow any external MOI-authorized system, capable of interfacing using Webhooks, JSON, XML, W3C XML Schemas, REST interfaces, etc. to integrate with OSS911 systems to:

1.6.1.16.1Receive data from OSS911 system as an Incident Record (with details to be agreed during detailed design following contract award)

1.6.1.16.2Update (Modify / Change / Delete) data to OSS911 system (with details to be agreed during detailed design following contract award)

1.6.1.17The BIDDER shall describe, explain and annotate all the integration capabilities of their proposed solution including sample interface control documents (ICDs) that include proposed definition of the different fields mentioned in the various integration capabilities in this document.

1.6.1.18The CONTRACTOR shall deliver an integrated set of OSS911 applications as defined in this RFP as a fully integrated enterprise application product.

1.6.1.19All components and subsystems of the OSS911 solution shall be fully Arabized following the official KSA Arabic language rules to include the date and time display that follows the official KSA Arabic calendar and which is based on the Umm Al Qura, unless otherwise notified by the MOI.

1.6.1.20The CONTRACTOR shall supply an integrated suite of applications that utilizes a Service Oriented Architecture (SOA) and integrates the different subsystems of the OSS911 solutions in this document including but not limited to Incident and Resource Management System (IRMS), Common Operational Picture (COP) Solution, IP Telephony solution, Recording solution, all of which shall be interlinked by an Enterprise Service Bus (ESB) and shall be delivered through a Software Defined Network (SDN) in a private cloud infrastructure.

1.6.1.21The CONTRACTOR shall deliver all OSS911 solutions that are aligned with the high-level concept of operations described for OSS911 in the respective section within this document and with the concept of operations that will be finalized by the MOI during the detailed design stage following the contract start.

1.6.1.22The CONTRACTOR shall supply and deliver an Enterprise Service Bus (ESB) for message brokering between various OSS911 systems, subsystems and applications, orchestration, mediation, transformation and routing of services and events.

1.6.1.23All the CONTRACTOR solutions shall be capable of independent deployment of services and components therein.

1.6.1.24All the CONTRACTOR solutions shall be automated for start, stop, recovery and restart of components.

- 1.6.1.25The CONTRACTOR solution shall be enabled for minimal service disruption and/or interruption during updates, and shall be capable of rollback in case of update failures.
- 1.6.1.26The CONTRACTOR shall deploy all the OSS911 applications with a secure web-based front-end that provides all the functionalities of a 'thick' client.
- 1.6.1.27The CONTRACTOR shall deliver all OSS911 applications as commercial-off-the-shelf (COTS) products with required features requiring customization in interfacing and integration with third-parties, custom reports, custom user interface and custom data representation among other things.
- 1.6.1.28The CONTRACTOR shall deliver all OSS911 COTS applications which have been in the market for at least five years and which are the latest Original Equipment Manufacturer (OEM) versions of the COTS application vendor and which shall not have OEM end-of-life and/or end-of-support for the next five years.
- 1.6.1.29The CONTRACTOR solutions shall comply with the data and transport security requirements for OSS911 center outlined in this document and with the MOI security standards and policies.
- 1.6.1.30The CONTRACTOR shall deliver OSS911 systems interfaces allowing the OSS911 applications to:
- 1.6.1.30.1Be able to locate transparently across the network, thus providing interaction with another service or application
 - 1.6.1.30.2Be independent from network services
 - 1.6.1.30.3Be reliable and always available
- 1.6.1.31The CONTRACTOR shall implement full lifecycle management for all OSS911 solutions including but not limited to versioning and backward compatibility, controlled deprecation of services and contract management (schemas, APIs).
- 1.6.1.32The CONTRACTOR OSS911 solutions shall be capable of synchronous and asynchronous integration patterns, depending on the use case finalization with the MOI.
- 1.6.1.33The CONTRACTOR shall deliver an OSS911 solution that shall be integrated with the Automatic Location Identification (ALI) information available from the sources defined by MOI which include but are not limited to the National Information Center (NIC), the National Center for Security Operations (NCSO) and any third-party service provider that include technologies such as Advanced Mobile Location (AML), Enhanced Location Services (ELS) and eCall Minimum Set of Data (MSD).
- 1.6.1.34The word "location" as mentioned in this document anywhere includes the location returned from any of the sources authorized by the MOI.
- 1.6.1.35The CONTRACTOR solution shall ensure that all the location data returned from any location source including the latitude, longitude, radius,

time of positioning and level of confidence, and the eCall MSD is decoded and presented on the IRMS and stored in all the OSS911 solution databases and tagged with the related calls, incidents and alerts.

- 1.6.1.36The CONTRACTOR solution shall ensure that all the location data is presented to the operator in IRMS within the incident capture/creation forms of the computer aided dispatch solution and within the softphone / software telephony console that receives the emergency call at the operator – this presentation shall include the location source, the location coordinates and all the location attributes.
- 1.6.1.37The CONTRACTOR shall deliver an OSS911 solution that shall be integrated with the Automatic Number Identification (ANI) information available from the telephony service providers and/or sources identified by the MOI.
- 1.6.1.38The CONTRACTOR shall be responsible for ensuring that the capability exists for collecting, storing and presenting or displaying the caller ID information from each of the external telephone service providers while identifying the service provider from which the call was received and the incoming source from which the call was received.
- 1.6.1.39The CONTRACTOR delivered OSS911 solutions shall retrieve and store call detail records information from the telephony service providers and/or sources identified by the MOI.
- 1.6.1.40The CONTRACTOR shall deliver an OSS911 solution that shall be integrated with the caller information database available from the National Information Center (NIC) and/or other sources identified by the MOI.
- 1.6.1.41The CONTRACTOR solution shall allow the population of the OSS911 systems' fields with relevant person information available from the different integrations – such as but not limited to the address, the national water company meter location identification data, and Saudi electricity company meter location data.
- 1.6.1.42The CONTRACTOR shall deliver an OSS911 solution that is integrated with MOI-authorized databases including but not limited to vehicle owner, vehicle license plate information, residence information, visitor information databases etc.
- 1.6.1.43The CONTRACTOR delivered solution for OSS911 shall include a gazetteer capability that provides free-form text-based search, spatial search, fuzzy search and matching, and retrieval of all the stored geospatial information around set of coordinates.
- 1.6.1.44The CONTRACTOR shall deliver the complete and detailed integration plans following award of contract.
- 1.6.1.45The CONTRACTOR shall ensure that status updates and messages from resources assigned to incidents shall automatically populate the OSS911 systems logs including the incident log.
- 1.6.1.46The CONTRACTOR shall configure the OSS911 solutions for integration of automatic resource location (ARL) such that the OSS911 solution receives and stores location data from any location service

enabled device and/or any automatic resource location system/service authorized by the MOI.

- 1.6.1.47The CONTRACTOR shall deliver OSS911 solution that has the capability to receive eCall information using the eCall MSD from the authorized mobile network operators (MNOs) and/or third-party service providers (TPS), and to route the voice part of the eCall through the IP telephony solution to the operators using the MOI-defined queue for emergency calls along with capturing the caller ID and the incident related information associated with this voice call that is contained within the related MSD and making all of these available to the operator through the various systems in the OSS911 solution (IP Telephony solution, the IRMS solution, etc.) within the various fields therein, and storing the voice, the caller ID and MSD within the emergency call and incident databases as part of the call and incident live and historical records.
- 1.6.1.48The CONTRACTOR shall deliver an OSS911 solution with open architecture and platform independent interfaces that allows MOI to perform future integrations with other systems.
- 1.6.1.49The CONTRACTOR shall deliver OSS911 solutions with the capability to serve on-the-fly the demand for data and databases through open standards and independent web services APIs utilizing Web Service Description Language (WSDL), XML and Simple Object Access Protocol (SOAP) based on most recent W3 XML standards and through APIs utilizing Representational State Transfer (REST), XML, JSON etc.
- 1.6.1.50The CONTRACTOR shall design the OSS911 systems integrations such that each interface and/or integration with the external systems may contain a maximum of fifty (50) bidirectional rewritable fields with combinations of primitive and custom data types including but not limited to String, Long Integers, Date-time, Double, Arrays, universally unique identifier (UUIDs), Globally Unique Identifier (GUID), etc.
- 1.6.1.51The CONTRACTOR shall consider the size limitation of these fields to the maximum supported for each of the data types and that each of these fields may contain alphanumeric Arabic and English characters, numbers, and special characters, and shall design the OSS911 databases and interfaces accordingly.
- 1.6.1.52The CONTRACTOR shall deliver a smart and AI-powered integrated recording solution that records in a single integrated repository all data, voice, video and text relating to operations across all OSS911 subsystems.
- 1.6.1.53The CONTRACTOR delivered integrated recording solution shall allow for linking of the stored information with related incidents and operational alerts, if any, for all OSS911 subsystems.
- 1.6.1.54The CONTRACTOR delivered integrated recording solution shall allow retrieval of related voice, video, data and text for a given incident, as a single record for all OSS911 subsystems which is retrievable through a web-based retrieval solution.
- 1.6.1.55The CONTRACTOR delivered integrated recording solution shall support the retrieval of recordings as a single integrated record based on any or a combination of the following parameters, including but not limited to:

- 1.6.1.55.1Date
- 1.6.1.55.2Time
- 1.6.1.55.3Calling line identity
- 1.6.1.55.4Called number
- 1.6.1.55.5Operator position (console ID)
- 1.6.1.55.6Operator ID
- 1.6.1.55.7Incident ID
- 1.6.1.55.8Camera ID
- 1.6.1.55.9Vehicle plate number
- 1.6.1.55.10Radio Talk group
- 1.6.1.55.11Radio ID

1.6.2 **مستودع بيانات متكامل وحل ذكاء الأعمال والتحليلات وإعداد التقارير (OSS911 SOLUTION BUSINESS INTELLIGENCE, DATABASE REPOSITORY, DASHBOARDS AND REPORTS)**

- 1.6.2.1The CONTRACTOR shall migrate existing operational data from existing datastores for Makkah 911 center into the single highly available structured relational database for the OSS911 solution.
- 1.6.2.2The CONTRACTOR shall implement all the requirements as below on the single highly available structured relational database for the OSS911 solution.
- 1.6.2.3The CONTRACTOR shall implement a highly available smart and AI-powered central structured information database solution that combines in a relational database the data from the different subsystems of the OSS911 solution and from any other relational databases identified by the MOI during the design stage.
- 1.6.2.4The CONTRACTOR shall deliver this highly available central structured database repository solution for all OSS911 systems covering:
 - 1.6.2.4.1Live data
 - 1.6.2.4.2Historical data (any data that is older than the current day)
- 1.6.2.5The CONTRACTOR shall consider for the central OSS911 database repository the expected amount of data related to the incident and different data from different systems and shall apply the best practices related to the database design, implementation, administration, and maintenance this shall include but not limited to (Indexing, Portioning, Indexed Views, Optimized queries, Period routine database checks and reports.).
- 1.6.2.6The CONTRACTOR shall consider the expected database expansion and provide capacity planning for this database.
- 1.6.2.7All OSS911 contractor supplied databases and any information received from the OSS911 connections to the legacy databases

connected to OSS911 shall be searchable against a configurable set of selection parameters and criteria which shall be displayed to operators for immediate recovery.

- 1.6.2.8The CONTRACTOR implementation of the OSS911 solution shall enable an authorized user to be able to query and generate smart and AI-powered reports, both on a regular and ad-hoc basis, based on a set of parameters, which may include, or exclude, incidents from more than one agency, and which may include or exclude, data relating to one or more subsystems of the OSS911 solution.
- 1.6.2.9The CONTRACTOR shall conduct workshops with the MOI for the CONTRACTOR to define the parameters to be queried, along with the search criteria, the views enabling the queries, the expected output and the visual and printed layout of the output.
- 1.6.2.10The CONTRACTOR shall implement the defined queries, views and reports from the different OSS911 databases.
- 1.6.2.11The CONTRACTOR OSS911 solution shall provide for the ability to create additional queries as needed with independent search parameters, filters, and alerts.
- 1.6.2.12The CONTRACTOR OSS911 solution shall provide for the ability to add, edit, and delete parameters from any query at any time on the fly.
- 1.6.2.13The CONTRACTOR OSS911 solution shall provide the ability to export and import parameters for inclusion in the query via a .CSV or another common file format.
- 1.6.2.14The CONTRACTOR OSS911 solution shall provide the ability to export the OSS911 solution reports in PDF and in Microsoft Word format.
- 1.6.2.15Reporting tools shall allow historical reporting of incidents on a periodic basis. It shall be possible to create short reports using a common Structured Query Language (SQL) based on operator role and privileges. More complex reports shall use a structured reporting tool or an enterprise reporting system linked to the middleware solution.
- 1.6.2.16The BIDDER shall indicate the format of reporting queries together with the required skills for their use. Typical types of reports may include:
 - 1.6.2.16.1Breakdown of incidents by type and severity
 - 1.6.2.16.2Incidents by location and time
 - 1.6.2.16.3Average incident resolution times
 - 1.6.2.16.4Average response times
- 1.6.2.17The CONTRACTOR shall provide a solution for the customer feedback to be included in business intelligence systems included in OSS911 solution, thus forming part of OSS911 performance statistics.
- 1.6.2.18The CONTRACTOR delivered OSS911 solutions shall provide all existing report types that are available in the existing Makkah 911 center using in-built reporting tools and/or leveraging existing report tools and AI-capabilities.

- 1.6.2.19The CONTRACTOR shall implement a smart and AI-powered single integrated reporting solution for all the subsystems of the OSS911 solution for all the OSS911 centers in scope.
- 1.6.2.20The CONTRACTOR shall conduct workshops with the MOI for the CONTRACTOR to arrive at the final number and type of reports that the OSS911 solution shall provide. The following are some examples of the reports and are provided here only for an indication of the types of reports – this is not the final list of reports:
- 1.6.2.20.1The number of rings that a caller hears before a call taker answered his call.
 - 1.6.2.20.2The number of calls presented to a call taker and the number of calls attended to by that call taker.
 - 1.6.2.20.3Which party terminated / cleared / ended the call - the call taker or the caller.
 - 1.6.2.20.4A history of the lost calls by the same caller.
 - 1.6.2.20.5The number of missed calls, calls transferred and annoying /malicious calls.
- 1.6.2.21In addition to being able to search and report on data from OSS911 center terminals as described in the above paragraphs, it shall also be possible to securely access this information from suitably authorized computers attached to the OSS911 network through a web browser.
- 1.6.2.22The CONTRACTOR shall provide a web browser-based business intelligence reporting capability that provides:
- 1.6.2.22.1Availability of data, statistics and information from OSS911 systems that shall help strategic decision making to authorized users.
 - 1.6.2.22.2Availability of all historical and current data, statistics and information from the OSS911 systems that shall help tactical decision making to authorized users.
- 1.6.2.23The business intelligence solution shall provide customizable reports, displayable at dashboards and printable in legible formats for all OSS911 systems.
- 1.6.2.24Based on their prior experience with public safety 911, emergency and crisis management and video management work, the BIDDER shall identify the predefined reports and dashboards that are available in the COTS products proposed by them for the different subsystems of OSS911 solution.
- 1.6.2.25Apart from the predefined reports within the COTS solutions for the various subsystems of OSS911 solution, the CONTRACTOR shall define forty (40) custom reports and twenty (20) dashboards per OSS911 center in scope, and shall implement these in the OSS911 solutions at different MOI nominated locations.

- 1.6.2.26 The CONTRACTOR shall provide solutions that enable reporting, statistics and dashboards for display over the various display devices available at the OSS911 center and at different MOI nominated locations.
- 1.6.2.27 Business Intelligence dashboards and displays to be delivered by the CONTRACTOR include at least the:
- 1.6.2.27.1 Live (real-time) Common Operational Picture
 - 1.6.2.27.2 Historical and current (real-time) performance statistics, trends and graphs identifying average, peak and transient incident traffic for all subsystems of the OSS 911 solution
 - 1.6.2.27.3 Historical and current (real-time) operator performance per shift for all functions of the OSS911 center
 - 1.6.2.27.4 Other historical and current (real-time) operational measures that will be mutually agreed between the CONTRACTOR and the MOI
- 1.6.2.28 The CONTRACTOR shall define reports matching the existing reports and dashboards in Makkah 911 center, along with other customizations as discussed with the MOI and shall implement these in the OSS911 solutions at different MOI nominated locations.
- 1.6.2.29 The various OSS911 systems shall make available primary incident and alerts information and related data as a web service so that other OSS911 and MOI applications may be able to accept this information for use through the middleware.
- 1.6.2.30 The CONTRACTOR shall include charts and graphical representation of the data in their reports, with the charts type and the representation to be agreed with the MOI as part of the detailed design.
- 1.6.2.31 In order to facilitate integration to the OSS911 solution:
- 1.6.2.31.1 The CONTRACTOR shall expose all the OSS911 systems and the related database and data structures, tables, fields and parameters of the OSS911 solution database repository as database-to-database interactions, Application Programming Interface (APIs), as REST interfaces, webhooks and as web services with access to these controlled by security mechanisms defined in the rest of this document.
 - 1.6.2.31.2 The CONTRACTOR shall develop, supply and implement the required adapters for the ESB so that external applications and solutions are able to access all the OSS solutions and related database and data structures, tables, fields and parameters of the OSS911 solution database repository.
- 1.6.2.32 The CONTRACTOR shall deliver an electronic unencrypted copy of the complete database repository to the MOI in relational database software format along with the corresponding definitions and descriptions of the different fields, tables, views, filters and reports, the database dictionary and the database models – all in Microsoft Office formats.
- 1.6.2.33 When a given OSS911 center is working as a failover center, the CONTRACTOR solution for business intelligence and dashboards shall

provide a single unified dashboard for the OSS911 centers that are being served, while maintaining identification and display of the parameters and statistics by province individually.

1.6.3 **التكامل مع أنظمة معلومات الموارد (INTEGRATION WITH RESOURCE INFORMATION SYSTEMS)**

1.6.3.1The CONTRACTOR shall deliver an OSS911 solution that shall be integrated through web services with the resources data available from the sources identified by the MOI and that is stored in the centralized database repository for the OSS911 solution.

1.6.3.2The CONTRACTOR shall deliver integration of the OSS911 solution with different resource information and data systems in the form of an integrated relational database-based information and a central data repository available to all OSS911 solution.

1.6.3.3The CONTRACTOR shall capture, populate, maintain and administer within OSS911 systems integrated relational database the data and information related to the different third parties involved.

1.6.3.4The CONTRACTOR delivered information repository shall allow all management and maintenance operations with the integrated relational database for authorized users including create, update, modify, query, insert, view, report, delete, print, export etc. along with the required tools for achieving these operations.

1.6.3.5The CONTRACTOR shall deliver this integration such that it allows OSS911 systems to interface with, consume services from, and exchange data with other external systems including but not limited to:

1.6.3.5.1Other AI-enabled tools and information systems

1.6.3.5.2Other communication systems utilizing SIP protocol and SIP protocol extensions

1.6.3.5.3Other computer aided dispatch systems

1.6.3.5.4Other Geographical Information Systems (GIS)

1.6.3.5.5Other video management systems

1.6.3.5.6Other hearing-impaired systems with video call capability

1.6.3.5.7Maidan platform and related systems from the MOI

1.6.3.5.8Central Information Datawarehouse (CID) solution from the MOI

1.6.3.5.9Kullunaamn system from the MOI

1.6.3.5.10Drones and IoT platforms

1.6.3.5.11Smart cameras

1.6.3.5.12Fire detection and management systems

1.6.3.5.13Alarm management systems

- 1.6.3.5.14Municipality systems
- 1.6.3.5.15Transportation Systems
- 1.6.3.5.16Traffic Management Systems
- 1.6.3.5.17Hospital management systems
- 1.6.3.5.18Saudi Red Crescent Authority systems
- 1.6.3.5.19Casualty management systems
- 1.6.3.5.20Weather monitoring and management systems
- 1.6.3.5.21Disasters monitoring and management systems
- 1.6.3.5.22Early warning systems
- 1.6.3.5.23Predictive modelling systems
- 1.6.3.5.24Civil Defense (CD) Systems
- 1.6.3.5.25Public Security Directorate (PSD) systems
- 1.6.3.5.26Supplies management systems
- 1.6.3.5.27National Center for Security Operations (NCSO) systems
- 1.6.3.5.28National Information Center (NIC) systems
- 1.6.3.5.29Shelters
- 1.6.3.5.30System of Entertainment Authority
- 1.6.3.5.31Management system for general sports authority
- 1.6.3.5.32Management system for educational institutions
- 1.6.3.5.33Automatic resource location systems
- 1.6.3.5.34Food and Drug Authority systems
- 1.6.3.5.35Ministry of Housing systems
- 1.6.3.5.36Tourism systems
- 1.6.3.5.37Civil Registry
- 1.6.3.5.38Mobile Network Operators CRM
- 1.6.3.5.39National Address

1.6.3.6The CONTRACTOR shall deliver information and data repository for the interfaces to the each of the systems as above that contains different fields for holding different resource attributes as below but which are not limited to:

- 1.6.3.6.1Type / classification

1.6.3.6.2Unique ID

1.6.3.6.3Subtype / classification

1.6.3.6.4Name

1.6.3.6.5Location Coordinates

1.6.3.6.6Street

1.6.3.6.7District

1.6.3.6.8Town

1.6.3.6.9City

1.6.3.6.10Governorate

1.6.3.6.11Region

1.6.3.6.12Date

1.6.3.6.13Time

1.6.3.6.14Severity / Priority

1.6.3.6.15Data / Information security classification

1.6.3.6.16Status

1.6.3.6.17Tags

1.6.3.6.18User information

1.6.3.6.19Owner name

1.6.3.6.20Contact email

1.6.3.6.21Contact telephone (landline)

1.6.3.6.22Contact mobile phone

1.6.3.6.23Contact web site

1.6.3.6.24Nickname

1.6.3.6.25Alternate contact email

1.6.3.6.26Alternate contact telephone (landline)

1.6.3.6.27Alternate contact mobile phone

1.6.3.6.28Time duration

1.6.3.6.29Description

1.6.3.6.30Model

1.6.3.6.31Access information

1.6.3.6.32 Total quantity / capacity

1.6.3.6.33 Current / used quantity / capacity

1.6.3.6.34 Available quantity / capacity

1.6.3.6.35 Available services

1.6.3.6.36 Completed action

1.6.3.6.37 Pending / next action

1.6.3.6.38 Schedules

1.6.3.6.39 Supporting file / plans

1.6.3.6.40 Direction of travel

1.6.3.6.41 Speed of travel

1.6.3.6.42 Update frequency

1.6.3.6.43 Miscellaneous

1.6.3.7 The interface descriptions, fields and third-party systems mentioned above are not the final and definitive list – the CONTRACTOR shall agree the final interface configurations, fields and third-party systems with the MOI during the detailed design and shall deliver OSS911 integrations accordingly.

1.6.3.8 The CONTRACTOR shall configure their solutions to handle partial and/or incomplete return of resources and resource attributes – in other words, it is not necessary that all of the resource attributes be fully defined for a resource to be managed through the OSS911 systems.

1.6.3.9 The CONTRACTOR shall deliver this integration such that the information stored is available to be consumed by other authorized users and/or applications in an open format, for example through web services defined with REST, XML, JSON, etc.

1.6.3.10 The CONTRACTOR shall deliver OSS911 solutions integration with the MOI CID solution, PSD Amn system, Kullunaamn and with MOI Maidan system through web services that delivers the capability of sharing of information with these solutions – examples of information to be shared include incident information records, reports, incident closure information, event types, data, location, responding agency, responding station etc. These are examples only and actual details of the interface configuration shall be agreed by the CONTRACTOR with the MOI prior to implementing the integration.

1.6.3.11 The CONTRACTOR shall deliver OSS911 solutions integration with the Saudi Red Crescent Authority (SRCA) system through web services that delivers the capability of automatically sharing the incident electronically with SRCA based on defined event types and subtypes, delivers the capability of updating the either systems when either OSS911 or SRCA creates, updates and/or cancel an incident and that delivers sharing of information between the two systems – examples of information to be shared include incident information records, reports, incident closure information, event types, data, location, responding

agency, responding station etc. These are examples only and actual details of the interface configuration shall be agreed by the CONTRACTOR with the MOI prior to implementing the integration.

1.6.4 **تكامُل حلول OSS911 مع الشبكات التابعة لوزارة الداخلية (OSS911 SOLUTION INTEGRATION WITH MOI RADIO COMMUNICATIONS NETWORKS)**

1.6.4.1The CONTRACTOR shall supply OSS911 systems that shall be integrated by the CONTRACTOR with the MOI radio communications networks in that province as determined by MOI.

1.6.4.2The BIDDER shall note that the details regarding existing MOI radio communication networks shall be shared with the CONTRACTOR after the award of the contract.

1.6.4.3The CONTRACTOR shall supply OSS911 systems that are integrated with MOI radio communications networks whether these networks are being served by their main or standby sites.

1.6.4.4The CONTRACTOR is responsible for the procurement, supply, installation, configuration, commissioning and testing of all equipment required to support such integration.

1.6.4.5The CONTRACTOR supplied interfaces for this integration at each OSS911 Center shall be sized to support the numbers of dispatchers, the joint operations and the crisis management team.

1.6.4.6Within their proposal, the BIDDER shall specify the detailed characteristics of this integration.

1.6.4.7The CONTRACTOR shall provide a full specification of this integration including details of the interfaces, APIs, etc.

1.6.4.8The CONTRACTOR shall provide OSS911 systems integration with MOI radio communications networks that shall include the following as a minimum:

1.6.4.8.1Emergency Alarm Notification

1.6.4.8.2Location data

1.6.4.8.3Status Codes received

1.6.4.8.4Short Data Messages

1.6.4.8.5Resource information and management

1.6.4.8.6Radio communications recording system

1.6.4.8.7Network Time Protocol (NTP) integration – the NTP source will be advised by MOI

1.6.4.9The BIDDER shall include in their proposal any other functions that their proposed integration provides.

1.6.5 **تصميم حلول أنظمة OSS911 (OSS911 SOLUTION DESIGN)**

1.6.5.1The BIDDER shall provide a preliminary high-level systems architecture and design blueprint for the OSS911 applications that

includes a conceptual data model for OSS911 systems based on their understanding of the RFP.

- 1.6.5.2 The BIDDER shall include a narrative that describes the conceptual data model along with a large, readable print of the conceptual data model.
- 1.6.5.3 The BIDDER shall confirm that the user interface, forms and reports design, layout and functionality is customizable, and explain this capability in their proposal.
- 1.6.5.4 Within the proposal, the BIDDER shall provide the necessary screenshots in their proposal which help the MOI to understand this capability.
- 1.6.5.5 Within the proposal, the BIDDER shall detail the mechanisms through which the MOI inputs will be accommodated in the system design, in the operator and supervisor user interface screens design and will include multiple system build stages such that MOI have flexibility in influencing the final design as late as possible within the project duration.
- 1.6.5.6 Before proceeding with any high-level or low-level design activities, the CONTRACTOR shall conduct a comprehensive predesign assessment to align their 911 solution stack with the MOI reference Architecture in a workshop with all relevant stakeholders. As an output from this, the CONTRACTOR shall submit a design document that includes the dimensions of resource reuse, integration with existing assets, and alignment with the MOI's efficiency and sustainability objectives.
- 1.6.5.7 The CONTRACTOR shall implement a high-level data architecture for the OSS911 solution that shall map to the following, identifying the different components therein in consultation and agreement with the MOI:
 - 1.6.5.7.1 Data sources including but not limited to voice calls, SIP-data, text & multimedia (images & video), location data from multiple sources, operational data systems from MOI agencies including AVL data, external systems feeding data (live traffic, weather, points of interest, fleet maps, incident information etc.).
 - 1.6.5.7.2 Secure data ingestion & integration layer including but not limited to telephony side SIP interfaces, middleware integration bus with API gateway and for orchestration, brokering and arbitration between different message queues and applications, data pass-through, and data validation and normalization in case of business logic application to data.
 - 1.6.5.7.3 Core business systems processing layer comprising an integrated suite of applications that utilizes a Service Oriented Architecture (SOA) and integrates the different functions and subsystems of the OSS911 solutions in this document including but not limited to Next Generation 911 (NG911) Incident and Resource Management System (IRMS), Common Operational Picture (COP) Solution, IP Telephony solution, recording solution, all of which shall be orchestrated by an Enterprise Service Bus (ESB).

- 1.6.5.7.4 Data storage layer supporting all the OSS911 solutions including but not limited to the operational databases for incidents and events data based on all data sources (voice, text, images, video), the lineup and units' data, users' information, geospatial data stores comprising the location data, maps, traffic and routes information, special operating zones, geofences, logs (operational and system logs), audit trails, data and analytics warehouse and the historical and analytical data stores.
- 1.6.5.7.5 Analytics and intelligence layer including but not limited to analytics insights and decision support supported by operational dashboards with real-time KPIs, concentration zones and hotspots, trend analysis, predictive analysis.
- 1.6.5.7.6 AI layer including but not limited to speech recognition and transcription, natural language understanding, decision support with recommendations, content generation, GenAI, conversational AI, etc.
- 1.6.5.7.7 Technical and Security infrastructure layer forming the backbone for the core business systems, data and analytics including but not limited to centrally management and monitored on-premises private infrastructure supporting the concept of operations outlined in this document.
- 1.6.5.8 The CONTRACTOR shall produce an enterprise system design that utilizes common the data architecture, data structures and consistent user interfaces.
- 1.6.5.9 The CONTRACTOR shall ensure that:
- 1.6.5.9.1 A common nomenclature and taxonomy shall be used across the different systems.
- 1.6.5.9.2 The delivered solution shall provide a dictionary of terms, nomenclature, and taxonomy available to users.
- 1.6.5.9.3 The dictionary shall provide look-up by name and by descriptive key words, and shall provide definition of each term and the usage of each term.
- 1.6.5.10 The CONTRACTOR shall develop a detailed system design blueprint / model. The structure and format of the detailed design blueprint / model shall be finalized in consultation with the MOI after the award of the contract.
- 1.6.5.11 The detailed systems design shall be delivered to the MOI in the form of a Detailed Design Review (DDR) document prior to the completion of the first month after the award of the contract.
- 1.6.5.12 This detailed design blueprint / model shall be available in to the MOI in an electronic version. The CONTRACTOR shall share with MOI one complete license at OSS911 Center with all functionalities required for the MOI to update the design blueprint beyond the life of the contract.
- 1.6.5.13 The detailed design blueprint / model for the OSS911 systems to be produced by the CONTRACTOR shall include:

- 1.6.5.13.1 Documentation, depiction and description of the workflows along with use cases
- 1.6.5.13.2 System standards manual
- 1.6.5.13.3 Hardware, software and middleware details
- 1.6.5.13.4 Unique identification of all system files
- 1.6.5.13.5 System architecture showing all logical layers
- 1.6.5.13.6 System architecture showing all modules, their relations to the various functionalities and their integration
- 1.6.5.13.7 Entity relationship diagrams
- 1.6.5.13.8 Detailed process maps, flow diagrams and description of all system modules including all inputs and outputs
- 1.6.5.13.9 Security design description for each functional area specifying various security attributes like access control, roles, rights, etc.
- 1.6.5.13.10 Concise description of each file, data table, forms and reports
- 1.6.5.13.11 On-screen and print layouts of each report by module
- 1.6.5.13.12 Detailed logical and physical data models linked with the CONTRACTOR supplied database along with the required tools to view and edit them
- 1.6.5.13.13 Detailed description of the workflow logic including all workflow processing rules for edits and decisions
- 1.6.5.13.14 Site maps for all web services and applications
- 1.6.5.13.15 Updated requirements traceability matrix that maps all the functional deliverables to requirements
- 1.6.5.13.16 A detailed data element dictionary in electronic and hard copy that contain at a minimum the data element names, data type(s), data structures, valid values and numbers, functional area definitions, sources of data elements and lists.
- 1.6.5.13.17 Interface control documents (ICDs)
- 1.6.5.13.18 Active Directory schema, network topology, storage diagrams, group and security policies and all configurations.
- 1.6.5.14 The CONTRACTOR shall deliver the systems Application Programming Interfaces (APIs) and the API toolkit associated with their systems.
- 1.6.5.15 The CONTRACTOR shall maintain the system design blueprint for the life of the contract with semi-annual updates.
- 1.6.5.16 The CONTRACTOR shall provide a single sign-on to the entire suite of applications that is based on the user privileges such that a single login

signs in the user to the system and a single logout logs out the user from all systems.

1.6.5.17 All the OSS911 applications shall be available to be accessed by a web browser from anywhere on the OSS911 networks, subject to authorization with the OSS911 directory.

1.6.5.18 The CONTRACTOR shall supply electronic and printed copies of all user manuals and technical documentation in both Arabic and English unless agreed with MOI for exceptions.

1.6.6 **توافر الأنظمة وموثوقيتها (SYSTEMS AVAILABILITY AND RELIABILITY)**

1.6.6.1 The CONTRACTOR shall deliver all the OSS911 solutions such that all the OSS911 systems deliver 99.999% availability, as a value per OSS911 center that will be measured, monitored and reported on a daily basis.

1.6.6.2 All essential components of all systems shall be hot swappable supporting online insertion and removal (OIR) allowing replacement of components whilst the system is active without affecting the operation of the remainder of the system, or the system will be configured to allow portions of the system to be turned off during a part replacement without degrading the delivery of services to users.

1.6.6.3 All systems shall be delivered with a level of spares that enable the required availability to be achieved for all OSS911 solutions.

1.6.7 **متطلبات أداء الأنظمة (SOLUTION PERFORMANCE REQUIREMENTS)**

1.6.7.1 The BIDDER shall provide a capacity and performance analysis for their proposed systems and technical infrastructure that shall include among other things:

1.6.7.1.1 Their approach to performance and capacity design and management.

1.6.7.1.2 Their definitions of what constitutes rated capacity, latency, response time, processing time, utilization, usage.

1.6.7.1.3 A matrix of the expected average transaction load and the expected peak transaction load along with the expected response times for both, for a defined set of operational scenarios.

1.6.7.1.4 Analysis of expected load based on a defined set of operational scenarios.

1.6.7.1.5 Benchmarking statistics from other implementations by the BIDDER of similar scale.

1.6.7.2 The BIDDER shall confirm the definition of the response time as the time between the depression of the last keystroke or pointing device activation (e.g., click) and the appearance on the workstation/terminal of the last character of the initial response (e.g., first page, pop-up window, etc.).

1.6.7.3 The BIDDER shall describe how their solution meets the response time requirements in this RFP.

1.6.7.4 The CONTRACTOR shall configure a batch processing window for jobs such as offline backups and replication which shall identify periods of low OSS911 operational activity and thus less load on the technical infrastructure.

1.6.7.5 The CONTRACTOR shall supply all systems, software and equipment such that:

1.6.7.5.1 All OSS911 activities for operations on live data are completed with a sub-second response time in 95 percent of all transactions.

1.6.7.5.2 When the system is loaded with transactions in the maximum busy peak hour volume, the response time shall be under two seconds 99 percent of the time for operations on live data.

1.6.7.5.3 None of the routine transaction operations on live data shall exceed three seconds.

1.6.8 **نظرة عامة على حلول الاختبارات الفنية (SOLUTIONS TESTING OVERVIEW)**

1.6.8.1 The BIDDER shall propose a comprehensive test plan and methodology that meets all the requirements set out in this document. This methodology shall be detailed enough to provide a complete understanding of the BIDDER's approach to testing and commissioning of all components of this RFP.

1.6.8.2 The BIDDER shall note that the MOI may require modification of the test plans and acceptance criteria.

1.6.8.3 The BIDDER shall note that the MOI reserves the right to participate in, and witness any of the stages of any testing.

1.6.8.4 Within their proposal, the BIDDER shall clearly document their proposed testing durations for the different types of testing during the different stages of the project duration.

1.6.8.5 The CONTRACTOR shall be responsible for the development of an Integrated Acceptance Test Plan (IATP) that clearly defines the requirements for all system commissioning and acceptance testing.

1.6.8.6 The CONTRACTOR shall develop an IATP that includes an overall approach and methodology for testing and acceptance and include rectification processes for acceptance testing issues.

1.6.8.7 The CONTRACTOR shall conduct Testing and Acceptance in the following stages with each stage having to be completed prior to the start of the next stage:

1.6.8.7.1 Preliminary Acceptance Testing (PAT)

1.6.8.7.2 Final System Acceptance Testing (FSAT)

1.6.8.8 The CONTRACTOR shall develop a requirements traceability matrix along with test cases for each MOI requirement, during the systems design, which shall be approved by MOI and shall be then used by the CONTRACTOR to verify and validate performance during all testing.

- 1.6.8.9The CONTRACTOR shall comply with the additional system-specific testing requirements noted in the technical systems sections, and that the CONTRACTOR is advised and required to review these requirements and include them in their Testing plans.
- 1.6.8.10At the end of each stage of testing, the CONTRACTOR shall provide a complete a report with the set of test results and its analysis to the MOI for review and for MOI approval.
- 1.6.8.11The CONTRACTOR shall conduct all testing in the production environment unless otherwise advised and approved by the MOI.
- 1.6.8.12Preliminary Acceptance Testing (PAT) shall be conducted by the CONTRACTOR for each OSS911 Center in scope with the schedule determined by the MOI.
- 1.6.8.13The CONTRACTOR PAT shall incorporate the functional and integration testing of all site-specific systems. Acceptance shall be based on the successful testing of the system as a complete system, demonstrating full integration with local systems and other interdependent systems.
- 1.6.8.14PAT inspections shall include at a minimum the following:
- 1.6.8.14.1 Delivered Bill of Material verification
 - 1.6.8.14.2All Documentation including project documentation
 - 1.6.8.14.3Configuration verification
 - 1.6.8.14.4SLA Compliance
 - 1.6.8.14.5Functionality implementation validation against the DDR
- 1.6.8.15The BIDDER shall include in their approach its proposed approach and metrics which will need to be approved by MOI prior to implementation.
- 1.6.8.16The CONTRACTOR shall include at least the following categories in their testing for all testing stages with clear and quantifiable metrics. This list is not exhaustive and will be worked in detail with the CONTRACTOR and MOI further to award of the contract.
- 1.6.8.16.1Functionality testing
 - 1.6.8.16.2Stress testing
 - 1.6.8.16.3System usability
 - 1.6.8.16.4Data Quality
 - 1.6.8.16.5End-to-end testing
 - 1.6.8.16.6Operational testing
 - 1.6.8.16.7Load testing
 - 1.6.8.16.8Performance testing

1.6.8.16.9Integration testing

1.6.8.16.10Interface testing

1.6.8.16.11Security testing

1.6.8.16.12Penetration testing

1.6.8.16.13Manageability testing

1.6.8.16.14Availability testing

1.6.8.16.15Resilience, fallback, failover and restore testing

1.6.8.16.16Technical infrastructure compliance

1.6.8.16.17Capacity verification

1.6.8.16.18SLA Reporting System and Compliance

1.6.8.17Locations where equipment is installed shall be the subject of formal inspections using checklists written by the CONTRACTOR which have been reviewed and approved by the MOI.

1.6.8.18Inspections of equipment and installations shall be conducted by the MOI and may be witnessed by authorized representatives of the CONTRACTOR.

1.6.8.19Testing shall be conducted by the CONTRACTOR and be witnessed by authorized representatives of the MOI.

1.6.8.20Testing shall be performed in accordance with test plans and procedures written by the CONTRACTOR which have been reviewed and approved by the MOI.

1.6.8.21CONTRACTOR shall not begin PAT until:

1.6.8.21.1All systems have been installed and individually and jointly tested to ensure they are operating properly.

1.6.8.21.2Written permission to begin testing has been received.

1.6.8.22As part of performance verification and testing, the CONTRACTOR shall test all components of system, and verify correct operation of the required system functionality as defined in these specifications.

1.6.8.23Further to the successful completion of CONTRACTOR testing, the MOI may undertake its own testing as part of the PAT for some or all of the test cases.

1.6.8.24As part of corrective actions, the CONTRACTOR shall correct all identified snags and deficiencies, and retest affected components.

1.6.8.25The CONTRACTOR shall mark performance verification test as complete when testing or retesting of each component has produced a positive result and has been approved in writing by MOI.

1.6.8.26The CONTRACTOR shall record in tabular form all test results, deficiencies, and corrective measures.

- 1.6.8.27 The CONTRACTOR shall provide a copy of all test results and data collected to the MOI in an agreed format at the end of each day's testing.
- 1.6.8.28 While the MOI will be responsible for the conduct of Operational Testing, the CONTRACTOR shall supply MOI with an approach and a plan for the related activities.
- 1.6.8.29 Operational testing shall take place during the PAT and shall be accepted before FSAT is allowed to commence.
- 1.6.8.30 The CONTRACTOR shall provide all necessary operational and technical support to the MOI during operational testing periods.
- 1.6.8.31 The BIDDER shall describe in its proposal what kind of technical and operational support will be provided, and how this support will be provided.
- 1.6.8.32 The BIDDER shall note that only the successful completion of Final System Acceptance Testing (FSAT) determines acceptance of the OSS911 project, per OSS911 center.
- 1.6.8.33 As a prerequisite to Final System Acceptance Testing, the CONTRACTOR shall undertake and complete full commissioning and complete testing on the entire system.
- 1.6.8.34 The CONTRACTOR shall deliver an FSAT that incorporates testing of all systems for the whole enterprise, including full integration and functional testing of all dependent systems throughout the entire OSS911 network and associated sites covering all testing mentioned in this document.

1.6.9 البيئات (ENVIRONMENTS)

- 1.6.9.1 The CONTRACTOR is responsible to deliver to the MOI the following physically and logically isolated environments which shall be assessed for compliance as part of the preliminary acceptance testing and as part of the final system acceptance testing:

1.6.9.1.1 Production environment

1.6.9.1.2 Training environment

1.6.9.1.3 Testing environment

- 1.6.9.2 Testing environment:

1.6.9.2.1 For the sake of testing, the CONTRACTOR shall provide an additional environment (systems, infrastructure, licenses, databases, network, cabling, Internet, circuits and any and all related components for all solutions within the OSS911 solution) within the OSS911 center for contractor testing and demonstration to MOI of new or maintenance releases of features and/or systems prior to their installation in the live/production and training environment.

1.6.9.2.2 The components (size, quantity, licenses) and their sufficiency for a test environment depends on the BIDDER

solution and the amount of testing necessary, and shall be provisioned accordingly by the CONTRACTOR.

1.6.9.2.3 The CONTRACTOR shall deploy the Testing environment which shall be isolated from the production and training environments.

1.6.10 برنامج الاختبارات والقبول (TESTING AND ACCEPTANCE PROGRAM)

1.6.10.1 The CONTRACTOR shall develop a testing and acceptance program to allow the MOI to accept the OSS911 service capability.

1.6.10.2 This program shall be contained within the Integrated Acceptance Test Plan (IATP) to facilitate the acceptance and approval process by both the CONTRACTOR and the MOI

1.6.10.3 The IATP shall clearly indicate the differing levels of testing, ownership and responsibilities for the entire acceptance testing program.

1.6.10.4 The tool / software used for testing shall be part of the OSS911 applications delivered to MOI by the CONTRACTOR.

1.6.10.5 Each service capability release shall be subject to acceptance by the MOI following an acceptance testing plan developed and executed by the CONTRACTOR.

1.6.10.6 Following formal acceptance of a new release of software or system from the CONTRACTOR, the MOI reserves the right to perform their own testing and/or training prior to roll out into the operational environment.

1.6.10.7 The CONTRACTOR shall include, as a minimum, the following requirements within the IATP:

1.6.10.7.1A Purpose Statement

1.6.10.7.2Statement of Scope

1.6.10.7.3Location(s) where the test will be performed

1.6.10.7.4Schedule of Activities

1.6.10.7.5Safety and Security statements (as appropriate)

1.6.10.7.6Roles and responsibilities

1.6.10.7.7Acceptance Checklist

1.6.10.7.8Test Procedures, Scripts and Test Cases

1.6.10.7.9Identify measurement and control procedures, test instruments, and supporting equipment required to support tests

1.6.10.7.10Retest Procedures containing a statement of what actions will be taken in the case of failure or need for retesting

1.6.10.7.11Documentation

1.6.10.7.12Standards

1.6.10.7.13Compliance Certificates

1.6.10.7.14Test results and reports format(s)

1.6.10.8 Provide how the test results will be interpreted and compared with performance requirements within RFP

1.6.10.9 The CONTRACTOR shall develop a reporting process that categorizes problems, faults and issues that arise during acceptance testing in a priority order depending on the severity of the anomaly and lists the steps necessary to be taken before further testing can be conducted (Irregularity Reports).

1.6.10.10 Acceptance criteria based on the quantity and nature of the Irregularity Reports shall be developed and agreed between the CONTRACTOR and the MOI.

1.6.10.11 Where an acceptance test fails, the CONTRACTOR shall assess the impact of the failures, propose a resolution plan to be agreed with the MOI and then implement the fixes to the irregularities that need fixing prior to the next release.

1.6.10.12 Further to the CONTRACTOR implementing fixes to any irregularities that need fixing in relation to a system that previously failed, the CONTRACTOR shall be obliged to perform the tests agreed with the MOI in the resolution plan.

1.7 التدريب (TRAINING)

1.7.1 لمحة عامة على متطلبات التدريب (TRAINING REQUIREMENT OVERVIEW)

1.7.1.1 The CONTRACTOR shall implement all requirements in this section with the understanding that:

1.7.1.1.1Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

1.7.1.1.2Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

1.7.1.1.3Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

1.7.1.2 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

1.7.1.3 The CONTRACTOR shall, as a minimum, shall deliver training for all roles listed in Appendix -07 (operational roles).

1.7.1.4 Through the training cycle, the CONTRACTOR shall follow the ISD /SAT model (Instructional System Design/System Approach to Training). All planning /design/Delivery endeavors,

deliverables, methodologies, reports and findings, evaluations and assessments shall be done in compliance with the ISD/SAT.

1.7.1.5 The CONTRACTOR shall size their training solutions and do the training delivery for each province as per the related Appendix#5.

1.7.1.6 The CONTRACTOR shall deliver knowledge transfer and workshops to the MOI – the location, schedule and duration of these will be decided in agreement with the MOI. The BIDDER shall submit the proposed knowledge transfer plan including the above details, as part of their proposals.

1.7.1.7 All training materials should be provided in Arabic language and in English if deemed require by the contractor. The training delivery language for all trainings should be Arabic or with an Arabic interpreter – at no additional costs to the ministry.

1.7.2 وثائق التدريب (TRAINING DOCUMENTATION)

1.7.2.1 The CONTRACTOR deliverables shall include the following:

1.7.2.1.1 Course Training Plans (CTP).

1.7.2.1.2 Plan of Instruction (POI).

1.7.2.1.3 Training needs analysis (TNA).

1.7.2.1.4 The content and supporting material for each training.

1.7.2.1.5 Training attendee information.

1.7.2.1.6 Relevant information on all recipients of that training- attendance, and pre-training and post-training assessment.

1.7.2.2 The CONTRACTOR shall systematically provide electronic and hard copies of all written training, course material and course instructions. Plan of Instructions (POI) shall include:

1.7.2.2.1 Course objectives, written in accordance with Bloom's taxonomy

1.7.2.2.2 Course target groups

1.7.2.2.3 Course prerequisites and other dependencies

1.7.2.2.4 Course duration and delivery method

1.7.2.2.5 Specific lab/experimental environment descriptions (if relevant)

1.7.2.2.6 Requirements for demonstrating trainee proficiency with the subject matter in each course

1.7.2.3 The CONTRACTOR shall provide a complete package of training modules comprising of a plan of instruction (POI):

1.7.2.3.1 Tailored training modules for each subject

1.7.2.3.2 Location of all training modules

1.7.2.3.3 Training schedules, including lesson plans and time allocated to each lesson

1.7.2.3.4 Skills matrices

1.7.2.3.5 Post training assessment

1.7.2.4 The BIDDER shall note and confirm that the MOI shall hold and retain the ownership and intellectual protocol for all training material, course content and data forming the contents of all training delivered by the CONTRACTOR for the training requirements in this RFP.

1.7.2.5 The CONTRACTOR shall deliver all of the above documentation in editable electronic format.

1.7.3 تحليل متطلبات التدريب (TRAINING NEEDS ANALYSIS (TNA))

1.7.3.1 The CONTRACTOR shall conduct Training Needs Analysis (TNA) for each role and function required to be trained.

1.7.3.2 The CONTRACTOR shall provide to the MOI a plan for the conduct of the TNA showing all phases, activities and resources to be assigned, as well as gates and deliverables.

1.7.3.3 The CONTRACTOR shall propose assessment methodologies to be used during the TNA which shall need to be approved by the MOI prior to the commencement of any TNA.

1.7.3.4 Following the TNA, the CONTRACTOR shall submit a TNA report to the MOI containing.

1.7.3.4.1 Assessment of current staff knowledge, skills and abilities (KS&As) and experience level.

1.7.3.4.2 Skills matrices for each role required to support the operations, maintenance and administration of the OSS911 Center (based on job tasks analysis) as the OSS911 Center will operate.

1.7.3.4.3 Skills gap analysis for the identified number of positions.

1.7.4 التدريب على رأس العمل (ON THE JOB TRAINING)

1.7.4.1 The CONTRACTOR shall deliver regular On the Job Training (OJT) in a structured method for all roles for any features/functions/updates/fixes for all OSS911 solutions.

1.7.4.2 The CONTRACTOR shall deliver an OJT plan shall that includes verification and validation (V&V) of the gained Knowledge, Skills and Abilities.

1.7.4.3 At the end of the training, the CONTRACTOR shall either:

1.7.4.3.1 Provide a certificate confirming that the individual has met the required standard, or

1.7.4.3.2 State why the individual failed to meet the required standard. In such cases the statement shall include a performance analysis indicating possible reasons of the failure and state whether the individual would be suitable for any other role within the OSS911 Center.

1.7.5 خطط التدريب (TRAINING PLANS)

1.7.5.1 As part of their proposals, the BIDDER shall include Course Training Plans (CTP) that demonstrate the BIDDER methodology, approach and schedule.

1.7.5.2 The CONTRACTOR shall revise the CTP in consultation with the MOI as part of the TNA.

1.7.5.3 The CONTRACTOR shall provide CTP for all OSS911 Center roles based upon the results and findings from the TNAs.

1.7.5.4 Training plans shall address at least the following groupings:

1.7.5.4.1 OSS911 systems Training for all OSS911 systems for all roles.

1.7.5.4.2 Administration and Technical training

1.7.5.5 The CONTRACTOR shall develop and deliver a comprehensive Master Training Plan and supporting documentation (e.g.: curricula, individual development plans) that addresses the full scope of Project training needs.

1.7.6 **تقييم التدريب والمتابعة (TRAINING ASSESSMENT AND FOLLOW UP)**

1.7.6.1 The CONTRACTOR shall conduct a pre-training assessment for each training course and/or batch that helps the evaluation of the initial impact and effectiveness of the Training Program on the skills, knowledge, and abilities of Training Program participants.

1.7.6.2 The CONTRACTOR shall use the pre-training assessment results:

1.7.6.2.1 To develop the training needs of each individual

1.7.6.2.2 To provide baseline data for use in later training course evaluations

1.7.6.3 Upon completion of a training course and/or batch, the CONTRACTOR shall provide the MOI with a post-training assessment report that documents the training activities and student pass/failure results, within seven days of the completion of the training course and/or batch.

1.7.6.4 The post-training assessment report shall include an executive summary, the methodology used, summaries of the course delivered, the names and total numbers of personnel trained, pre and post training scores, a written narrative of training activities including any significant activities, discoveries, and issues encountered during the practical training exercises.

1.7.6.5 In addition, the post-training assessment report shall include conclusions and a list of recommendations of on-going activities and/or additional training.

1.7.7 **فريق إدارة التدريب من المقاول (THE CONTRACTOR TRAINING MANAGEMENT TEAM)**

1.7.7.1 The CONTRACTOR shall supply a training specialist who will be fully responsible for the overall training processes and who shall meet the qualification and experience requirements as outlined in the Arabic section of this RFP.

1.7.8 **المدرسين من جانب المقاول (THE CONTRACTOR TRAINERS)**

1.7.8.1 The contractor shall provide a list of their proposed trainers to the MOI along with details of their training qualifications and experience.

1.7.8.2 The MOI reserves the right to reject any resources deemed to have unsuitable qualifications or relevant prior experience.

1.7.8.3 Only suitably certified trainers shall be used to develop and deliver training courses. The certification must be provided by the technology vendor.

1.7.9 **OSS911 مدربين أنظمة (THE MOI OSS911 TRAINERS)**

1.7.9.1 The CONTRACTOR shall train MOI-identified personnel as trainers, as part of the Train-the-Trainers deliverable for the subjects and courses to be agreed with the MOI.

1.7.9.2 The CONTRACTOR shall provide a Methods of Instruction (MoI) for all the MOI OSS911 Trainers to allow the MOI trainers to fully understand the training methodology and the training/teaching material.

1.7.10 **مرافق التدريب (TRAINING FACILITIES)**

1.7.10.1 The CONTRACTOR shall equip the training rooms in the OSS911 center with OSS911 solutions with full project features.

1.7.10.2 THE CONTRACTOR shall provide a complete dedicated centralized training network and platform for OSS911 solutions that shall be installed in the OSS911 Center, and that is isolated from any of the OSS911 production network and platform.

1.7.10.3 At all times during the contract, the CONTRACTOR shall ensure that the training environment is up to date with the releases, patches and updated versions of all software that is at that time running in the live/production environment.

1.7.10.4 The CONTRACTOR shall ensure that the training platform shall utilize only non-live incident data and shall have no access to any live operational system used in the OSS911 center.

1.7.10.5 The CONTRACTOR shall be responsible for all costs associated with training, including any to be carried locally and outside the country. These costs shall include but not be limited to such things as:

1.7.10.5.1 For training within the OSS911 center: All materials required to conduct the training – reference material and books, paper, pens, pencils, whiteboard in each training room, smart projector in each training room, refreshments and meals for all, etc. Such training will use the building audio system. The CONTRACTOR is responsible for any & all works required to install the equipment supplied by them.

1.7.10.5.2 For training at any other site, local or abroad: All materials required to conduct the training – reference material and books, paper, pens, pencils, whiteboards, smart projectors, monitors, lecterns, table's chairs, audio systems, refreshments and meals, etc.

1.7.11 **مجالات التدريب الإرشادية (INDICATIVE SUBJECT AREAS)**

1.7.11.1 In conjunction with the MOI, the CONTRACTOR shall define the subject areas to be included within each training that shall be delivered by the CONTRACTOR.

1.7.11.2 The following is a list of subject areas considered appropriate to the roles to be performed within the OSS911 Center. The list shall not be regarded as exhaustive until the design of all the OSS911 systems is completed.

1.7.11.2.1 IRMS for all functions and capabilities therein

1.7.11.2.2 Common Operational Picture

1.7.11.2.3 IP Telephony for all functions and capabilities therein

1.7.11.2.4 Integrations

1.7.11.2.5AI capabilities for all solutions

1.7.11.2.6Systems, network and security administration,
management, monitoring and troubleshooting

1.7.11.2.7Private cloud and Software Defined Networks (SDN)

2 الأنظمة والبنية التحتية (SYSTEMS AND INFRASTRUCTURE)

2.1 Geographical Information System (GIS)

2.1.1.1 The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.1.1.1.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.1.1.1.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.1.1.1.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

2.1.1.2 The CONTRACTOR shall deliver a GIS solution covering all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

2.1.1.3 The CONTRACTOR shall deliver OSS911 solution integration with GIS to deliver smart and AI-powered capabilities for OSS911 use.

2.1.1.4 The CONTRACTOR OSS911 solutions shall be delivered as integrated with the GIS solution for all the requirements in this RFP.

2.1.1.5 Should the MOI require, the CONTRACTOR shall deliver the OSS911 911 solutions as integrated with other GIS platforms to deliver all the requirements in this RFP.

2.1.1.6 For the contract duration, the CONTRACTOR shall be responsible for the processing of the maps, map data, and satellite imageries, and related updates, into the OSS911 solution to be available in the OSS911 systems and solutions.

2.1.1.7 Through the integration with GIS solution maps, map data and imagery, the CONTRACTOR shall provision as a minimum the following spatial data layer and features within the OSS911 systems and solution:

2.1.1.7.1 Roads and Street Layers with:

- a. Digital navigable Streets Network
- b. Streets' centerlines
- c. Streets names and aliases
- d. Streets' classifications
- e. Rights of way
- f. Streets' directions

- g.** Streets' intersections
- h.** Turns
- i.** Speed
- j.** Bridges
- k.** Overpass
- l.** Underpass
- m.** Ramps
- n.** Roundabouts
- o.** Exits
- p.** Street edges
- q.** Road number
- r.** Road islands
- s.** Highways and Kilometers markers

2.1.1.7.2 Railways and public transport paths and stations

2.1.1.7.3 The administrative boundaries layers and place-names for all:

- a.** Region / province boundaries
- b.** Governorates' boundaries
- c.** Cities/towns boundaries
- d.** District's boundaries
- e.** Centers, Villages, Hamlets
- f.** Municipalities boundaries

2.1.1.7.4 All landmarks and categorized points of interests (POIs) including banks, hotels, restaurants, mosques, diplomatic areas, special security areas, parks, schools, malls, hospitals and health services, universities, police stations, civil defense stations, police check points, and natural landmarks etc.

2.1.1.7.5 Saudi Post - SPL National Address points with both traditional and short address.

2.1.1.7.6 Buildings' footprints with 3D model for the momentous.

2.1.1.7.7 Land parcel boundaries with land use.

2.1.1.7.8 Water-bodies layer including waterways.

2.1.1.7.9 Other layers and data that may support the 911 center operations and capabilities

- 2.1.1.7.10 Live Traffic service Online and Offline that includes real-time traffic flow information and incident data.
- 2.1.1.7.11 This live traffic information shall include, but not limited to, traffic speeds and flow, travel times, color coded traffic flow conditions, and congestion alerts (where available) - all of which will be integrated by the CONTRACTOR to be viewable in the Common Operational Picture on the OSS911 systems.
- 2.1.1.7.12 This Traffic information shall be integrated and represented in the OSS911 systems with color coded traffic flow conditions (Roads, where traffic flows freely, shall be marked in green, roads where congestion is moderate shall be in orange, while those that are congested shall be shown in red).
- 2.1.1.7.13 This Traffic service shall have the capability to support unlimited number of transactions to support 911 operations.
- 2.1.1.8The CONTRACTOR shall integrate and provide all geospatial data described above (including the traffic service) into all OSS911 systems.
- 2.1.1.9The CONTRACTOR shall ensure that all the geospatial data layers listed above are spatially related, hierarchically linked, and have the full location-centric details.
- 2.1.1.10The CONTRACTOR shall apply all the standard topology rules and geometric network.
- 2.1.1.11The CONTRACTOR shall supply all data in Arabic language along with additional attributes in English language, where available.
- 2.1.1.12The CONTRACTOR shall:
- 2.1.1.12.1Analyze and process the raw data provided by MOI in any formats.
 - 2.1.1.12.2Compile the processed data to have an integrated data source in the OSS911 systems.
 - 2.1.1.12.3Ensure that the maps representations including labels and annotations shall be in Arabic language.
 - 2.1.1.12.4Provide a detailed data analysis report for each dataset provided by MOI.
 - 2.1.1.12.5Ensure that the boundaries of the operational layers shall cover the entire target province without gaps or overlaps between the different polygons. Boundaries of the operational layers shall be programmed by the CONTRACTOR in the system ONLY after validation by MOI - dispatch for events will be based on MOI-validated data only. Further to this, if gaps exist in the operational areas, then these need to be resolved by the CONTRACTOR with the help of MOI through workshop with the concerned agencies. The CONTRACTOR shall be responsible for programming the updates and fixes of the boundaries of the operational layers within the system at no additional cost to the MOI.
 - 2.1.1.12.6Provide and ensure optimal map performance on all OSS911 systems and shall apply the best practices that affect the

performance including (optimizing spatial data, optimizing layers and roasters, optimizing maps ...etc.).

2.1.1.12.7 Comply with the address system, territory, GCS standards, UTM grid zones and datum of the Kingdom of Saudi Arabia.

2.1.1.12.8 The authorized users shall be able to control maps and layers permissions and access and shall be able to modify the default geographic layers that are displayed at each zoom scale for each group and user.

2.1.1.13 The CONTRACTOR shall implement and provide all OSS911 solutions integrated with Geographical Information Systems to provide smart and AI-powered capabilities with region-wide coverage and which meets all of the requirements detailed within this document.

2.1.1.14 The CONTRACTOR shall deliver an OSS911 solution that shall be able to directly integrate with any sources that MOI may desire to connect to and / or any GIS systems including maps and data and/or any other GIS systems or geospatial data identified by MOI.

2.1.1.15 The CONTRACTOR shall deliver an OSS911 solution that shall be able to integrate with Google maps and APIs.

2.1.1.16 The CONTRACTOR's approach shall conform to the Open Geospatial Consortium (OGC) international standards and practices.

2.1.1.17 The CONTRACTOR shall implement OSS911 systems and solutions that shall integrate with different types of map services and layers (WebGL/MapGL, WMS, WFS, WMTS, and KML... etc.).

2.1.1.18 The CONTRACTOR shall implement OSS911 systems and solution that shall integrate with different kinds of GIS maps and data including (e.g., Vector Basemaps, Demographic maps, Topographic maps, Ortho-photographic maps, Imagery, panoramic maps, Digital Elevation models, Environmental data, Aeronautical Charts, Transportation, Traffic, etc.).

2.2 COMMON OPEARATIONAL PICTURE (COP)

2.2.1.1 This RFP defines OSS911 Common Operational Picture (COP) as the "Use of advanced technology for evolving a single identical, integrated, and combined geographic display of all the information required for planning, and incident response by collecting, displaying, and sharing multi-dimensional information of the incident, resources, assets, alarms, tasking lists,...etc. on maps to support operators and decision makers across disciplines with real-time situation views to achieve situational awareness and to facilitate collaborative planning and combined execution, which can support the daily operations, planned events and crisis situations as well."

2.2.1.2 The BIDDER shall confirm their understanding of this definition and its application to the OSS911 solution and concept of operations.

2.2.1.3 The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.2.1.3.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.2.1.3.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.2.1.3.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

2.2.1.4 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

2.2.1.5 The CONTRACTOR shall deliver a dynamic, multi-layered AI-enabled COP for instant visualization of incidents, units, routes, assets and points of interests among others, for contextual, comprehensive and continuous situational awareness.

2.2.1.6 The CONTRACTOR shall deliver an OSS911 COP that is based on and integrated with MOI-supplied GIS.

2.2.1.7 The CONTRACTOR shall deliver an OSS911 COP that is available to authorized personnel via all access modalities – desktop, multi-touch device and mobile devices.

2.2.1.8 The CONTRACTOR shall deliver an OSS911 COP that provides “real-time” information to all users.

2.2.1.9 The COP shall include the following capabilities:

- 2.2.1.9.1The display of different static and dynamic geographic information and layers (i.e., maps, overlays, Imageries)
- 2.2.1.9.2Jurisdictional boundaries and areas of responsibilities for the different agencies
- 2.2.1.9.3Incident location and status information
- 2.2.1.9.4The display, within those maps, of the resource information including location and status of the resources
- 2.2.1.9.5The map shall dynamically update showing the position and status of resources.
- 2.2.1.9.6The map shall show available resources (with status) within a certain pre-defined distance from the incident that is appropriate to the operator's role.
- 2.2.1.9.7Information gathered from various integrations as defined in this RFP
- 2.2.1.9.8Information gathered for the incident under scrutiny
- 2.2.1.9.9Location data to be overlaid on the maps.
- 2.2.1.9.10User customizable views for specific user roles
- 2.2.1.9.11The rules used for displaying available resources shall be configurable.
- 2.2.1.9.12The ability for compiling data from multiple sources to ensure the same understanding and awareness of the incident status
- 2.2.1.9.13The display of standard operating procedures in relation to an incident once qualifying details for an incident are available in the system.
- 2.2.1.9.14The ability to provide routing information
- 2.2.1.9.15The determination of what response assets are closest for deployment and provide recommended "closest" responders to operators.
- 2.2.1.9.16Represent the location, properties, and capabilities of all video surveillance cameras integrated within the CAD solution.
- 2.2.1.9.17The capability to indicate to the user those cameras that are able to be viewed at that location, or within a distance of incident location.
- 2.2.1.9.18The ability to control sharing of incidents, resources, and incident related information between local, regional, national and private sector entities to create a common operating picture.
- 2.2.1.9.19The ability to collect and compile information at subordinate levels (municipal, regional, national, and agency levels)
- 2.2.1.9.20Allow display of all layers, maps, data, and map services.

- 2.2.1.9.21 Provide access to the layers, maps and data to networked and mobile users through the network.
- 2.2.1.9.22 Provide capability for resource location updates to be updated on the map dynamically.
- 2.2.1.9.23 The resources icons shall dynamically change on the map whenever the object/ resources data has been updated and the color of icons themselves shall also be configurable using the status field or other predefined field(s) related to the resource. (Example: if the status of the shelters changed to be closed the color / symbol shall be changed to be another color – say change to red instead of green).
- 2.2.1.9.24 Provide the capability for clustering of objects for visualization on maps.
- 2.2.1.9.25 Provide the user with the ability to customize his view to display icons and/or symbols as individual icon and/or symbol, as clusters and as a combination of individual icons and/or symbols, and clusters.
- 2.2.1.9.26 Map Icons that are near each other, or are hidden by overlapping icons, shall have the capability to be grouped together as a single “cluster” of icons.
- 2.2.1.9.27 User shall be able to zoom into any single data point by selecting the cluster and a pop-up list shall be displayed. In addition, the user shall be able to zoom to the extent of the entire cluster for faster navigation. When zoomed out, icons are clustered. The size and color of the cluster become more emphasized as the number of data points included in the cluster increase.
- 2.2.1.9.28 Provide a cluster icon for multiple resources at one site. Each icon must uniquely represent the presence at the site of multiple resources, and, by mouse click, cause a window to pop up which will display data about all resources represented by the icon.
- 2.2.1.9.29 Provide a "cluster" icon for multiple incidents at one site. Ideally, different icons will represent multiple events at one site, and, by mouse click, cause a window to pop up which will display data about each incident represented by the icon.
- 2.2.1.9.30 All structural icons shall display the facility name and resources either within, immediately above, or immediately below the icon. The icon shall also indicate the existence of particular hazards or other information a user should be aware of.
- 2.2.1.9.31 Provide a user-friendly control to help the users change Basemaps, imagery and manage the visibility (Switches On/Off), change opacity of the different spatial layers and imageries re-order layers, exclude layers/base-maps from displaying, with the detailed Legend of the different layers identifying the meaning for the colors or symbols used.

- 2.2.1.9.32 Provide an Identify/Info tool that allows the users to select any objects or features interactively from the map to get the list of their attributes data. An Info pop-up window shall open and display the record information. The set of features/ layers /attributes that are identifiable shall be able to be defined and configured based on the integration with the provided GIS, maps, map data and imagery.
- 2.2.1.9.33 Provide a capability to find a location by specifying the Longitude, Latitude or go to XY, the user shall be able to choose the units in which to enter the coordinates (Decimal Degrees, Degree Minute Seconds, etc.).
- 2.2.1.9.34 Capability of enabling the Geo-location, which makes the map zoom to the current location, draw a circle to show the accuracy and track any movement or change in the position.
- 2.2.1.9.35 Provide a Geo-bookmarks tool that allows the users to create a place holder on a map that can be quickly accessed later.
- 2.2.1.9.36 The Geo-bookmark tool shall allow a user to create bookmarks for providing shortcuts to interesting places on a map, share drawings, text, or other pertinent annotations associated with a specific view.
- 2.2.1.9.37 Provide the capability that when the user selects a bookmark; the map zooms to that location.
- 2.2.1.9.38 The authorized users shall be able to manage, remove, or share the bookmarks with other users.
- 2.2.1.9.39 Provide a capability for panning and zooming around the map display. Users shall always be able to zoom and pan to a specific location at a specified zoom scale or to interactively indicate a rectangular area that shall occupy the entire map.
- 2.2.1.9.40 Provide a capability to zoom and pan around the jurisdiction by the use of mouse drag on slide bar or mouse click on appropriate directional icons.
- 2.2.1.9.41 Provide a default set of zoom scales with defined layers to be displayed in each scale.
- 2.2.1.9.42 Provide a capability to authorized users to be able to modify the default geographic layers that are displayed at each zoom scale for each user.
- 2.2.1.9.43 Enables the authorized users to manage different map layers to better control how various information and external data is being consumed and displayed.
- 2.2.1.9.44 Provide a measurement tool for distance and area that lets the users to measure linear distance and areas on the map.
- 2.2.1.9.45 Allow the measurement tool to be used for drawing a line or polygon on the map and generating its length or area.

- 2.2.1.9.46 Allow a user to click directly on a feature and get measurement information.
- 2.2.1.9.47 Provide the navigation tool that provides the controls that allow the users to navigate, pan, zoom-in, zoom-out, go to last extent (back), go to next extent (Forward), full extent, magnification bar, map scale bar.
- 2.2.1.9.48 The user also shall be able to zoom in or out using the scroller button in the mouse, if such a button is available.
- 2.2.1.9.49 Provide a Print tool to allow authorized users to print the area of the map that is currently visible in the browser window.
- 2.2.1.9.50 The user shall be able to add a title, subtitle, or copyright and set the scale option for the map that he is being printing.
- 2.2.1.9.51 Provide feature highlighting as a tool for user interaction with the map.
- 2.2.1.9.52 Features shall be highlighted when the mouse cursor is placed above them.
- 2.2.1.9.53 The user shall be able to click on the highlight in order to open up the popup containing information about the feature.
- 2.2.1.9.54 Provide support for displaying the latitude and longitude with the mouse pointer on the map.
- 2.2.1.9.55 Provide a capability for the authorized users to consume and access any OGC complaint services including but not limited to GIS Web Map Services, Web Feature Service, Image services, etc. whether online or on premises using their REST endpoints (URLs.) and credentials.
- 2.2.1.9.56 Provide a capability for geo-referenced database that converts addresses into location information
- 2.2.1.9.57 Provide a capability of Geo-fencing:
- To be able to mark-up and / or draw pre-determined or dynamic virtual boundaries on maps, either as a radius around a given point or as a set of drawn up boundaries.
 - To be able to generate visual and audio notification to authorized operators in case of a tracked resource movement in to and out of the defined boundary.
 - To be able to generate a visual and audio notification to authorized operators in case of specific events/incident types or alarms happened in the defined boundary.
- 2.2.1.9.58 Provide a capability for generating automatic Geo-messages/Geo-alerts in multiple languages for mass notification in case of public emergencies and/or announcements.
- 2.2.1.9.59 Provide support for viewing, accessing, interacting with 2D, 3D, and panoramic maps.

- 2.2.1.9.60 Allow authorized users to perform structured and spatial queries of data interactively on the map and to analyze and report on incidents, crisis, assets, features, traffic, and resources.
- 2.2.1.9.61 Provide a selection tool that allows users to query the data layers in a dynamic map for an area of interest, and view the results either in a table format and/or see it graphically displayed in the map, the set of features/ layers / attributes that are selectable can be defined and configured.
- 2.2.1.9.62 The user can select which layer(s) to perform the query against.
- 2.2.1.9.63 The user can define the area of interest by drawing i.e., rectangle, point, line, or polygon, etc. on the map.
- 2.2.1.9.64 User can expand or define the area of interest displayed on the map by using the Buffer tool this will draw a radius around the area selected, the user can enter the distance and select the distance unit (Miles, Kilometers, Meters, Feet...etc.).
- 2.2.1.9.65 The selection results should be classified based on the features/Layers types; the user can view the attributes of the associated features; zoom to the selected feature(s); the selected feature(s) should be highlighted on the map.
- 2.2.1.9.66 The user can deactivate the selection tool and clear the query results
- 2.2.1.9.67 Provide the capability of point-to-point routing with directions and the ability to create road blocks interactively from the map for alternate routing as described in the below section.
- 2.2.1.9.68 The solution shall be capable for generating routing and travel instructions and transmitting them to users. This section describes some requirements for that component of the system. The BIDDER shall support the claim that its solution will fulfil these requirements.
- 2.2.1.9.69 If a user is accessing the routing system via web browser, then the routing system will be able describe routes to users via map, text, and voice.
- 2.2.1.9.70 If a user is accessing via mobile device application, then the routing system will be able to describe routes to users via map, text, and voice.
- 2.2.1.9.71 The user may select how the routing system will describe a route (i.e., via map and/or text and/or voice).
- 2.2.1.9.72 The user will be able to select the starting point of a route. This location may be entered:
- As the user's current location, if the GIS has that information
 - As latitude and longitude
 - As a street address

- d. As cross streets
- e. As a facility name (e.g., Princess Nora University)
- f. By indicating a location on a map

2.2.1.9.73A user will be able to select the destination of a route. This location may be entered:

- a. As latitude and longitude
- b. As a street address
- c. As cross streets
- d. As a facility type (e.g., gas station, hospital, etc.)
- e. As a facility name (e.g., KAI Airport)
- f. By indicating a location on a visible map
- g. By identifying or selecting a locatable object (i.e., a resource, asset, etc.), either by name, or by indicating object on a visible map

2.2.1.9.74The user will be able to require that the generated route pass through any number of intermediate locations. These locations may be entered as:

- a. As latitude and longitude
- b. As a street address
- c. As cross streets
- d. As a facility type (e.g., gas station, hospital, etc.)
- e. As a facility name (e.g., KAI Airport)
- f. By indicating a location on a visible map

2.2.1.9.75The routing system will generate routes appropriate for the mode of travel selected by the user. These modes will include: walking, driving, or driving followed by walking.

2.2.1.9.76The user will be able to place other restrictions on the routing system, e.g., the user may indicate a vehicle weight, vehicle width and height, cargo type, etc.

2.2.1.9.77The routing system will generate a route with the shortest time (considering the live traffic and road conditions) or shortest distance, as selected by the user, appropriate for the selected mode of travel, intermediate points, and other restrictions.

2.2.1.9.78The user will be able to view the conditions and properties of features along the route (e.g., lane widths, overpass heights, bridge weight limits, etc.)

- 2.2.1.9.79 Any time after the user has received a route, he will be able to forbid the routing system from generating a route that passes through a particular point, and the routing system will generate a new route that does not pass through that point.
- 2.2.1.9.80 At any time, the user will be able to move the route's starting point; move, add or remove intermediate points; and move the route's destination; place further restrictions on the route, without starting the routing request afresh.
- 2.2.1.9.81 If the routing system cannot generate a route between the starting location and the destination, it will indicate the reason to the user.
- 2.2.1.9.82 If the user asks for a "driving route", and there is no route from the starting location to the destination that is appropriate for a vehicle, but there is one that includes exiting the vehicle and completing the route on foot, then the routing system will generate this route. The route information will prominently indicate that a "dismount" will be required.
- 2.2.1.9.83 The routing system will not direct the user to perform illegal activities, unless the user instructs it to do so. For example, the routing system will not direct the user to travel the wrong way on a one-way street, but the user can over-ride this.
- 2.2.1.9.84 The routing system will incorporate an obstruction feature: any authorized user may indicate within the CAD solution that a route section is no longer available, and the routing system will not include that section in any routes. For example, if a user reports that a bridge is no longer passable, the routing system will not direct a user to cross that bridge.
- 2.2.1.9.85 The routing system will be able to provide real-time instructions to a user: when a user completes a section of the route, the routing system will provide the user with instructions for the next section.
- 2.2.1.9.86 If an obstruction necessitates an update to a user's route, the routing system will immediately incorporate that update into the user's real-time instructions. If the user is not receiving real-time instructions, then the routing system will alert that user to the update.
- 2.2.1.9.87 The solution shall include an implementation for smart search control to allow the operators seamlessly search and locate the different spatial layers, locations and their related information and data on the same interface of the CAD solution, this control shall provide as a minimum the following capabilities:
- 2.2.1.9.88 Shall include a bilingual (Arabic and English) Gazetteer search with auto complete capability (like Google Search), on all available spatial layers and data such as (Administrative Boundaries layers, Streets' layer, POIs layers, etc....).

- 2.2.1.9.89The search control shall support the capability of a type-ahead-search behavior of Google Maps search field, when the operator starts typing the search keywords the “Auto complete” will fill the rest and minimize the populated results list.
- 2.2.1.9.90The Gazetteer search over the gazetteer feature shall allow, by simple entry keywords / characters, to search over these layers and return the results in a ranked manner based on the result matching score.
- 2.2.1.9.91The Gazetteer search shall return the full spatial address of the features and its related information.
- 2.2.1.9.92The Gazetteer search shall identify and return the layer type (whether the returned matching results belongs to the POIs layers, Streets’ layer...etc.) this help the operators to select the right category, if multi matching results from different layers been returned.
- 2.2.1.9.93The Gazetteer shall support the capabilities for both exact match and fuzzy match search.
- 2.2.1.9.94the user shall have the capability to select either search options for exact match only or including the fuzzy matching results, with the default exact matching option been selected.
- 2.2.1.9.95The search control shall support sorting of the returned results (The exact matching results to be shown first and hence the fuzzy matching results)
- 2.2.1.9.96The search control shall be smart enough to return the search results in a ranked manner based on the result matching score.
- 2.2.1.9.97The smart search control shall provide the capability for the users to select specific layers to search against, for example (the user may select searching over the streets and/or landmarks layers only). If the user doesn't specify any selection, all layers considered selected by default.
- 2.2.1.9.98The smart search control shall be smart enough to fetch only the database records that belong to specific layers based on the user search strings, for example (If the user type in the search field "Street Kind Khaled" it should go searching in the streets layer only).
- 2.2.1.9.99In addition to the Gazetteer search the solution shall include an advanced search control for the location identification in a manner of hierarchical drop-down lists filtered based on the user selection.
- 2.2.1.9.100Each drop-down list is filtered based on what is selected on the parent level drop-down list
- 2.2.1.9.101The advanced search control shall have and use all the spatial layers made available for the CAD solution in hierarchical order such as (Region, Governorates, Cities, Districts, Streets, Landmarks, National Address... etc.)

2.2.1.9.102The advanced search control shall provide the user the flexibility to stop by any level of the hierarchy and the selected features shall be displayed on the map.

2.2.1.9.103The search control for (Gazetteer search and advanced search) shall provide the capability to auto zoom to the map scale / level in which the selected features by the users from the search result are visible on the map; center the map to the selected feature/object, the feature shall be highlighted.

2.2.1.9.104The display of assets and resources should be appropriate for the map scale.

2.2.1.9.105The display of assets and resources should be appropriate for the (local) density of assets and resources.

2.2.1.9.106The display of assets, resources, and properties thereof, should readily and intuitively provide the user with necessary information about each asset and resource.

2.2.1.9.107This informational display should be appropriate for the map scale and (local) density of assets and resources.

2.2.1.9.108This informational display should be appropriate for the current usage of the system (emergency response, strategic planning, etc.)

2.2.1.9.109The authorized user should be able to customize this information display on the user interface.

2.2.1.9.110The authorized user should be able to save, restore, and share these customizations.

2.2.1.9.111The selection of map symbol, feature representations, overlays, labels, shapes, sizes, icons, colors, and legend shall conform to widely accepted standards.

2.2.1.9.112The selection of map symbol, feature representations, overlays, labels, shapes, sizes, icons, colors, and legend shall accommodate cultural sensitivities of users.

2.3 ((IRMS) نظام إدارة الأحداث والموارد INCIDENT AND RESOURCE MANAGEMENT SYSTEM (IRMS))

2.3.1 لمحة عامة عن المتطلبات الوظيفية (OVERVIEW AND FUNCTIONAL REQUIREMENTS)

2.3.1.1The MOI requires that the OSS911 solution incorporate an integrated Incident and Resource Management System (IRMS) to enable the recording and management of all relevant information relating to reported incidents – from call receipt to call closure.

2.3.1.2The BIDDER shall propose primarily “off-the-shelf” IRMS software functionality, requiring the minimum number of modifications necessary in order to support necessary functions and interfaces.

2.3.1.3The CONTRACTOR shall implement all requirements in this section with the understanding that:

- 2.3.1.3.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.
- 2.3.1.3.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.
- 2.3.1.3.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.
- 2.3.1.4 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.
- 2.3.1.5 The CONTRACTOR shall deliver a smart and AI-powered IRMS solution delivering IRMS capabilities for OSS911 use.
- 2.3.1.6 The IRMS software shall be such that it makes no distinction between different positions – call taker, dispatch. Appropriate user controls and interface shall be provided based on the privileges at the login.
- 2.3.1.7 The IRMS shall be deployed within the OSS911 Center and shall also be available to mobile users.
- 2.3.1.8 Field users shall be able to access information in, and interact in a bidirectional way with, the IRMS via the IRMS integration with MOI field systems / platform / apps.
- 2.3.1.9 The CONTRACTOR shall deliver an IRMS that shall be based on open architecture and shall provide an Application Programming Interface (API) for third parties to connect to and establish integration with the IRMS functions.
- 2.3.1.10 The IRMS shall automatically send a query to NIC and MOI databases for registration and wants and warrants checks when a license plate and/or person’s name is entered.
- 2.3.1.11 A typical IRMS operator and supervisor shall be configured to utilize two display monitors – one for the incident details display and management, and another for the map display.
- 2.3.1.12 The IRMS shall have a capability to be able to receive and disperse emergency calls and incidents information embedded in the OSS911 solution and also using SMS.
- 2.3.1.13 The CONTRACTOR shall include a solution for an SMS to be automatically sent from the IRMS to nominated officials based on incident details as part of incident-based alerting. For example, once a particular incident type / particular incident severity / particular incident detail is entered / selected by call taker and/or dispatcher, then an SMS shall be sent automatically to nominated officials.
- 2.3.1.14 The IRMS shall include an SMS solution for authorized OSS911 staff (call takers, dispatchers, supervisors etc.) to update the callers

periodically and/or arbitrarily, about the status of their emergency calls that have been registered in the IRMS with an incident ID.

2.3.1.15The IRMS-SMS solution shall include capability to deliver regular status updates to callers at an MOI approved frequency.

2.3.1.16The IRMS-SMS solution shall include capability to deliver an SMS-based customer satisfaction check service with the caller(s) once an incident is resolved.

2.3.1.17The CONTRACTOR shall deliver operator status SMS-based alerts through IRMS solution with all calculations needs to be done per shift and with the capability to define the thresholds to have a default value per shift with the ability for authorized personnel to be able to change them along with the required reporting. Such alerts include the following but not limited to:

2.3.1.17.1Number of operators available / busy at any time.

2.3.1.17.2Send SMS if number of Idle operators is less a threshold

2.3.1.17.3Send SMS if number of busy operators (Status in Rest) is greater than threshold

2.3.1.17.4Operator availability/busy time in a time period (example: in a shift).

2.3.1.17.5Send a message When the Call taker reaches the Defined Rest (Busy) limit

2.3.1.17.6Send a message when call taker is being Idle for more the threshold time

2.3.1.17.7Number of calls in the queues waiting more than a time period (example: 10 seconds).

2.3.1.17.8Send SMS if Certain Number of Calls X is waiting in the Y Queue for a period of Z Seconds

2.3.1.17.9Number of dropped calls greater than a threshold/time period.

2.3.1.17.10Non-receipt of any new emergency calls on the floor greater than X minutes (this will likely indicate to operations management a SIP failure or backend failure or campus network failure, thus requiring immediate attention and possible mitigations).

2.3.1.18The CONTRACTOR shall include the in the central OSS911 database repository all interactions with SMS including the necessary associations with the related incidents to be thus treated as part of the incident record.

2.3.1.19The CONTRACTOR shall deliver reports and dashboards relating to the SMS solution with details of these reports and dashboards to be agreed with the MOI during the implementation planning.

2.3.1.20The CONTRACTOR shall deliver an OSS911 solution that shall be able to directly integrate with any GIS systems including maps and data or open sources that the MOI may desire to connect to.

2.3.1.21The CONTRACTOR shall deliver an IRMS that shall integrate with video management system to receive alarms such as those from automatic number plate recognition, face recognition, crowd management, people counting, tripwire, geofencing etc., enabled cameras and then automatically create incidents in the IRMS.

2.3.1.22The IRMS shall:

2.3.1.22.1Create, manage, and archive Incidents using a variety of inputs and data sources, including automatic data acquisition and population into Incident Records and Incident Reports.

2.3.1.22.2Identify to an agency dispatcher, incidents that require management and/or resolution

2.3.1.22.3Identify and track resources that have been allocated to an incident

2.3.1.22.4Support virtual partitioning by:

- a. Agencies
- b. Departments
- c. Geography
- d. Security classifications
- e. Dispatchers

2.3.1.22.5Allow incidents to be accessed from one or more operating positions simultaneously

2.3.1.23The IRMS shall have the capability to display tickers with incident data, alerts and alarms on:

2.3.1.23.1All operators' and supervisors' screens

2.3.1.23.2Selected non-operators' screens

2.3.1.23.3All video walls screens

2.3.1.23.4All MDCs and handheld terminals

2.3.1.23.5Updates to tickers shall be configured to be performed by authorized OSS911 personnel.

2.3.1.24The IRMS shall have the capability to display alert messages on authorized operators' and supervisors' screens in case incident service levels threshold are exceeded.

2.3.1.24.1The alert message shall be configured to stay on screen until action is acknowledged by an authorized operator or the supervisor.

2.3.1.24.2 For example, if in a shift, when number of operators on duty fall below a threshold (threshold to be customizable by authorized personnel), automatic alerts and SMS to be sent to authorized officers' terminals and phones respectively.

2.3.1.25 The IRMS solution shall provide authorized OSS911 employees the capability to monitor in a log, all events pertaining to operator actions on the systems.

2.3.2 تسجيل الأحداث (INCIDENT REGISTERING)

2.3.2.1 The IRMS shall fully support multi-agency unified call handling. Multi-agency unified call handling implies that any call taker is able to receive calls for any of the agencies under the scope of this RFP.

2.3.2.2 The IRMS shall fully support receiving and registering emergency calls from a single emergency number or from multiple emergency numbers.

2.3.2.3 The IRMS shall fully support receiving and registering emergency calls through video calls.

2.3.2.4 The IRMS shall support the following:

2.3.2.4.1 A new and unique electronic record shall be created when an incident is first reported.

2.3.2.4.2 The incident record shall remain open and active through the life of the incident.

2.3.2.4.3 Once an incident record has been created it shall not be possible to delete that record.

2.3.2.4.4 Alert operator that an incident could be a duplicate and provide details for confirmation to prevent a duplicate record creation.

2.3.2.4.5 Changes to records shall be limited to specific fields. No changes shall be allowed to the incident log.

2.3.2.5 The IRMS shall support incidents regardless of how the information is initially received. Registering mechanisms will include but not be limited to the following:

2.3.2.5.1 The general public through the emergency service numbers

2.3.2.5.2 The general public contacting the operations rooms of the OSS911 Center through a general admin number or central Telephony number

2.3.2.5.3 A walk-in report at an emergency service facility (example: local police station)

2.3.2.5.4 A radio report from field staff

2.3.2.5.5 General public contacting the OSS911 center IRMS from SMS, and/or Smartphones application that is integrated with the IRMS

- 2.3.2.6 For incidents registering through applications on other MOI platforms like Maidan and through other MOI-advised Smartphone applications, integrations will be provided by the CONTRACTOR as per MOI concept of operations.
- 2.3.2.7 The IRMS will be required to support the following actions with regard to telephone calls accepted by or delivered to a call taker:
- 2.3.2.7.1 Operator can do an immediate transfer to: A language translator within or at any other OSS911 Center if required; A call taker with language translator qualification performing the translator role ; alternatively, if required, a language translator may be a part of the call and involved in a multi-party call with the call taker and the calling party.
 - 2.3.2.7.2 Transfer caller using conference call capability to a non-emergency point of contact if the call is classified as not an emergency and does not require response through OSS911 solution.
 - 2.3.2.7.3 Transfer caller using conference call capability to the relevant Agency when the call is not dispatched by OSS911 agencies – for example an ill person who requires a Saudi Red Crescent Authority response.
 - 2.3.2.7.4 Maintain a recording of received inbound call and call transfers.
- 2.3.2.8 The IRMS shall deliver the functionality of business intelligence (with reports, statistics and dashboards) and decision support.
- 2.3.2.9 The interface to the telephony system shall allow call details to be passed automatically to the IRMS.
- 2.3.2.10 The interface to MOI radio communications networks will allow silent (text) dispatching and provision of background information to support incident response. It will also allow for auto updating through the use of status messaging and field access to incident records.
- 2.3.2.11 Where an incident is automatically created as a result of an incoming call via the telephone system, on receipt of the call details, the IRMS shall generate a new unique record, date and time stamp the call and give the input screen to the operator.
- 2.3.2.12 The system shall allow the operator to commence filling the input forms and provide:
- 2.3.2.12.1 Visual alert to an open incident record, or a closed record created within a user configurable time frame which already exists from the calling number
 - 2.3.2.12.2 Visual alert of an open incident record, or a closed record created within a system configurable time frame which already exists within a system configurable distance from the incident location.

2.3.2.12.3 Access to such an incident record and a summary of those record(s) shall be viewable to the operator. The operator shall be able to interrogate full record details.

2.3.2.13 If the incident is confirmed by the operator to be a duplicate the details from the new incident are copied to the master where different and the duplicate is closed.

2.3.2.14 The IRMS shall:

2.3.2.14.1 Pass the phone number data from the CONTRACTOR provided telephony solution to the IRMS on reception of an emergency call.

2.3.2.14.2 Automatically query the ANI/ALI records and the caller profile information through the integration provided by the CONTRACTOR to the caller location and caller information service providers as determined by the MOI.

2.3.2.14.3 Populate distinct fields in the incident record (event) form and all the databases with returned caller location information from this query such that the operator is able to determine and use the best caller location from the different caller location sources and compare it with the information provided by the caller.

2.3.2.14.4 Provide two caller location fields within the IRMS solutions in each of the applications in the IRMS solution including but not limited to the computer aided dispatch solution and IP telephony softphone / software telephone console.

2.3.2.14.5 Populate distinct fields in the incident record (event) form and all the databases with returned caller profile information returned from this query.

2.3.2.14.6 Interface with MOI identified web services to obtain the emergency information (relative name, relative telephone number, addresses, blood group, known medical history etc.) for each person and populate the corresponding fields in IRMS and store the obtained data as part of the incident record within the databases to be utilized by concerned personnel including dispatch, field responders, quality team, center management among others.

2.3.2.14.7 Perform an association of the Incident Record to the call so that if the call is transferred, the incident record with ANI/ALI information and caller profile information can be automatically associated with the call at the receiving operator's workstation

2.3.2.14.8 In a separate window, plot the location of the caller on an IRMS map display

2.3.2.14.9 Based on the incident type and location, the IRMS shall automatically determine which agency, and if relevant the geographic area within that agency that will be responsible for handling the incident.

2.3.2.14.10 Integrate with the record management system in the NIC and the MOI through the related web service provided by them.

2.3.2.14.11 Integrate with different agencies information systems through IRMS web services to share the incident and/or event information, statistics and reports.

2.3.2.15 The IRMS shall allow a call taker to send updates related to the incident to dispatch immediately in some cases while continuing to engage the call taker and take additional details. It shall include update details in the call record. The OSS911 solution shall have the capability to report the use of this feature.

2.3.2.16 For incidents registered through integration of IRMS with video management system (VMS) and video analytics (VA), within the IRMS the received video is tagged (linked) to the incident (event) created and a pointer to the video is stored as part of the incident (event) record in all the databases for live and for historical reporting.

2.3.3 **قدرات المشغل (OPERATOR CAPABILITIES)**

2.3.3.1 It shall be possible for an operator to manually create an incident.

2.3.3.2 It shall be possible for the operator to create and forward incident to a dispatcher for action whilst retaining the template on screen to capture further information from the caller.

2.3.3.3 It shall be possible for an operator to manually geographically re-locate the incident that had previously been automatically identified by the system.

2.3.3.4 The operator shall be able to separately record details of the caller location and the incident location in different, searchable fields within the incident record.

2.3.3.5 Access and view of an incident record shall be limited to authorized operators and operator roles.

2.3.3.6 Within the IRMS, it shall also be possible for the supervisors to apply limits to what period of incident-related action reports an operator can view on the incident record – for example, the supervisor may decide that an operator for a given shift is only able to see the updates of the previous two shifts.

2.3.3.7 It shall be possible for high priority or critical incidents, that they should automatically be displayed in a portion / the entire video wall in the main operation room, and the solution shall be such that the video feed from the camera nearest to the incident scene and within 200 meters is automatically selected and displayed.

2.3.3.8 The IRMS shall provide the ability to record information from callers about particular situations or incidents that do not require the dispatching of any public safety resources. These incidents shall be recorded and retrievable from the history files for later access and information analysis.

2.3.4 معلومات الموقع (LOCATION INFORMATION)

2.3.4.1 Once sufficient location information has been entered into the incident report (either manually or automatically), the location of the incident shall be displayed on a map.

2.3.4.2 Once the location of the caller has been automatically identified by the telephony system or has been selected by an operator from the map, all the location details including (Region, City, District, Street, Parcel, Postal number, and Coordinates, etc.) shall be generated automatically from whatever spatial layers available.

2.3.4.3 The IRMS shall be capable of receiving this information and updating the incident record with the all the location data received from any source.

2.3.4.4 The CONTRACTOR shall deliver an IRMS solution that shall be integrated with the Automatic Location Identification (ALI) information available from the sources defined by MOI which include but are not limited to the National Information Center (NIC), the National Center for Security Operations (NCSO) and any third-party service provider that include technologies such as Advanced Mobile Location (AML), Enhanced Location Services (ELS) and eCall Minimum Set of Data (MSD).

2.3.4.5 The word "location" as mentioned in this document anywhere includes the location returned from any of the sources authorized by the MOI using any of the technology mentioned above.

2.3.4.6 The CONTRACTOR solution shall ensure that all the location data returned from any location source including the latitude, longitude, radius, time of positioning and level of confidence, and the eCall MSD is presented on the IRMS and stored in all the databases and tagged to the related calls and incidents in the call and incident records.

2.3.4.7 The CONTRACTOR solution shall ensure that all the location data as above is presented to the operator in IRMS within the incident forms of the computer aided dispatch solution and within the softphone / software telephony console that receives the emergency call at the operator.

2.3.4.8 The CONTRACTOR solution shall ensure that all the location data as above is presented to the operator in IRMS within the incident forms of the computer aided dispatch solution and within the softphone / software telephony console that receives the emergency call at the operator.

2.3.4.9 In the case of operational OSS911 Center workstations where two screens are installed on a PC the map shall be displayed on a separate screen to the Incident Record.

2.3.4.10 The Call Taker or Dispatcher shall be able to correct the location by updating the location fields in the incident form and in the incident record and incident report.

2.3.4.11 The telephone operator shall be able to correct the location by identifying the location on the map. In this case the mapping application

shall be capable of sending data back to the IRMS. The IRMS shall be capable of receiving this information and updating the record.

2.3.5 مصادر البيانات وعمليات البحث (DATA SOURCES AND SEARCHES)

2.3.5.1 Where entries are made in any fields relating to data that is held in external databases, e.g., car license numbers, then the system shall validate the entered data against this data from available databases and highlight to the operator any conflicts and automatically populate the other related attributes that shall be captured in the incident log.

2.3.5.2 The IRMS shall also search for previous incidents relating to the callers' telephone number or post code and bring these to the attention of the call taker.

2.3.5.3 The IRMS shall also be able to perform supplementary searches of the NIC and other MOI databases with the same caller information to provide additional and relevant cross reference information to the call takers. Any resulting warnings or indicators shall be attached to the incident log.

2.3.6 مؤقتات الأحداث (INCIDENT TIMERS)

2.3.6.1 When an incident is created, timers, configurable through system configuration, shall be started to ensure that entry of incident information and dispatching occurs within set parameters.

2.3.6.2 It shall be possible to set a start timer at some time in the future so that the incident does not appear to be outstanding.

2.3.6.3 Timers shall be automatically reset once the required action has been carried out, i.e., incident information has been entered or dispatching has occurred.

2.3.6.4 On expiry of each timer a message with visual and audible alert shall be automatically sent by the system to a supervisor in order to alert them to a potential problem. It shall be possible to turn on / turn off the alerts for specific users.

2.3.6.5 The alert shall continue until acknowledged by the Supervisor. (Operator performance related alerts should continue to stay until acknowledged by the supervisor. Unit status alerts are not necessarily required to continue to stay until acknowledged by the supervisor).

2.3.7 وظيفة دعم قرار المشغل (OPERATOR DECISION SUPPORT FUNCTION)

2.3.7.1 As an aid to call-taking, the system shall provide a configurable set of questions and prompts which shall relate to incident types and incident response procedures and plans. This is particularly important with regard to incidents that appear to be serious or involve high profile people or locations.

2.3.7.2 The system shall be able to present further questions, depending on the data that is input as a result of the previous set of questions to ensure all available and appropriate incident information is obtained and recorded.

2.3.7.3 The system shall be configured by the CONTRACTOR to contain questions and prompts for all incident types. These questions and

prompts shall be based on the SOPs and response plans approved by the MOI.

2.3.7.4 This list shall identify a series of tasks or actions (Incident Response Procedures) which must be carried out before an incident can be closed.

2.3.7.5 The BIDDER shall confirm their understanding of the SOPs definition as being the reference checklists showing how each organization or agency will accomplish its assigned tasks for pre-defined incident from call receipt to call closure. SOPs and response plans shall also describe who will be responsible for dealing with incidents, how they shall be informed, what information is required, what escalation measures are available and what performance targets have been set for responding and dealing.

2.3.7.6 The IRMS shall be supported by a decision support function based upon the MOI approved Standard Operating Procedures (SOPs) and response plans. The CONTRACTOR shall work with the MOI to create the workflow of the decision support function using the roles configured in the system. The procedures to be followed by operational personnel will be provided by the MOI.

2.3.7.7 The CONTRACTOR shall deliver SOPs and response plan databases as described above and already populated in the IRMS for each agency operating in the OSS911 Center. These SOPs and response plan databases shall be integrated with the CONTRACTOR supplied IRMS.

2.3.7.8 The CONTRACTOR shall liaise with the MOI to obtain, update and program the SOPs and response plans that provide the optimum decision support for all OSS911 staff involved in the OSS911 process. Sources of this information will include, but not be limited to:

2.3.7.8.1 Existing agency policies & procedures defined to support MOI strategies which if available, will be shared by the MOI

2.3.7.8.2 Processes or common practices that have evolved institutionally over the years which if available, will be shared by the MOI

2.3.7.8.3 Best practice observed or reported in other PSSA environments. The CONTRACTOR shall evaluate the SOPs provided by MOI against best practices observed by them in other public safety environments and shall advise to the MOI any feedback that they may have.

2.3.7.9 The CONTRACTOR shall update the locally developed SOPs and response plans in the system, when such updates are shared by the MOI.

2.3.7.10 The CONTRACTOR shall document the SOPs and the response plans such that they are included in the IRMS decision support function.

2.3.7.11 The CONTRACTOR shall configure the SOPs and response plans for all incident types and for all incident priorities and integrate them into the IRMS.

2.3.8 **SECURITY RELATED CALLS HANDLING** التعامل مع المكالمات المتعلقة بالأمن

2.3.8.1 The CONTRACTOR shall create a secure and isolated solution for handling national security related calls.

- 2.3.8.2The CONTRACTOR's solution for handling such calls shall ensure that all related information is handled with utmost security. There shall be no sharing of any information unless authorized by MOI.
- 2.3.8.3All such call information shall be saved in a secure and isolated database that is accessible only by authorized operators. This database shall have the highest security and shall have highly restricted access that is determined by MOI.
- 2.3.8.4The CONTRACTOR's solution shall capture the location of any caller and record it in this secure database.
- 2.3.8.5For the people who are determined by the MOI to have participated in suspicious and / or criminal activity, the CONTRACTOR solution shall ensure that it shall be possible for specialist security-related call handlers to update such people's personal records in the security database to reflect the same.
- 2.3.8.6The BIDDER shall dimension this solution for up to ten positions.
- 2.3.8.7This supervisor position shall include the capability to monitor these security related incidents at all levels and for ensuring that actions are happening as required.
- 2.3.8.8The system functionality on the computers and the equipment shall be the same as that of the OSS911 call takers and Supervisors.
- 2.3.8.9The call retention guidelines for such security-related calls are the same as other OSS911 calls.
- 2.3.8.10The CONTRACTOR delivered activities for OSS911 SOPs and response plans development and deployment shall cover all procedures for these positions and their related functions.
- 2.3.8.11The CONTRACTOR shall configure the workflow for such calls to include the following:
- 2.3.8.11.1Once it is determined by a call taker that the call pertains to national security, the call information shall be passed on to a team of specialist security-related call handlers, which shall handle all such calls.
- 2.3.8.11.2Further, the call taker who received the call at OSS911, shall no longer be in the call or have access to the call card after transfer.
- 2.3.8.11.3The caller's identity shall not be displayed when the call appears on the screens of the specialist security-related call handlers. The caller identity that may have been recorded by a call taker who received the call at OSS911, is visible only to extremely restricted and highly placed MOI officials. No one else shall be able to see the caller identity. For any operator who cannot see the caller's name, the call shall be identifiable by the call number only.
- 2.3.8.11.4Information about callers and about what caller-related information is seen by call takers, dispatchers and other staff will be classified. If there is any information associated with any caller

that he has previously provided security information to OSS911, then such associated information about that caller shall only be available to these security-related call handlers and will not be available to anyone else in OSS911. When a subsequent non-security related call is made by the caller to OSS911, then the OSS911 call takers, dispatchers and other OSS911 staff shall be able to view all details about the caller except any security calls related information linked to that caller.

2.3.8.11.5 It shall be possible for the specialist security-related call handlers to transfer the call back for action by police or any other agency if they determine that the call does not come under the jurisdiction of the MOI agencies handling security related incidents.

2.3.8.11.6 It shall be possible for the specialist security-related call handlers to call back a caller to get specific information. As such, the CONTRACTOR shall provide a solution for them to call back the callers.

2.3.8.11.7 If it is determined by the specialist security-related call handlers that the call is a "positive" call, they shall determine the call type and that whether the call can be handled locally or if it is needed to be dispatched to other security-related MOI centers in the province or in any other provinces.

2.3.8.11.8 The specialist security-related call handlers attending to the call shall research if more information is required to proceed with handling that call. In case of more information required, the specialist security-related call handlers shall research this information.

2.3.8.11.9 If required, the call is escalated appropriately based on the urgency of the situation.

2.3.8.11.10 An SMS shall be configured to be sent to the caller updating him/her appropriately about the action done and its status.

2.3.8.11.11 Once the security related call has been attended, a solution shall be provided by the CONTRACTOR for a report to be prepared by the specialist security-related call handlers team and to be sent to an appropriate MOI authority.

2.3.9 المتطلبات المتعلقة بالقطاعات المتعددة (MULTI-AGENCY REQUIREMENT)

2.3.9.1 Each incident shall be allocated to one PSSA. It shall be possible to later change this allocation when further information has come to light.

2.3.9.2 In addition to the unique IRMS incident number, the system shall generate an incident number which is unique to the agency assigned to the incident.

2.3.9.3 The IRMS shall provide a capability to have different types of incidents configurable for different agencies.

2.3.9.4 The IRMS shall provide specific formats to meet the requirements of each agency such as the inclusion of vehicle registration and type fields,

search criteria and the capability to select pre-determined attendance and facilitate automatic alerts.

2.3.9.5The IRMS shall present partitioning options based on standard operating procedures to ensure that, when the call taker inputs information, the incident log is identified as falling under the jurisdiction of a particular agency.

2.3.9.6The IRMS shall support all an interface capability in the IRMS that shall allow provision of the incident details in electronic format so that external systems can create an incident on their own.

2.3.9.7Once an incident involving multiple agencies is resolved, the systems shall have the capability that:

2.3.9.7.1A single common report is sent to all relevant agencies and organizations that is viewable by authorized people.

2.3.9.7.2The system shall have the capability that if defined by the concept of operations, an agency shall be able to view on the report only such information that it is permitted to view – for example, it could be decided that CD views only its' resources while Police Patrol views its' resources.

2.3.9.7.3The system shall have the capability that if warranted by the concept of operations, the details of the number and type of resources used by an agency may only be visible to the respective authorities and not be visible between the agencies.

2.3.10 **نشر حدث (POSTING AN INCIDENT)**

2.3.10.1In the case where an incident record is being created as the result of a telephone call, the operator shall be able to manually release the call.

2.3.10.2The call taker screen shall provide a display of dispatchers who are logged onto the IRMS system and their areas of responsibility to facilitate the expedient manual routing of incidents.

2.3.10.3The IRMS shall support “supervisory position routing” of particular user-defined incident types to designated supervisors.

2.3.10.4The operator shall be able to assign the incident to an incident dispatch queue for dispatching. Incidents requiring a deferred response shall not enter an incident dispatch queue until their start timer is reached.

2.3.10.5In other cases, it may be necessary to post the incident and transfer the call with the assistance of a call taker (an assisted transfer) or without the assistance of the call taker (a blind transfer). Assisted transfer is a tactic frequently employed when dealing with an on-going or developing situation such as a robbery.

2.3.10.6Incidents assigned to an agency shall be visible to dispatchers assigned to that agency and the voice part of the call (if relevant) be transferred into a voice queue. Each agency may be sub-divided (for example by geographic area) and visibility controlled by those divisions.

- 2.3.10.7 The interface to the telephony shall support the initiating of call release and call transfer from within the Incident Management application.
- 2.3.10.8 Each incident shall be assigned a priority.
- 2.3.10.9 Assignment of incident priority shall be either automatic based on the entered parameters, or manual based on the operator's choice.
- 2.3.10.10 The selection of the incident dispatch queue(s) shall be automatic based on parameters which will include service required, location of the incident and pre-determined responsible area.
- 2.3.10.11 It shall be possible for the operator or the supervisor to override the choice of incident dispatch queue and dispatch to locations. In such cases, the supervisor shall be notified with visual and audible alerts that will continue, until acknowledged by Supervisor. These locations may be identified either by type or unique address.
- 2.3.10.12 The incident shall be capable of being posted or transferred at different priority levels automatically.
- 2.3.10.13 The operator shall be able to manually assign a different priority level on the system than that automatically assigned.
- 2.3.10.14 The system shall provide at least three (3) levels of incident priority which are described by the use of Arabic alphanumeric identification. Incidents must be classified in such a way as to identify those requiring urgent attention (high priority), a standard response (normal priority) or those where a response could be delayed if necessary (low priority) and can be able to be assigned a different priority by agencies for the same incident.
- 2.3.10.15 The BIDDER shall detail the solution for handling of malicious calls within the IRMS.
- 2.3.10.16 The BIDDER shall detail the solution for management of the transferred calls in their proposal.
- 2.3.10.17 For the scenario where more than one call is received for the same incident which requires a response, the call taker shall create a single record for the multiple calls related to the incident. This single call record shall have a unique incident number identification for each permitted agency for that incident.

2.3.11 **ترحيل البلاغات (DISPATCHING)**

- 2.3.11.1 As part of the IRMS integration with MOI radio communications networks, for each dispatch position and dispatch supervisor position, MOI will be provisioning a MOI radio communications network computer along with its ancillary equipment including push-to-talk, for voice communications over the MOI radio communications networks.
- 2.3.11.2 A dispatcher shall be able to monitor the content of multiple incident dispatch queues at the same time.
- 2.3.11.3 New incidents shall be placed in the appropriate incident dispatch queue(s).

- 2.3.11.4 Queued incidents shall be able to be sorted and filtered by a user configurable process which will include, as a minimum, priority, location (where incident locations have been pre-defined as areas on a map) , administrative area, type.
- 2.3.11.5 The system shall provide a mechanism to ensure that those incidents, which are deemed to have the highest priority, are immediately brought to the notice of the dispatcher.
- 2.3.11.6 A full or partial listing of an incident dispatch queue or queues shall always be visible on the electronic logging screen irrespective of other tasks being performed.
- 2.3.11.7 Where there is more than one dispatcher monitoring an incident dispatch queue, work queues shall be available into which the incident can be moved for management by an individual dispatcher.
- 2.3.11.8 Where a dispatcher moves an incident into their own work queue, this shall not remove the association of the incident from the dispatching queue it was placed.
- 2.3.11.9 The system shall tag the incident in such a way as to indicate it is in one or more work queues.
- 2.3.11.10 The system shall be configurable on a user-by-user basis as to the way dispatching queues are viewed. For example, they could show:
- 2.3.11.10.1 All open incidents with an indicator as to whether or not the incident is being managed in a work queue
 - 2.3.11.10.2 All incidents in a dispatch queue that are unacknowledged
 - 2.3.11.10.3 Incidents sorted by priority or by other parameters such as longest time since incident updated
- 2.3.11.11 The system shall be configurable to allow triggers to generate reminders. For example, if an incident is not actioned in an incident dispatch queue within a certain period of time, an audible and visual alert notification shall be generated in front of authorized operational positions.
- 2.3.11.12 Supervisors shall be able to monitor all incident dispatch logs and work queues for their specific area of responsibility and shall be able to update in real time.
- 2.3.11.13 Users with sufficient privilege shall be able to monitor all incident dispatch queues in the system.
- 2.3.11.14 The view of an incident that a user is given when they open an incident shall be configurable based on the parameters set by the agency and role of user.
- 2.3.11.15 Different fields shall be available to a user within an incident record, depending on their role with in an agency.
- 2.3.11.16 A dispatcher shall be able to manually (by pressing a button on the incident form) open the electronic map. This feature shall not be automatic as a dispatcher might also be monitoring another incident at the same time.

2.3.11.17The map shall show the location of the incident.

2.3.11.18The map shall show available resources (with status) within a certain pre-defined distance from the incident that are appropriate to the operator's role. Resources can include personnel, vehicles and specialist equipment.

2.3.11.19The rules used for displaying available resources shall be configurable.

2.3.11.20The map shall dynamically update showing the position and status of resources.

2.3.11.21It shall be possible for a dispatcher to call up a set of standard operating procedures (possibly in the form of a checklist) to follow based on the incident type. It shall be possible to update these procedures subject to authorization.

2.3.11.22Dispatchers shall be able to select and assign resources to the incident. This process will also require the dispatcher to communicate with the relevant resources.

2.3.11.23The dispatcher shall be able to dispatch to Field Units by voice (radio or telephone) or by transmitting the relevant aspects of the incident log to a mobile device.

2.3.11.24It shall be possible for the IRMS to automatically update the status of resources, based on a status update made by the Field Units on radio or MDC.

2.3.11.25It shall be possible to manually update resource status.

2.3.11.26The system shall be capable of making these updates using status information messages that are sent by the field.

2.3.11.27Multiple dispatchers (possibly from different agencies) shall be able to have the same incident and or instance of the same incident record open at the same time.

2.3.11.28Multiple dispatchers (possibly from different agencies) shall be able to make simultaneous updates to the same incident and/or instance of the same incident. Ownership of the incident remains with the appropriate agency and the CONTRACTOR shall provide a solution that prevents data loss in case of simultaneous updates.

2.3.11.29A dispatcher shall be able to re-assign the incident to a different incident dispatch queue within:

2.3.11.29.1The same agency

2.3.11.29.2Another agency

2.3.11.30Authorized users shall be able to re-assign the incident to a different incident dispatch queue within:

2.3.11.30.1The same agency

2.3.11.30.2Another agency

2.3.11.31 The IRMS shall maintain a file of the emergency contacts for locations, containing, at a minimum:

2.3.11.31.1 Name of the contact

2.3.11.31.2 Relationship of the contact to the location

2.3.11.31.3 Home and work phone numbers

2.3.11.31.4 Free form comments

2.3.11.32 Whenever an incident location has emergency contacts, an indicator shall be displayed to the operator advising them of the existence of the emergency contact information.

2.3.11.33 The detailed information on the emergency contact shall be displayed in a separate area or window on the screen, allowing the incident to be displayed at the same time as the contact information.

2.3.11.34 All information concerned with an incident record shall be recorded onto an action log, together with date and time of the user making the changes. This applies equally to information recorded by call takers, dispatch operators and from spatial and intelligence database searches.

2.3.11.35 A dispatch operator shall be able to select any item or items on an action log and transmit them as text messages to suitably equipped radio or field users.

2.3.11.36 Likewise, a dispatch operator shall be able to paste any text received from such users directly into incident action logs.

2.3.11.37 A user shall be able to view a single historical log for that record that lists all the individual changes to the record in chronological order.

2.3.11.38 It shall be possible to configure the rules that must be enforced when an operator requests to close an incident, for example the IRMS may prevent closure if resources still assigned (for safety events) or when an operator closes an incident with resources assigned, they are all released (non-safety related events).

2.3.11.39 When an incident is closed it shall move out of the current incident dispatch queues and move to an associated closed incident queue.

2.3.11.40 No changes shall be possible to a closed incident, other than to add additional closure information.

2.3.11.41 The Incident Management System shall allow dispatchers to display on a map and in a list, before seeking, assigned resources and/or assigned and available resources.

2.3.11.42 The IRMS shall automatically provide the dispatcher with a suggested unit recommendation. This recommendation shall be composed of a specific unit identifier and shall be derived by taking into account the following basic elements:

2.3.11.42.1 Unit staffing and unit types.

2.3.11.42.2 Unit types and unit equipment capability.

- 2.3.11.42.3 Incident type to determine the type and number of units to recommend.
- 2.3.11.42.4 Incident geographic location.
- 2.3.11.42.5 The IRMS shall allow the dispatch of different types of resources for the same incident type when the incidents are located in different geographic sub-areas (i.e., geographically sensitive dispatch policies).
- 2.3.11.42.6 Real-time unit status to determine unit availability.
- 2.3.11.42.7 Unit recommendations shall correspond to the current, real-time status of all resources.
- 2.3.11.42.8 The IRMS shall never recommend a unit that is on another assignment or otherwise unavailable for dispatch.
- 2.3.11.42.9 The IRMS shall facilitate the definition and recommendation of second, third, etc., level units in the event primary recommended response unit is in an unavailable status.
- 2.3.11.42.10 Flexible patrol district plans and the currently active patrol district plan. The IRMS shall support multiple patrol district plans. The BIDDERS shall indicate how many different plans may be entered by the user agency.
- 2.3.11.42.11 The IRMS shall provide the Dispatcher with ability to override patrol district assignments on a temporary basis.
- 2.3.11.42.12 The IRMS shall support increasing the normal response based on "locations".
- 2.3.11.42.13 The IRMS shall be able to dispatch the nearest appropriate unit based on its actual location from vehicles equipped with Automatic Vehicle Location (AVL) capabilities.
- 2.3.11.42.14 The IRMS shall have the capability to accept the system-provided unit recommendations with a single keystroke or action of a point and click device, or override the recommended units and replace them with one or more other units.
- 2.3.11.42.15 The dispatcher shall have the capability to select a unit that is on a lower priority incident and reassign to a higher priority incident.
- 2.3.11.42.16 A single keystroke shall remove the unit from the previous incident and assign it to the new incident.
- 2.3.11.42.17 If a unit is the only unit or the last unit assigned to an incident, the incident shall be automatically placed in the pending incident queue and either held (stacked) for that unit or available for the dispatch of another unit.
- 2.3.11.42.18 When the unit clears the incident to which it was assigned, the unit shall be recommended to the incident from which it was pre-empted, if incident is still pending.

- 2.3.11.42.19 If a different unit is assigned to the incident in the pending queue, the incident shall no longer be stacked (held) for the original unit and the system shall not automatically recommend it when the unit becomes available again for dispatch.
- 2.3.11.42.20 All times associated with assignment and re-assignment shall be kept in the incident history file.
- 2.3.11.42.21 The dispatcher shall have the ability to change the primary unit at the time of dispatch or at any time during the handling of the incident.
- 2.3.11.42.22 The IRMS shall provide the ability to stack, or assign low priority incidents to a busy unit. These incidents shall be time stamped, and displayed in the pending incident display with an indication that the incident has been stacked to a unit.
- 2.3.11.42.23 When a unit clears from one incident, the IRMS shall provide an indication that the unit is now available.
- 2.3.11.42.24 The IRMS shall time stamp when the unit is en route to the new incident.
- 2.3.11.42.25 Upon acceptance of a unit dispatch recommendation or input of a dispatcher's own unit recommendation, the applications software shall automatically and dynamically update the status of all affected units throughout the IRMS and the updated status shall be shown at all IRMS operators.
- 2.3.11.42.26 The IRMS shall provide the ability to automatically transmit "pre-defined" messages to MOI authorized field application and/or SMS, based on the type of incident.
- 2.3.11.42.27 The IRMS shall have the ability to transmit specific information from the incident to specified users via MOI authorized field application and/or SMS.
- 2.3.11.42.28 The IRMS shall provide the use of drag-and-drop functionality for dispatching units and for performing status changes.
- 2.3.11.42.29 The IRMS shall provide an indication as to whether a Patrol unit is a single or multi-person unit. The unit icon displayed on the map shall indicate this also.
- 2.3.11.42.30 The IRMS shall provide unit status timers that will advise the dispatcher if a unit has exceeded the predefined amount of time in a status condition.
- 2.3.11.42.31 The time interval for each status timer shall be defined and set by the system administrator. The time of the original assignment shall be maintained as part of the incident record.
- 2.3.11.42.32 The application shall provide secondary check-back times which shall be the defined time periods, after the initial check-back, that the IRMS shall continue to prompt the dispatcher to check on a unit's condition.

- 2.3.11.42.33 When a defined check-back period has expired, the system shall visually and aurally alert the dispatcher assigned to the unit to make contact with the unit.
- 2.3.11.42.34 Once contact is made with a unit whose check-back timer has expired, and the unit advises that its status is fine, the dispatcher shall be provided with an appropriate means to cancel the check-back alert and reset the unit's check-back timer.
- 2.3.11.42.35 In addition to incident related timers, the proposed IRMS system shall include a unit welfare check function.
- 2.3.11.42.36 The IRMS shall be configured to track the elapsed time since the last welfare check or since the unit went on duty and to prompt the controlling dispatcher to check on the unit's welfare if it hasn't called in.
- 2.3.11.42.37 The timers shall be table driven, and have the ability to be configured by shift, such that more frequent checks may be performed at night.
- 2.3.11.42.38 Dispatchers shall be capable of updating unit status through keyboard input of appropriate unit identifiers and a single function key, by a pointing device or via command line entry.
- 2.3.11.42.39 The IRMS shall allow dispatchers to update a unit's status while performing any call taking or dispatching function within the IRMS system by providing easy accessibility to an interactive command line at all times. This command line shall allow multiple units to have the same status update simultaneously.
- 2.3.11.42.40 Updating unit statuses must be accomplished without losing the incident information displayed on the screen. If the cursor is repositioned to perform the command, it must be automatically returned to the correct screen and cursor position where the user left off, without losing any information.
- 2.3.11.42.41 The IRMS shall provide dispatchers with the capability to clear any number of, or all units, assigned to an incident with a single command.
- 2.3.11.42.42 From the field, equipped vehicles shall be able to update their own statuses through its integration with the IRMS.
- 2.3.11.42.43 The system shall track these status updates as if they were entered by system operators, but indicate that they were updated by the unit itself (e.g., track the time and the ID of the person/device completing the status update)
- 2.3.11.42.44 The IRMS shall allow both call takers and/or dispatchers to review an active incident and update the incident with corrections or additions.
- 2.3.11.42.45 All corrections or additions shall contain the time, date, and operator ID.

- 2.3.11.42.46 When a call taker forwards updated information to the appropriate dispatcher, the dispatcher shall be visually/aurally alerted by the system to the presence of the update.
- 2.3.11.42.47 In addition, there shall be a clear indication of which information is new or changed.
- 2.3.11.42.48 The IRMS shall allow units to be added as assisting units to an incident after it has been dispatched.
- 2.3.11.42.49 If another agency response is required, such as adding an ambulance to an incident, the IRMS shall automatically copy the active incident and route the new incident to the appropriate dispatcher.
- 2.3.11.42.50 The BIDDER shall provide an explanation of their system's ability to add additional public safety agency resources to an active incident.
- 2.3.11.42.51 The dispatcher shall be able to escalate the priority of an event.
- 2.3.11.42.52 The IRMS shall make additional unit recommendations based on the new priority level. Dispatchers must be able to "preview" the next priority level assignments for an active incident.
- 2.3.11.42.53 The IRMS shall only recommend the "remainder of the required units" for an incident if units have already been assigned and dispatched to the incident.
- 2.3.11.42.54 The IRMS shall allow units to be reassigned from one incident to another or to be easily "exchanged" from one active incident to another.
- 2.3.11.42.55 The previous incident shall be returned to the pending queue if the reassigned unit is the last or only unit assigned to the incident.
- 2.3.11.42.56 The BIDDER shall provide an explanation of how these functions are accomplished by their systems.
- 2.3.11.42.57 The IRMS shall allow users to clear either single units or all units on an incident with a single command, function key, or point-and-click device action.
- 2.3.11.42.58 If the last unit on an incident is cleared, IRMS shall require a disposition code.
- 2.3.11.42.59 The IRMS shall provide for the capture of a user-maintainable incident disposition code, an indicator that a report is or is not required and incident completion comments.

2.3.12 IRMS المتكاملة الجغرافية المعلومات النظام قدرة (IRMS INTEGRATED GEOGRAPHICAL INFORMATION SYSTEM CAPABILITY)

- 2.3.12.1 The IRMS with its integration with GIS, maps, map data and imagery shall all the requirements mentioned in sections 2.1 and 2.2 of this RFP.
- 2.3.12.2 The IRMS system shall have the following capabilities:

2.3.12.2.1 Address the needs of first responders in regular incidents and crisis and emergencies, including:

- a. Providing situational awareness interactively on the map
- b. Providing access to mission-critical information like previous incidents in the area, estimated hazards etc.
- c. Providing routing information, via map and voice, from current location to destination
- d. Provide history of routing information

2.3.12.2.2 If the IRMS system is unable to provide an exact location match, a list of potential matches, shall be displayed to the user. An operator shall be able to select the correct location from the displayed list, scroll forward or backward for other potential locations, or restart the location lookup with a new location.

2.3.12.2.3 Resources / units shall be displayed on the maps by means of different shaped, colored and expressive symbols.

2.3.12.2.4 The display of resource / unit group, type and status shall be configurable by authorized department, agency and respective managers.

2.3.12.2.5 It shall be customizable if such display about a resource is always available or available when hovering over with the mouse or available on a click.

2.3.12.2.6 The CONTRACTOR shall submit legend, asset, and resource symbols to the MOI; the MOI may make changes at their discretion; the CONTRACTOR shall incorporate these changes into the solution.

2.3.12.2.7 When the map is displaying the entire dispatch area, all vehicles shall be clearly displayed. Each icon shall display the resource / unit identification number within, either immediately above, or immediately below the icon.

2.3.12.2.8 The IRMS solution shall support as minimum the following types of map services and APIs- WebGL/MapGL, WMS, WFS, WMTS.

2.3.12.3 The BIDDER shall explain in their proposal how their solutions deliver support to the following operational scenarios:

2.3.12.3.1 Caller and incident location

2.3.12.3.2 Caller and incident validation

2.3.12.3.3 Incident response resource allocation

2.3.12.3.4 Incident response routing

2.3.12.3.5 Incident management

2.3.12.3.6 Resource monitoring

2.3.12.3.7Responder route planning

2.3.12.3.8Selecting exclusion zones

2.3.12.3.9Selecting evacuation routes and zones

2.3.12.3.10Deployment of security checkpoints

2.3.12.3.11Automatic resource location

2.3.12.3.12Spatial incident analysis

2.3.12.3.13Geo-fencing

2.3.13 (ESCALATED INCIDENTS AND EMERGENCIES) إدارة الأحداث الطارئة والمُصعِّدة (MANAGEMENT)

2.3.13.1The MOI requires that the CONTRACTOR supplied IRMS solution incorporate handling emergency and crisis to support the OSS911 Center response to major emergency incidents.

2.3.13.2The BIDDER shall confirm the understanding of the MOI definition that an “Emergency Incident” is defined an occurrence or event, natural or man-made that requires an immediate response to protect life or property.

2.3.13.3The IRMS shall combine contingency plans, standard operating procedures and the data to provide support for commanders, incident managers and communications dispatchers for emergencies.

2.3.13.4The IRMS tool shall include a spatially referenced database of contingency plans integrated with the maps.

2.3.13.5The BIDDER shall provide a list of all the hazards and events the IRMS shall support and shall supply the necessary tools and training for the MOI to add more hazards and events to the system in future, should the need arise.

2.3.13.6The IRMS shall have the capability to assist commanders and operators to ascertain the nature and magnitude of an emergency or crisis situation so that appropriate resources can be deployed to limit the extent of disruption (Protect) and begin the Response and Recovery phases without delay.

2.3.13.7The IRMS shall be able to:

2.3.13.7.1Link an incident to the appropriate contingency plans.

2.3.13.7.2Prompt the dispatcher in line with prepared plans and documented SOPs.

2.3.13.7.3Identify associated hazards related to a location based on the provisioning of such information from the MOI.

2.3.13.7.4Locate resources, log incident activities, track requests and tasks, generate situation reports and communicate critical information across local and wide area networks and the internet.

2.3.13.7.5 Identify resource availability, including their locations and specialist skills.

2.3.13.7.6 Support task allocation to individuals, departments or Agencies.

2.3.13.7.7 Maintain duty logs for recording all significant activities and actions taken during a fixed period of time (duty shift).

2.3.13.8 The CONTRACTOR shall provide a solution for the IRMS to be able to disseminate through SMS solution warnings related to emergencies and crisis to agencies, and to people at large.

2.3.13.9 The CONTRACTOR shall provide a solution for the IRMS to be able to disseminate through SMS alerts to agency staff and officials informing them about an event and for eliciting their participation in event resolution.

2.3.13.10 The IRMS shall support OSS911 operators and supervisors by providing the following capability:

2.3.13.10.1 Assess and Control Hazards

- a. Analyse the hazard
- b. Display links for relevant SOPs based on type and location of the crisis event,
- c. Determine and disseminate the public warnings,
- d. Display links for relevant SOPs based on type and location of the crisis event.

2.3.13.10.2 Ensure that the solution will deliver and display links for relevant SOPs based on type and location.

2.3.13.10.3 Ensure that the solution will deliver and display links for relevant SOPs based on type and location and capture the relevant task and activity data.

2.3.13.10.4 Ensure that the solution will deliver and display links for relevant SOPs based on type and location and capture the relevant task and activity data.

2.3.13.11 Information that shall be available through the IRMS shall include, at a minimum:

2.3.13.11.1 Initial Notification - Recognizing that an emergency has occurred and then notifying the proper agencies and authorities to respond to the emergency.

2.3.13.11.2 Incident Assessment - The procedures to be followed by those who arrive on the scene first and identify the risks posed by the incident.

2.3.13.11.3 Incident Command - The processes used by the OSS911 to implement an Incident Command and manage emergency operations.

2.3.13.11.4 Communications - The processes for providing reliable and effective communications among organizations participating in an emergency operation.

2.3.13.11.5Warning - The system that provides reliable, timely, and effective warnings to the public at the onset and throughout a disaster.

2.3.13.11.6Resource Management - The resource management mechanisms used to identify and acquire resources during the response to a disaster.

2.3.13.11.7Critical Infrastructure and Key Resource Restoration

2.3.13.12The CONTRACTOR shall supply integrated IRMS solutions to take care of escalated incident and emergencies management for the following operational functions as described in this document:

2.3.13.12.1Crisis management

2.3.13.12.2Joint Operations Dispatch

2.3.13.12.3Joint Operations Liaisons

2.3.14 إدارة الموارد من قطاعات متعددة (MULTI-AGENCY RESOURCE MANAGEMENT)

2.3.14.1The MOI requires that the IRMS incorporate integrated multi-agency resource management which can be used to identify and track resources and display their availability for allocation in response to incidents.

2.3.14.2The BIDDER shall confirm their understanding that a resource can be defined as personnel, a deployable asset or combination of assets. Assets may include vehicles and equipment, among other items.

2.3.14.3The IRMS will hold data about resources for use by authorized personnel in the OSS911 Center.

2.3.14.4The IRMS shall provide map-based tracking of assets.

2.3.14.5The IRMS shall integrate to the MOI-identified resources to provide GPS-based positional status of assets and personnel.

2.3.14.6The IRMS shall integrate with MOI systems / platform / apps in the field for bi-directional transfer of data, voice and video.

2.3.14.7The IRMS shall support the management, allocation, assignment, and monitoring of MOI resources.

2.3.14.8The system shall support the display of resource availability information for dispatchers and managers. This shall include current availability, short term planned availability, and mid-term availability to identify planned periods of unavailability of resources.

2.3.14.9Configurable historical resource record search formats shall be provided in the IRMS.

2.3.14.10The IRMS shall provide for 'resource designation', which will typically be alphanumeric. These resource designations will be determined by each user group.

2.3.14.11The IRMS will provide a translation tool for alphanumeric designations that provide a description of the resource, including resource type, name, etc.

- 2.3.14.12 The IRMS shall have the flexibility to embody resource designation changes throughout the system life, with the changes, including additions and deletions, available to be made by authorized users and system administrators.
- 2.3.14.13 The IRMS shall provide facilities for resource groups to be created from individual resources. Individual resources may appear in several groupings.
- 2.3.14.14 The IRMS shall allow for the editing and modification to resource groups by authorized users and system administrators.
- 2.3.14.15 When any list is displayed, whether automatically generated or called for, an individual resource shall only appear once and be located in the most appropriate group.
- 2.3.14.16 Proposed naming conventions and designations shall be presented to the MOI for review and approval as part of the detailed design review.
- 2.3.14.17 The IRMS shall provide for the resources to be input for an agreed period of time in advance of availability, typically one month.
- 2.3.14.18 The BIDDER shall propose solutions for populating the IRMS with advance resource data example shift working patterns and regular attachments.
- 2.3.14.19 The IRMS shall provide an ability to quickly change the working status of a single unit, including on or off duty, area of coverage, personnel assigned, and whether it can be recommended for dispatch.
- 2.3.14.20 The IRMS shall also provide the ability to build a shift roster at least 30 days prior to the shift.
- 2.3.14.21 Changes to the shift roster shall be able to be accomplished remotely through a web interface to the IRMS.
- 2.3.14.22 All changes shall be logged and record the user ID making the change.
- 2.3.14.23 The shift roster shall be maintained in the system for later access and analysis.
- 2.3.14.24 Historical roster information should be available for a minimum of one year and available for archive to other media.
- 2.3.14.25 Resources shall show their current deployment state and conditional availability. These attributes will include, but not be limited to:
- 2.3.14.25.1 Available
 - 2.3.14.25.2 Enroute to incident (including the incident number or identifier)
 - 2.3.14.25.3 Assigned
 - 2.3.14.25.4 Engaged but available
 - 2.3.14.25.5 At local station

2.3.14.25.6Break (Prayer, Refreshments, etc.)

2.3.14.25.7At Court

2.3.14.25.8At Hospital

2.3.14.25.9Off duty

2.3.14.25.10Out of Service

2.3.14.26These deployment states shall be represented by a numeric code, which can be manually input by an operator against the individual, or resource, code, or retrieved from information input into the electronic logging system.

2.3.14.27It shall be possible to update the status of resources remotely through the use of MOI radio communications networks Status Messages.

2.3.14.28The resource state shall be available with the resource record, for so long as it is valid, wherever that resource appears in the OSS911 system.

2.3.14.29The system shall be able to provide details of operational resources and the availability status and state of those resources, such as available, allocated to incident, day off, on leave.

2.3.14.30The systems shall be able to record the activity of a resource and attach additional information such as the incident that the resource is allocated to.

2.3.14.31Changes to resource information must be reflected in near real time for all users accessing the data, including the update of resource availability information.

2.3.14.32The IRMS shall provide for supervisors and managers to make resources available for deployment by manually assigning assets to the resource.

2.3.14.33The IRMS shall provide for supervisors and managers to input details of individuals and classify each by a series of parameters, such as availability times, skill sets, equipment used, and geographical constraints.

2.3.14.34The IRMS shall be capable of being partitioned in such a way that managers and users from one agency or department may be prevented from inputting or accessing information relating to other agencies or departments.

2.3.14.35The IRMS access shall be tracked and audited, with a history of date, times of access, changes to data and system configuration.

2.3.14.36The IRMS shall allow administrators, managers, and supervisors to assign user rights based on individual, job function, location, and agency. Assignable user rights shall include but not be limited to:

2.3.14.36.1Register information associated with an incident for further action, dissemination or analysis

2.3.14.36.2 Identify and record all elements of a resource separately and/or collectively, as required by various SOPs and response plans

2.3.14.36.3 Manage, group and present this information in a logical form as configured by the various SOPs and response plans

2.3.14.36.4 Record and timestamp actions, decisions and events

2.3.14.36.5 Provide record keeping in support of accountability and traceability

2.3.14.36.6 Record on-going changes to resource availability and allocation to improve efficiency and availability

2.3.14.36.7 Delineate a series of levels and authorities, grouped into functional areas and roles, access to each being determined by individual password

2.3.14.37 The display of available resources shall be restricted by agency type.

2.3.14.38 Additional, configurable, filters, such as equipment levels and skill sets shall be made available for use by particular agencies and departments.

2.3.14.39 Resource availability shall be displayed by resource type, individual resource, and resource status (available, location, unavailable, engaged, etc.). Only current available resources shall be displayed on an incident log.

2.3.14.40 It shall be possible to populate incident logs with additional information, for example details of individual officers who make up a resource.

2.3.14.41 It shall be possible, through the IRMS, to request lists of available resources.

2.3.15 موقع الموارد التلقائي (AUTOMATIC RESOURCE LOCATION)

2.3.15.1 The MOI requires that the IRMS incorporates integrated Automatic Resource Location (ARL) to be used for Automatic Vehicle Location (AVL), Automatic Personnel Location (APL) and any other resource that needs to be tracked or located within a defined area.

2.3.15.2 The CONTRACTOR shall deliver the ARL solution (AVL and APL) that is integrated with multiple MOI-identified AVL systems.

2.3.15.3 BIDDERS shall propose a solution whereby the IRMS servers collate and store IRMS data from the sources and make this information available to other applications and queries in an open format, for example XML, JSON, etc.

2.3.15.4 The IRMS shall allow suitably authorized users shall be able to view AVL data and manage unit update rates and displayed data at all operator and supervisor workstations within OSS911 Center

2.3.15.5 Designated supervisor and manager users within the OSS911 Center shall be authorized to configure user access rights that include:

2.3.15.5.1Read only, including which resource groups can be accessed

2.3.15.5.2Authority to management of update rates

2.3.15.5.3Authority to create/edit/delete resources

2.3.15.5.4GUI management.

2.3.15.6 The IRMS shall be dimensioned to support variable refresh rates depending on ARL network limitations and system capacities. Updates shall primarily be event driven, either caused by the resource moving a pre-set distance or activation of an alert. These would include:

2.3.15.6.1Automatically activation, for example based on speed, use of sirens or collision.

2.3.15.6.2User activation, through use of the radio emergency button or specific status messages.

2.3.15.6.3Incident Management System activation, to locate patrols available to attend a new incident, or

2.3.15.6.4Dispatcher or operator request.

2.3.15.7 The CONTRACTOR shall configure the timer refresh rates to come in periodically when no other event or action has triggered an update.

2.3.15.8 The BIDDER shall indicate in their proposal the refresh rates that they propose for their solution.

2.3.15.9 The CONTRACTOR shall engage with the MOI to define update triggers that provide useful data whilst keeping data rates within usable boundaries as permitted by the MOI.

2.3.15.10 The IRMS shall provide the functionality to partition the automatic location information delivered in accordance with the requirements of each agency and department.

2.3.15.11 Based on the user access privilege, the IRMS shall:

2.3.15.11.1Provide selectable display of individual units on a map

2.3.15.11.2Provide selectable display of units on a map by group

2.3.15.11.3Provide selectable display of units on a map by unit type (e.g., rescue vehicle, patrol vehicle, pursuit vehicle)

2.3.15.11.4Center map on selected unit and scroll display to follow that unit

2.3.15.11.5Polling of units for real time updates (all units or selected categories)

2.3.15.11.6Display history of a resource/vehicle status and position.

2.3.15.11.7Confirmation of a resource/vehicle's status (i.e. in motion, stopped, alarm, etc.).

2.3.15.11.8Continuous monitoring of a resource/vehicle's itinerary with the use of a continuous line of dots

2.3.15.11.9Create points of interest

2.3.15.11.10Ability to search on the map for points of interest

2.3.15.11.11 Report on a resource/vehicle's itinerary in relation to the points of interest

2.3.15.11.12 Location of the resource/vehicle closest to a selected point of interest

2.3.15.11.13 Location of the point of interest closest to a selected resource/vehicle

2.3.15.11.14 Creation of geographical boundaries for zones

2.3.15.11.15 Creation of geographical boundaries for routes

2.3.15.11.16 Geo-fencing - allocation of units to geographical boundaries (i.e. if a resource moves into or out of the zone or from a route to which it has been allocated or excluded, an event pop-up appears on the screen to alert the operator)

2.3.15.11.17 Select the information layer(s) to be displayed on the map

2.3.15.11.18 Zoom-in and Zoom-out

2.3.15.11.19 Pan within each direction of the map boundary

2.3.15.11.20 Transmission of status and messages to individual resources equipped with a visual display

2.3.15.11.21 Transmission of status and text messages to groups of resources equipped with a visual display

2.3.15.11.22 Transmission of location co-ordinates to individual users or groups of users

2.3.15.11.23 Select the volume level of audible event pop-up alarm

2.3.15.12 The IRMS shall include management capabilities available for automatic resource location that allow authorized users to:

2.3.15.12.1 Make changes to ARL system configuration

2.3.15.12.2 Configure access rights

2.3.15.12.3 Create fixed zones and routes (Geo-fencing)

2.3.15.12.4 Print historical reports using pre-determined report templates

2.3.15.12.5 Create new report templates

2.3.15.13 The IRMS shall have digitally record all ARL activity simultaneously to retain that data online for up to three months.

2.3.15.14 In addition to real-time location of assets that carry AVL capability and resources, the ARLS shall support history and playback modes as follows:

2.3.15.14.1 History mode shall provide access to previous updates through structured enquiries and the provision of pre-formatted history reports.

2.3.15.14.2 Playback mode shall allow the selection of one or more resources, or the selection of group resources, for playback on the map view. It shall be possible to select elapsed time views from a point in time or show the cumulative tracks between two specified times up to three months before from the current day.

2.3.15.15 The ARL data older than three months shall be available in archive form such that it can be restored for access within one hour.

2.3.15.16 The IRMS shall have the ARL history and playback available on all authorized OSS911 operator terminals and through web browsers for authorized users.

2.3.16 طرق الوصول إلى نظام إدارة الأحداث والموارد (IRMS) (IRMS ACCESS MODALITIES)

2.3.16.1 The IRMS solution shall support the following access mechanisms:

2.3.16.1.1 Via dedicated OSS911 computers at all operator and supervisor positions.

2.3.16.1.2 Via browsers, The BIDDER shall specify supported browsers and browser technology.

2.3.16.1.3 Via integration with applications running on suitably authorized handheld and vehicle-mounted mobile devices.

2.3.16.2 The BIDDER shall specify information and data security protocols for this access modality, and support the selection of these protocols.

2.3.16.3 The CONTRACTOR shall provide the OSS911 applications for the most recent, tested and stable versions of the contractor-proposed desktop operating systems for desktops.

2.3.16.4 Within their response, the BIDDER shall clearly indicate the sets of IRMS functionalities and features available in each mode of access, along with a detailed explanation as to why functionality or a feature is differently-abled and/or disabled in a particular mode of access.

2.3.16.5 The BIDDER shall clearly state the dimensioning principle for all the components of the OSS911 solutions.

2.3.16.6 The OSS911 solution shall provide a recorded audit trail of all key operator actions with access to it protected by authorized roles. This shall be accessible via a provided viewing package which will be capable of producing user configurable report files.

2.4 التكامل مع الأنظمة الميدانية / المنصات / التطبيقات التابعة لوزارة الداخلية (INTEGRATION WITH MOI FIELD SYSTEMS / PLATFORMS / APPS)

2.4.1 لمحة عامة على المتطلبات (REQUIREMENT OVERVIEW)

2.4.1.1 The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.4.1.1.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.4.1.1.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.4.1.1.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

2.4.1.2 The CONTRACTOR shall deliver a smart and AI-powered integration with MOI field systems/platforms/apps all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

2.4.1.3 The MOI requires that the CONTRACTOR integrate the OSS911 solution with MOI systems / platforms for access through:

2.4.1.3.1 Vehicle fitted or hand held ruggedized Mobile Data Computers (MDC) containing systems / platforms / applications that shall be supplied by others.

2.4.1.3.2 Smartphones containing apps that shall be supplied by others.

2.4.1.4 The CONTRACTOR shall deliver this required integration with any of the following available connectivity channels including but not limited to any of the MOI radio communications networks or the commercial wireless APN SIM card services from the telecom carriers.

2.4.2 المتطلبات الوظيفية (FUNCTIONAL REQUIREMENTS)

2.4.2.1 The CONTRACTOR shall provide this integration solution such that the field users shall have the ability to self-assign to an event that has not been reported to the OSS911 Center. When an event has been initiated by the mobile user, the IRMS shall automatically create an event log in the IRMS at the OSS911 Center.

2.4.2.2 The CONTRACTOR shall provide the following functions for authorized users through this integration:

2.4.2.2.1 Receipt and presentation of incident data relating to incidents the respective field resource has been dispatched to, or added as a recipient to.

2.4.2.2.2 Data entry fields for updates to incident logs.

2.4.2.2.3 Presentation of additional historical information regarding any previous related incidents.

2.4.2.2.4 Enabling the prompt input of updates directly from the scene of incidents.

2.4.2.2.5 Enabling access via field to central OSS911 information resources and to the NIC information resources through the OSS911-NIC interfaces.

2.4.2.2.6 Enabling commencement of crime, incident and traffic reports via field.

2.4.2.2.7 Support mapping and route-finding applications.

2.4.2.2.8 Provide for sending and receiving messages, data, reports and information from the IRMS.

2.4.2.2.9 Provide for the ability to send and receive location information.

2.4.2.2.10 Provide for the ability to send and receive resource information.

2.4.2.2.11 Provide status updates on one or more units, list of active incidents, etc.

2.4.2.2.12 Generate queries, view query responses, and manage the response information for person, vehicles, license plates and other authorized types of queries that are available through OSS911 and NIC and other MOI databases that the OSS911 center interfaces to.

2.4.2.2.13 Send, receive and display images and video feeds.

2.4.2.2.14 The ability to transmit and receive incident information to assigned units and authorized people without having to utilize voice (RF) communications.

2.4.2.3 All data messages between dispatch operators and the field shall be logged and tracked (time stamped) and have the receiving and transmitting parties identified and recorded. These text message log files shall be accessible to authorized MOI personnel.

2.4.2.4 A solution shall be provided that shall allow the creation and updates of an incident report in to the IRMS from the field.

2.5 INTEGRATION WITH VIDEO MANAGEMENT SYSTEM (VMS) AND VIDEO ANALYTICS (VA)

2.5.1.1 The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.5.1.1.1 Unless otherwise explicitly stated, the phrase "OSS911 center" refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.5.1.1.2 Unless otherwise explicitly stated, the phrases "OSS911 solution" or "OSS911 systems" or "OSS911 subsystems" refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.5.1.1.3 Unless otherwise explicitly stated, the word "data" in this document includes all types of data including voice, text, video and SMS.

2.5.1.2 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

- 2.5.1.3The CONTRACTOR shall deliver OSS911 solutions that are integrated with the MOI provided VMS and VA to provide the following capabilities:
- 2.5.1.3.1Display to the user which cameras were viewing a location at a given time
 - 2.5.1.3.2Lock a camera before a user can control the pan, tilt, or zoom of a camera in order to prevent another user from controlling it
 - 2.5.1.3.3Keep historical record of camera locks
 - 2.5.1.3.4Identify to other users which user has locked a camera
 - 2.5.1.3.5Request a camera lock
 - 2.5.1.3.6Notify to users of another user requesting a camera lock
 - 2.5.1.3.7Unlock a camera, or override the lock on a camera
 - 2.5.1.3.8A capability such that the mapping window shows the installed cameras on the map, further to the availability of the camera location data.
 - 2.5.1.3.9A capability such that an operator is able to click on one or more cameras in the vicinity of an incident, whether the camera is fixed or is mobile, and able to obtain see the field of view of the cameras on the map shown in the mapping window.
 - 2.5.1.3.10A capability such that an operator is able to click on one or more cameras in the vicinity of an incident, whether the camera is fixed or is mobile, and able to obtain live feed at various resolutions from the cameras, based on the resolutions supported by the cameras.
 - 2.5.1.3.11A capability such that an operator is able to click on one or more cameras in the vicinity of an incident, whether the camera is fixed or is mobile, and able to control pan, tilt and zoom of the cameras if the cameras allow pan, tilt and zoom.
 - 2.5.1.3.12A capability such that an operator is able to click on one or more cameras in the vicinity of an incident, whether the camera is fixed or is mobile, and able to playback and capture video feeds stored in cameras should the cameras have video recorders installed with them.
 - 2.5.1.3.13A capability such that an operator is able to click on one or more cameras in the vicinity of an incident, whether the camera is fixed or is mobile, and able to obtain live feed from the cameras equipped with number plate recognition feature that then populates the OSS911 systems with the vehicle number plate information in the appropriate field.
 - 2.5.1.3.14The capability to play live video from any camera by clicking on the camera icon on the displayed map.
 - 2.5.1.3.15A capability such that the mapping window shows the installed cameras on the map.

2.5.1.3.16A capability to integrate such that there is a trust for users between the systems so that the authorized end-users are able to access both the systems and network connections such that all the traffic that is required for this integration is permitted.

2.5.1.3.17Once sufficient location information has been entered into the incident report (either manually or automatically), the location of the incident shall be displayed with all cameras located within a configurable incident radius on a map in all directions.

2.5.1.3.18The capability to indicate to the user those cameras that are able to viewed at that location.

2.5.1.3.19The capability to display the active “field-of-view” (FOV) of each camera, that is within a configurable distance of the incident location.

2.5.1.3.20The capability to display the available “field-of-view” (FOV) of each camera, that is within a configurable distance of the incident location. The available field-of-view is that area which the camera could view, even if it is not viewing that area now.

2.5.1.3.21Selection of a location and a time, and indication to an authorized user as to which cameras were viewing that location at the selected time.

2.5.1.3.22Whenever an incident occurs, to be all cameras covering the incident provide a visual indication to authorized user (Example: Flashing Camera Icon).

2.5.1.3.23Capability to control through user rights the roles that have the ability to control the Video Wall.

2.5.1.3.24The ability to share this video through web browser services that shall support decision makers across disciplines with real-time situation views.

2.5.1.3.25If a VMS and/or VA operational alarm turns out to be a false alarm, the OSS911 solutions to VMS integration shall include a solution such that the false alarm details are captured and recorded in a database, along with any voice conversations that have been done by the operator as part of the investigation.

2.5.1.3.26If a VMS or VA operational alarm turns out to be a positive alarm (implying, it relates to an event that requires a dispatch), the OSS911 solutions to VMS integration shall include a solution such that:

- a) The location of the trigger for the alarm (i.e., the location of the incident) and the location of the camera reporting the alarm is captured automatically using the mapping function.
- b) The location is populated in the corresponding field within the OSS911 systems.
- c) The incident is recorded by the operator using OSS911 system interfaces.
- d) Based on the defined event types, priorities and location, the incident gets transferred to the appropriate dispatch group.

- e) The received video is tagged (linked) to the incident (event) created and a pointer to the video is stored as part of the incident (event) record.
- f) From then on, the regular dispatching process takes over the workflow (as it would be any other call or demand that requires a dispatch, and that which has been detailed in this document).

2.5.1.3.27VMS or VA operational alarms shall be handled by the OSS911 solutions as they would handle voice calls. All process steps, rights and responsibilities, and archiving requirements for a positive alarm at the video operator are similar to the steps for emergency calls.

2.5.1.3.28The capability to request and play video recorded such that an authorized user is able to select a camera from within the OSS911 map interface, request a time frame for the video, and then view that video.

2.6 IP TELEPHONY (VoIP) **نظام هواتف الأي بي**

2.6.1 **REQUIREMENT OVERVIEW** **لمحة عامة على المتطلبات**

2.6.1.1The MOI requires that the OSS911 Center is equipped by the CONTRACTOR with Voice over IP (VoIP) telephony solution to provide internal and external telephony services within the OSS911 Center buildings.

2.6.1.2The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.6.1.2.1Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.6.1.2.2Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.6.1.2.3Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

2.6.1.3 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

2.6.1.4The CONTRACTOR shall deliver a smart and AI-powered IP Telephony solution delivering the IP telephony and recording capabilities for OSS911 use.

2.6.1.5The OSS911 telephony and recording solution shall be deployed by the CONTRACTOR and hosted within the OSS911 center in the scope.

2.6.1.6The CONTRACTOR shall deliver telephony and recording solutions that include capability to integrate with, consume services from, and exchange data with other communication systems utilizing SIP protocol and SIP protocol extensions.

- 2.6.1.7The CONTRACTOR shall deliver telephony and recording solutions that provide voice and data exchange in real-time and for recorded voice and data, with AI-enabled tools supplied by the CONTRACTOR and with other third-party AI-enabled tools.
- 2.6.1.8The BIDDER shall analyze the operational requirements of the OSS911 Center to determine the actual number and type of telephone handsets and trunk lines that are required and shall include this analysis in their proposal.
- 2.6.1.9When the unified single emergency number call taking is implemented, the CONTRACTOR shall deploy IP telephony system such that it shall present all incoming calls on that single emergency number to the call takers.
- 2.6.1.10Within their proposal, the BIDDER shall detail the actual amount of scalability that is available for immediate use within the OSS911 contract per OSS911 center, without adding additional equipment and software.
- 2.6.1.11The CONTRACTOR delivered telephony system shall include redundant, geographically diverse trunk connections to the PSTN, to provide a resilient architecture.
- 2.6.1.12Emergency calls will be delivered into the center using:
- 2.6.1.12.1SIP connections over fiber circuits
 - 2.6.1.12.2SIP connections over microwave
 - 2.6.1.12.3Power independent analog connections over copper
- 2.6.1.13The BIDDER shall perform their own analysis and justify their calculations based on likely operational requirements. The BIDDER shall present this analysis in their response.
- 2.6.1.14It is the responsibility of the CONTRACTOR to provide for any gateways or other hardware that is required for interfacing to the telecommunication networks and for integrating with them for the SIP and analog connections and to connect the MDF to the analog cables.
- 2.6.1.15It is the responsibility of the CONTRACTOR to provide for any interfacing equipment (software and/or hardware) to connect to mobile network operators and/or third-party service providers for receiving emergency calls including eCall and to receive, record, process, route and report any and all calls and/or data transferred from KSA telecommunication network providers for single-SIM and/or multi-SIM using MSISDN/IMSI/IMEI or any other numbers.
- 2.6.1.16The CONTRACTOR shall ensure that the hardware IP telephone is fully integrated with the softphone application provided by the CONTRACTOR.
- 2.6.1.17The CONTRACTOR shall provide any and all devices and software necessary for ensuring that the voice communications through softphone application switch to hardware IP telephone automatically without manual operator intervention while still maintaining voice playback through the operator headsets.

- 2.6.1.18 The CONTRACTOR shall provide any and all devices and software necessary for ensuring that the voice communications through hardware IP telephone switch to softphone application automatically without manual operator intervention while still maintaining voice playback through the operator headsets.
- 2.6.1.19 The CONTRACTOR shall provide complete control of the telephony and recording solution, and all components therein, from a centralized web-browser based management and monitoring system.
- 2.6.1.20 All the CONTRACTOR OSS911 solutions shall be compatible with each other and shall integrate among them seamlessly without compromising any feature or capability of OSS911 solutions.
- 2.6.1.21 The BIDDER shall include a preliminary design with their response. The preliminary design schematic shall be provided as a Microsoft Visio drawing and shall depict the overall system architecture and connectivity requirements and on award of contract the CONTRACTOR shall carry out a detailed design for the telephony solution. The detailed design shall include an overall numbering plan using common numbering pattern.
- 2.6.1.22 The detailed design shall include a configuration specification for the telephony solution including, but not limited to, aspects such as:
- 2.6.1.22.1 Detailed system interconnectivity diagram
 - 2.6.1.22.2 Failover and failback workflow and SOPs
 - 2.6.1.22.3 Equipment wiring diagrams
 - 2.6.1.22.4 Equipment rack elevations
 - 2.6.1.22.5 Power calculations
 - 2.6.1.22.6 Identification of equipment along with product datasheets
 - 2.6.1.22.7 Network configuration information, including identification of security settings, Quality of Service (QoS) settings, bandwidth requirements, VPN clients, IP schema, VLAN schema, etc.
 - 2.6.1.22.8 Groups and hunt groups
 - 2.6.1.22.9 Individual extension configurations
 - 2.6.1.22.10 Numbering Plan
 - 2.6.1.22.11 Night service
 - 2.6.1.22.12 Trunk and line configurations
 - 2.6.1.22.13 Feature access codes
 - 2.6.1.22.14 Short code dialing lists
- 2.6.1.23 The CONTRACTOR solution shall be configurable to allow / disallow direct dialing outside numbers based on role, location, telephone numbers, etc.

- 2.6.1.24The network and system design shall support automatic transfer of relevant IP voice and data communications between the paired OSS911 Centers including assisted transfer, and/or blind transfer – with the actual transfer mechanism agreed with the MOI as part of project design.
- 2.6.1.25The CONTRACTOR solution shall include delivery of analog telephony solution as emergency backup for all operators and supervisors in the event of a failure of the SIP circuits(s) including the supply of related analog gateways and telephone sets.
- 2.6.1.26The CONTRACTOR shall integrate the telephony and recording solution with the IRMS solution and the Business Intelligence and Reporting Solution through the CONTRACTOR supplied middleware/enterprise service bus.
- 2.6.1.27The telephony solution integration with IRMS shall provide a solution for the handling of all emergency calls including eCall, irrespective of the eCall being received directly from the Mobile Network Operators OR through third-party service providers (TPS).
- 2.6.1.28The telephony solution shall provide communication solution within the OSS911 Center and with external entities.
- 2.6.1.29The CONTRACTOR shall deliver a telephony and recording solution that:
- 2.6.1.29.1Is scalable and modular.
 - 2.6.1.29.2Shall support all open system industry standards including H.323, SIP, 802.1p, and 802.1s.
 - 2.6.1.29.3Includes a voice messaging system that shall be a scalable solution that supports industry standards including LDAP, and IMAP.
 - 2.6.1.29.4Includes a web-based management and monitoring solution.
 - 2.6.1.29.5Includes abbreviated dialing pattern
 - 2.6.1.29.6Includes the ability to be monitored via SNMP or other protocol compatible with the CONTRACTOR solutions for OSS911 systems management.
 - 2.6.1.29.7Includes the ability to support the automatic and dynamic identification of phone locations.
 - 2.6.1.29.8Includes device assignments via VLAN by MAC Address.
 - 2.6.1.29.9Includes the ability to relocate phones without requiring configuration changes.
 - 2.6.1.29.10Includes defense mechanisms to protect from security attacks like telephony denial of service.
 - 2.6.1.29.11Includes the ability to detect unauthorized IP stations and prevent systems access.

2.6.1.29.12 Shall support Dynamic Host Control Protocol (DHCP).

2.6.1.29.13 Includes the ability to pass Caller line Identification (CLI) into and out of the telephony and recording solution.

2.6.1.29.14 Includes fast call setup.

2.6.1.30 It is the responsibility of the BIDDER to liaise with the telecom providers to understand the required details about SIP trunk services and all other connection types.

2.6.1.31 The BIDDER shall clearly describe in the proposal response the security measures that are proposed to be included in their overall security solution that shall help the telephony solution to be secure against attacks or intrusions that would impact service availability or compromise integrity and for defense against telephony denial of service attacks.

2.6.1.32 The equipment shall conform to the following:

2.6.1.32.1 Minimum 6 Centrum Call Seconds (CCS) per station port/36 CCS per trunk port

2.6.1.32.2 Minimum 15,000 BHCA (guaranteed under full load)

2.6.1.33 The BIDDER shall provide the following information regarding the telephony solution:

2.6.1.33.1 Minimum trunk groups

2.6.1.33.2 Minimum classes of service

2.6.1.34 The CONTRACTOR shall ensure that all their technical infrastructure fits within the space designated as per the datacenter designs.

2.6.2 خصائص النظام (SYSTEM FEATURES)

2.6.2.1 LINE GROUPS

2.6.2.1.1 The telephone lines to be connected to the system shall be divided into a number of distinct groups. The grouping given here is indicative and the system shall be capable of supporting further or different groups.

2.6.2.1.2 A line group shall be made up of a number of characteristics of the line. These are:

- a. The type of the line
- b. The service that is delivered over the line
- c. The direction or directions in which the lines can be used

2.6.2.1.3 The Telephony system shall support the ability to further divide a group into sub-groups.

2.6.2.1.4 Indicative line groups shall be as follows but not limited to:

- a. Public generated emergency calls of line type trunk group and/or direct exchange lines (DEL) with

direction inbound only for usage for delivery of emergency 9XX calls from MOI defined numbers.

- b. Hotlines of line type 'private wire' with direction both-ways (inbound and outbound) for usage as emergency hotlines from different MOI and non-MOI agencies. Each private wire shall have the capability to be identified on a unique sub-group.
- c. Direct Exchange Lines (DELs) for urgent calls from other MOI and non-MOI agencies with direction both-ways (inbound and outbound).
- d. Emergency calls transferred internally for line type IP telephony extensions used to transfer emergency calls internally.
- e. General use IP telephony extensions with direction both-ways (inbound and outbound) with the capability to have a number of sub-groups to allow individual positions or functional areas to be addressed.
- f. Outbound overflow for line type IP telephony extensions with direction outbound to ensure some outbound lines are always available and would be automatically selected by the system in the event that all lines are busy.
- g. Direct Exchange Lines (DELs) for general use with direction both-ways (inbound and outbound).

2.6.2.1.5 The system shall be capable of supporting a minimum of 30 line groups.

2.6.2.1.6 Each line group to which the operator has access shall be presented separately to the operator.

2.6.2.1.7 Emergency Calls arriving into the system shall be automatically assigned a unique Call ID that shall remain with the call until such time as it is cleared.

2.6.2.1.8 Emergency Calls arriving into each group or sub-group shall be queued pending answer by an operator. The call that has been waiting the longest shall be at the top of the queue.

2.6.2.1.9 There shall be a separate queue for each group.

2.6.2.2 تقديم البلاغات (CALL PRESENTATION)

2.6.2.2.1 Emergency Calls in queue shall be presented to operators using a priority allocation system.

2.6.2.2.2 Emergency Calls shall always be presented to an available operator with the highest priority for a call type.

2.6.2.2.3 Where more than one operator is available with the same highest priority, Emergency calls shall be assigned to the operator who has been waiting the longest

- 2.6.2.2.4 An operator shall be available to receive Emergency calls by going into a “Ready” state on the system.
- 2.6.2.2.5 The system shall be able to be configured to allow Emergency calls to be automatically presented to the operators i.e. the operator has no opportunity to accept or decline the Emergency call.
- 2.6.2.2.6 The system shall be able to be configured to allow the operator to manually accept or decline Emergency calls.
- 2.6.2.2.7 If an Emergency call is not accepted by an operator in a configurable time period that call shall be presented to the next available operator and the first operator shall go into a “Not Ready” state. A notification of failure to respond shall be presented to the supervisor.
- 2.6.2.2.8 The system shall be configurable so as to allow a change in the audio/visual presentation of an Emergency call to an operator to change when calls remain unanswered after predetermined time periods.
- 2.6.2.2.9 When the Emergency call is presented to an operator, the CLI of the call shall be displayed on the operator’s terminal in the telephony software console, and in the application that caters to requirements for event creation and event dispatch.
- 2.6.2.2.10 The system shall be able to present multiple queues in the form of a common queue to an operator. Emergency Calls shall be queued based on the operator’s priority for each group and then the length of time that calls within each group have been queuing.
- 2.6.2.2.11 The system shall be able to permanently display the number of Emergency calls queuing for each call group that are assigned to a call group. This facility shall be available on every console.
- 2.6.2.2.12 When an Emergency call is presented to an operator there shall be both a visual and audible alarm.
- 2.6.2.2.13 The operator shall be able to mute the audible alarm for certain call groups for a predetermined time period.
- 2.6.2.2.14 The system shall support the ability of an authorized user to define, through system configuration, which non-emergency call groups shall be queued in a “silent queue”, where, if an operator has not answered the call after a configured time, the system shall generate an audible alarm.
- 2.6.2.2.15 There shall be at least two different types of audible alert that can be associated with a call group. For example, public generated emergency calls may have a different audible alert to other call groups.
- 2.6.2.2.16 The audible alert associated with a mutable call group shall be automatically suppressed when the operator is already handling a call.

2.6.2.2.17 If a call has arrived or arrives for a non-mutable call group then the appropriate audible alarm shall sound.

2.6.2.2.18 The audible alert that sounds shall be the one that is associated with the call group of the highest priority waiting call.

2.6.2.2.19 All incoming Emergency calls shall be presented to all operators who have that group or sub-group associated with their profile.

2.6.2.2.20 The action taken in the event that an Emergency call is missed/dropped/lost/clears before it is answered by an operator shall be configurable. Possible actions include:

- a. Clear the call from the system, removing it from the call queue
- b. Leave the call in the queue and present to the operator as a dummy call providing a feature for the operator to attempt to call back the caller in a queue that presents these calls for configurable number of call back attempts along with related statistics, reports, monitoring etc.

2.6.2.2.21 The system shall drive wall displays with configurable information concerning call statistics and parameters such as:

- a. Number of calls in each queue
- b. Longest waiting time

2.6.2.2.22 Dashboards shall be delivered by the CONTRACTOR to be able to be displayed on the wall displays and will be configured for parameters to be defined with the MOI per OSS911 center.

2.6.2.2.23 Priority messages for the display shall be configurable. For example, if public generated emergency calls are queuing it shall be possible to display a message which takes priority over other scheduled messages.

2.6.2.2.24 The system shall be able to provide a graphical display for display on one or more flat panel displays.

2.6.2.2.25 The system shall support multiple displays for showing different information in different functional areas across OSS911 center.

2.6.2.3 الرد على البلاغات (ANSWERING CALLS)

2.6.2.3.1 The operator shall be able to answer the highest priority Emergency call by means of a single key.

2.6.2.3.2 The operator shall be able to select an Emergency call from a queue to answer, even if that call is not of the highest priority.

2.6.2.3.3 When an Emergency call is answered by an operator it shall be removed from the queue and any alerting associated with that call shall be stopped on all relevant operator consoles.

2.6.2.3.4 When an Emergency call has been answered, the CLI (if available) of the call shall be clearly displayed on the console and shall remain present until such time as the operator has cleared the call.

2.6.2.3.5 An operator shall be able to answer an Emergency call, even if they are already handling a call, by a single key press, providing they do not already have more than a configurable number of calls on hold.

2.6.2.3.6 If an operator answers an Emergency call whilst they are already handling another call then the first call (the one they were handling) shall be automatically put on hold and given appropriate on hold treatment.

2.6.2.3.7 The system shall support the ability to configure an alarm that can be raised on the supervisors' console if an Emergency call has remained unanswered for more than a pre-configured period of time. This time period shall be configurable for each call group.

2.6.2.3.8 For some call groups (e.g., public generated emergency calls) it shall be possible to commence an incident record (in the incident management system) shall be automatically opened for each call that arrives.

2.6.2.3.9 In order to provide efficiency in the generation of the incident record and reduce the opportunity for error, the system shall support the electronic exchange of data associated with an Emergency call with an external application.

2.6.2.3.10 An interface shall be provided that allows Emergency call details of each call answered at operators' console to be passed to an external application.

2.6.2.4 **إغلاق المكالمات أثناء اتصال طوارئ CALL CLEARING DURING EMERGENCY CALLS HANDLING**

2.6.2.4.1 Call clearing shall be invoked by either the operator or the called/calling party. Within their proposal, the BIDDER shall identify the support for call clearing.

2.6.2.4.2 In the event that the called/calling party invokes call clearing, all information associated with that call shall remain on the screen until the operator performs call clearing of their half of the connection.

2.6.2.4.3 It shall be configurable by call group whether operator-initiated call clearing is allowed.

2.6.2.4.4 For calls in a call group for which operator-initiated call clearing is permitted, if the operator invokes call clearing, the call shall be released and cleared.

2.6.2.4.5 For calls in a call group for which operator-initiated call clearing is NOT permitted, if the operator invokes call clearing then that call shall automatically be re-queued and its priority in the call queue will be based on the length of time the call has been in the system (This would normally cause it to go to the head of the queue).

2.6.2.4.6 The next operator who receives the call shall be given an indication that the call was re-queued and for authorized users

like the supervisors, the identity of the operator who released the call.

2.6.2.4.7 In the event that the called/calling party clears whilst the call is being re-queued then its treatment shall depend on how that call group is configured.

2.6.2.4.8 It is possible that due to some type of fault condition a call that cannot be cleared by an operator from the system. In this case it shall be possible for the supervisor to release this call and clear the line. This action shall be recorded in a fault log for the system as a fault condition.

2.6.2.5 المكالمات الصادرة (OUTGOING CALLS)

2.6.2.5.1 The operator shall be able to make outgoing calls on any group that is configured for both-way or outbound calls.

2.6.2.5.2 An on-screen keypad shall be available for the operator to dial from.

2.6.2.5.3 One touch speed dial key shall be configurable on the console. It shall be possible to configure these keys through system configuration, where the system administrator will identify both the group to be used and the number to be dialed.

2.6.2.5.4 The number being dialed by the operator using the keypad shall be displayed.

2.6.2.5.5 The operator shall be able to correct the number being dialed via the keypad during the dialing process.

2.6.2.5.6 The operator shall be able to cancel outbound dialing at any time.

2.6.2.5.7 A last number redial facility shall exist allowing the last number dialed from that console to be displayed and redialed.

2.6.2.5.8 The keypad shall be available to the operator while the call is in progress.

2.6.2.5.9 Before the call is answered the keypad shall be used to send additional dialed digits.

2.6.2.5.10 Once the call has been answered, pressing any number (0-9, *, #) key shall result in the associated DTMF tone being generated on the line.

2.6.2.5.11 Outgoing calls on hotlines shall require no more than 2 key presses.

2.6.2.6 وضع البلاغات قيد الانتظار (HOLDING CALLS)

2.6.2.6.1 An operator shall be able to place a minimum of 3 calls on hold and to later retrieve them.

2.6.2.6.2 Once a call has been placed on hold the operator shall be able to answer another call or place an outgoing call.

2.6.2.6.3Held calls shall be displayed in a separate area of the console display.

2.6.2.6.4Where an operator answers a call whilst already handling a call, the call they are handling shall automatically be placed on hold until the limit for the maximum number of calls on hold is reached.

2.6.2.6.5Audible Reminder of Held Calls: Provides an audible tone as a reminder that calls are on hold on a user's station.

2.6.2.7 تحويل المكالمات (CALL TRANSFER)

2.6.2.7.1The operator shall be able to transfer a call as follows:

- a. To a specific operator
- b. To a queue
- c.To an external PABX extension number

2.6.2.7.2Where the operator transfers the call to a specific operator, that call shall be presented to the operator in a separate queue for transferred calls.

2.6.2.7.3The operator shall be able to perform a blind transfer where they are able to release the call before the called operator answers.

2.6.2.7.4The operator shall also be able to perform an assisted transfer where the called operator answers the call allowing the calling operator to speak to the called operator with the caller excluded from the conversation.

2.6.2.7.5During an assisted transfer, after consulting with the called operator the calling operator shall be able to:

- a.Connect the caller to the called operator
- b.Release the called operator and reconnect to the caller

2.6.2.7.6The operator shall be able to transfer a call via a direct exchange line or a PABX extension. The operator shall be able to monitor this call until either party clears and if necessary, re-connect to the call.

2.6.2.7.7In the event that any blind transferred call is unanswered within a configurable time period, that call shall be re-queued into the queue for its call group.

2.6.2.7.8It shall be possible to customize by authorized OSS911 personnel, the number of unanswered rings before a call gets transferred.

2.6.2.8 المراقبة (MONITORING)

2.6.2.8.1The operator or supervisor shall be able to monitor the call of another operator. This facility shall be non-intrusive and the microphone of the monitoring operator(s) shall be muted.

2.6.2.8.2 Monitored audio shall not be played through the operator's loudspeaker but only routed to a headset (if connected).

2.6.2.8.3 An indication shall be given to the operator being monitored that monitoring is taking place.

2.6.2.9 **كتم صوت الميكروفون (MICROPHONE MUTING)**

2.6.2.9.1 The operator shall be able to mute their microphone at any time so that the other party/parties cannot hear them.

2.6.2.10 **التعامل مع المكالمات قيد الانتظار (ON HOLD TREATMENT)**

2.6.2.10.1 The system shall provide a treatment for calls on hold.

2.6.2.10.2 The treatment shall be configurable for each call group.

2.6.2.10.3 On hold treatment shall consist of one of the following:

- a. Silence
- b. Comfort tone
- c. Music on hold
- d. Announcement

2.6.2.11 **واجهة صوت المشغل (OPERATOR AUDIO INTERFACE)**

2.6.2.11.1 Each operator shall be provided with a capability to allow the connection of two headsets so as to allow an operator and supervisor or operator and facilitator to work together.

2.6.2.11.2 Each operator shall be provided with a sound arbitration box that shall be connected to the PC, the IP phone and to the headset to arbitrate the sound source to the headset irrespective of where the sound source.

2.6.2.11.3 The arbitration box and the headset shall be mounted in such a way that they are easily accessible and conveniently located for use by the operator.

2.6.2.11.4 The arbitration box and the headset shall be mounted in such a way that they are not susceptible to damage from chair movements or other accidents that could occur during normal use.

2.6.2.11.5 The arbitration box and the headset shall be robust and designed for high frequency insertion and removal.

2.6.2.11.6 The arbitration box shall have digital controls that allow the configuration and adjustment of the audio controls for microphone and speakers of the headset per each arbitration box and headset.

2.6.2.11.7 The headsets shall consist of a microphone and earpiece(s) / speakers.

2.6.2.11.8 The headsets model that the BIDDER proposes shall be based on global PSSA command and control room best practices

and shall support 911 Public Safety Communications and Dispatch Applications. Within their proposal, the BIDDER shall identify and propose headsets which are most appropriate for use by the different types of operators.

2.6.2.11.9The cable between the headset and arbitration box shall allow free movement without having to remove the headset or unplug it.

2.6.2.11.10Unplugging the headset from the arbitration box socket shall have no impact on call handling.

2.6.2.11.11Headsets shall have noise cancellation built-in.

2.6.2.11.12Audio levels at the headset shall be limited and incorporate acoustic shock protection so that they cannot exceed safe working levels.

2.6.2.11.13The telephony solution shall be able to receive and relay audio from third party systems enabling an operator to operate audio feeds associated with them.

2.6.2.12 الاتصال الآلي والدليل (AUTOMATIC DIALING AND DIRECTORIES)

2.6.2.12.1The system shall provide a system wide directory of telephone numbers. These shall be capable of being administered by a supervisor or system administrator. This system directory shall include information on all systems.

2.6.2.12.2The system shall provide a user defined directory for telephony numbers. The user directory shall be associated with the user ID or profile. It shall be available to that operator irrespective of which position they use to log on.

2.6.2.12.3A minimum of two levels shall be available in the directory to allow numbers to be grouped into sub-directories.

2.6.2.12.4The operator shall be able to search the directory by means of an on-screen search page.

2.6.2.12.5The operator shall be able to select an entry in the phone book and then initiate dialing with one further key press.

2.6.2.12.6An authorized supervisor shall be able to control and pre-determine which groups and telephone lines are available to which roles and user accounts.

2.6.2.13 الاتصال الداخلي (INTERNAL INTERCOM)

2.6.2.13.1An internal intercom facility shall be provided between the operator positions.

2.6.2.13.2If an operator attempts to use the intercom to contact another operator, and that operator is busy, they shall receive an engage notification.

2.6.2.14 ساعة النظام (SYSTEM CLOCK)

2.6.2.14.1The system shall provide a clock showing time, date and year, displayed at all operator positions.

2.6.2.14.2The clock shall be synchronized to the NTP timing sources identified by MOI.

2.6.2.15التسجيل الصوتي المستمر (CONTINUOUS VOICE RECORDING)

2.6.2.15.1The recording solution shall be able to continuously record all voice conversations securely with AI capabilities.

2.6.2.15.2Both sides of all the voice conversations shall be recorded.

2.6.2.15.3Details of the call-in progress, e.g., operator positions, operator id, talk-group, calling line identity of the telephone call, shall be able to be recorded against each call.

2.6.2.15.4The recording solution shall support the retrieval of recordings based on any or a combination of the following parameters:

- a. Incident ID
- b. Operator position (console ID)
- c. Operator ID
- d. Radio Talkgroup
- e. Radio ID
- f. Calling line identity
- g. Date and time

2.6.2.15.5The CONTRACTOR shall provide secure server storage to allow voice recordings to be kept on-line for a period of not less than twelve (12) months, followed by an off-line storage for five years.

2.6.2.15.6It shall be possible for a suitably authorized operator to randomly select recorded calls for quality assurance purposes.

2.6.2.15.7The solution shall have the capability to archive onto removable media any recording older than a certain date.

2.6.2.15.8The process of archiving recordings shall not impact on the current recording.

2.6.2.15.9The CONTRACTOR shall integrate the recording solution with the business intelligence and reporting solution through the enterprise service bus.

2.6.2.15.10The CONTRACTOR recording solution shall have the APIs available to be integrated with other solutions including but not limited to other data analysis solution, AI-solutions, etc.

2.6.2.16إمكانية التشغيل الفوري (INSTANT PLAYBACK FACILITY)

2.6.2.16.1The system shall provide an instant playback facility for each console the facility shall be fully integrated into the console and use digital technology.

2.6.2.16.2 Recording shall begin on the initiation or reception of a telephone call

2.6.2.16.3 Recording shall cease when the telephone conversations have ceased.

2.6.2.16.4 Each recorder shall be able to store at least the most recent 30 minutes' worth of voice activity at the operator's position.

2.6.2.16.5 The system shall have the ability to export the voice recording to a file that is playable on PC without the installation of additional CODECS.

2.6.2.16.6 The recording system shall be capable of continued recording whilst simultaneously playing back a recorded conversation.

2.6.2.17 وظائف المشرف (SUPERVISOR FUNCTIONS)

2.6.2.17.1 An operator shall be able to log in as a supervisor for the purpose of monitoring a functional area.

2.6.2.17.2 A supervisor shall be able to monitor more than one functional area at the same time.

2.6.2.17.3 A supervisor shall be able to configure all the users that are assigned to the functional area that they have authority for.

2.6.2.17.4 Supervisors shall be able to access all the features available to the operators.

2.6.2.17.5 A supervisor shall be able to monitor the status and configuration of all operator positions (consoles) within their functional area(s).

2.6.2.17.6 A supervisor shall be able to monitor the console display for any console position in their functional area.

2.6.2.17.7 The supervisor shall be able to communicate with an operator in their functional area in such a way that the communication is inaudible to the telephone caller.

2.6.2.17.8 A supervisor shall be able to override any operator's position. This shall include seizing control of any telephone line to which the operator is connected, and exclude the operator.

2.6.2.17.9 Where a supervisor takes control of a telephone call from an operator, the operator shall be given a clear visible notification.

2.6.2.18 إدارة النظام (SYSTEM MANAGEMENT)

2.6.2.18.1 The telephony and recording solution shall be provided with a web-based management and monitoring system.

2.6.2.18.2 The telephony and recording solution shall provide real time alarms notification for the different faults within the solution.

2.6.2.18.3The telephony and recording solution management and monitoring system shall be integrated with the CONTRACTOR solutions for OSS911 management and monitoring.

2.6.2.18.4The telephony and recording solution shall be configured to provide an audible and visual alert for alarms to authorized personnel.

2.6.2.18.5The telephony and recording solution shall support configurable variable messaging displays that can display information such as number of calls queuing, calls answered, longest waiting time and a free text message facility.

2.6.2.18.6The telephony and recording solution shall be supplied with a graphical user interface-based tool for updating the phonebook. This tool and the phonebook shall be integrated with the OSS911 solutions and LDAP so that there is one phone directory across all solutions.

2.6.2.18.7The telephony and recording solution shall support multiple variable message displays that can be configured independently according to functional or physical area.

2.6.2.19 تسجيل المكالمات (CALL AND TRAFFIC LOGGING)

2.6.2.19.1The CONTRACTOR shall provide a call and traffic logging solution that shall record the details of all calls into, out of and within the telephony solution.

2.6.2.19.2Details of all telephony calls shall be able to be recorded in the solution including but not limited to:

- a. CLI or extension number of the Calling Party
- b. Dialed number
- c. Start time of the call
- d. End time of the call
- e. Reason for call clearing

2.6.2.19.3The CONTRACTOR shall provide a telephony and recording solution that includes a reporting package that allows standard and custom reports to be designed and run, and shall implement the required APIs to integrate the solution with the OSS911 business intelligence and reporting solution delivered by the CONTRACTOR.

2.6.2.19.4The CONTRACTOR shall configure the telephony and recording solution to schedule reports to be produced on a periodic or ad-hoc basis.

2.6.2.19.5The CONTRACTOR shall configure the telephony and recording solution to log and monitor all the groups that a functional area is responsible for monitoring.

2.6.2.19.6The CONTRACTOR shall deliver an alarms page on the supervisor display that displays all active alarms on the system relevant to the supervisor's functional area.

2.6.2.19.7The recording solution shall be configured for retaining call records in accordance with the retention guidelines in the section on 'Technical Infrastructure'.

2.6.2.20The CONTRACTOR shall deliver a telephony system that includes the following features as a minimum:

2.6.2.20.1Autodial: Allows users to dial a number by pressing a single key.

2.6.2.20.2Automatic Answerback: Allows an incoming call to be answered automatically via the hands-free option on the telephone.

2.6.2.20.3Automatic Line Selection: Allows the user to automatically answer a predetermined line by lifting the handset.

2.6.2.20.4Authorization Code: Allows selected users to temporarily override the access restrictions assigned to any station or trunk by dialing an authorization code.

2.6.2.20.5Automatic Route Selection: Allows an outgoing call to be automatically completed by the least expensive route available.

2.6.2.20.6Automatic Call Distribution: Distributes incoming calls among a designated group of telephone sets on a "most available station" basis.

2.6.2.20.7Call Detail Recording: System must provide the ability to report CDR information and/or be exported to a CDR reporting package.

2.6.2.20.8Call Forward All Calls/No Answer: Allows a station's incoming calls to be automatically forwarded to a pre-selected destination.

2.6.2.20.9Call Forward Reminder Tone: Provides IP telephone users a visual reminder to remind the user that their station is in the call forward mode.

2.6.2.20.10Call ID: System allows a reference number or tag associated with each call during its entire lifetime on the system (not to be confused with Caller ID).

2.6.2.20.11Call Join: Allows calls already existing on separate lines on a telephone to be conferenced together.

2.6.2.20.12Call Party Name Display: Identifies the calling or called party by name.

2.6.2.20.13Call Park: Places a call in a parked state, similar to hold, where it can be retrieved by any attendant console or by another telephone.

- 2.6.2.20.14Call Pickup: Allows a group of telephones to answer a ringing station in its group through the use of either an access code or a programmed pickup button.
- 2.6.2.20.15Call Back: Upon receiving a busy signal from another station, allow callers to be notified that the line is now available.
- 2.6.2.20.16Conference: Allows users to add additional parties to an established call. State maximum allowed conference parties.
- 2.6.2.20.17Distinctive Ringing: Provides a difference in the cadence of the set ringing to distinguish internal calls from outside trunk calls.
- 2.6.2.20.18Do Not Disturb: Allows the station user to place their station in a “Do Not Disturb” mode as defined in a typical analogue telephone system (calls go straight to voice mail, etc.)
- 2.6.2.20.19History File: System must automatically keep a record of maintenance problems it uncovers during its diagnostic process.
- 2.6.2.20.20Hunt: The maximum number of steps available in the system’s hunt sequence. All call information shall stay with the call for the duration of the sequence.
- 2.6.2.20.21Malicious Call Trace: Allows users to activate a call trace while engaged in a conversation by solutions such as tagging / identifying potential malicious calls for possible investigation.
- 2.6.2.20.22Message Waiting Indication: Provides for Message Waiting Indication to be activated within the system for analogue and IP Phones.
- 2.6.2.20.23Multi-Language: Arabic and other multiple languages shall be supported by the system for telephone displays.
- 2.6.2.20.24Music on hold: plays music while a call is placed on hold. The system shall support multiple music sources.
- 2.6.2.20.25Override/Barge: Allows a user to enter into an established “bridged” connection. A warning tone is sent to the established connection.
- 2.6.2.20.26Priority Calling: Allows users on a class of service basis to have priority when placing calls through a busy trunk condition.
- 2.6.2.20.27Ring Again: Provides the station user the opportunity, after encountering a busy station, to have the system monitor the station and notify when it becomes free.
- 2.6.2.20.28Station Camp-On/Call Waiting: Allows the attendant and/or a station user to transfer a call to an active station. Caller must hear ringing while the station to which a call is placed hears an audible tone indicating the call waiting.
- 2.6.2.20.29Station Priority Indication: Allows an attendant to selectively answer internal attendant calls on a priority basis.

2.6.2.20.30 Stored Number Redial: Allows telephones and attendant consoles to store one previously dialed number for automatic re-dialing.

2.6.2.20.31 Time and Date: shall be displayed on all display telephones.

2.6.2.20.32 Time and Date Routing: system must have the ability to automatically change the routing of calls and/or the Class of Service of a station on a time and/or date basis, on a class of service basis,

2.6.3 **الاتصال الداخلي المباشر (DIRECT INWARD DIAL (DID))**

2.6.3.1 The system shall support DID which allows individual extensions or hunt groups to be contacted by external parties without the need for operator assistance or through the dialing of an extension.

2.6.3.2 The systems shall support DID numbers in contiguous or non-contiguous blocks.

2.6.3.3 Where a call is received to a number in a DID range that belongs to the system but has not been assigned to an extension or group it shall be possible to configure how that call is treated. The following configurations shall be possible, as a minimum:

2.6.3.3.1 Busy

2.6.3.3.2 Presentation to an operator

2.6.3.3.3 Presentation on an extension or group

2.6.4 **أجهزة الهاتف (TELEPHONE HANDSETS)**

2.6.4.1 The BIDDER shall propose two types of handsets – basic and advanced.

2.6.4.2 The CONTRACTOR shall deliver basic handsets that shall have the following features and capability as a minimum:

2.6.4.2.1 Keypad in order to place calls

2.6.4.2.2 A single line display for the display of a name and/or CLI

2.6.4.2.3 Support for the connection of a headset

2.6.4.3 The CONTRACTOR shall deliver advanced handsets that shall have the following features and capability in addition to those of the basic handset:

2.6.4.3.1 Hands free operation

2.6.4.3.2 A minimum of 8 configurable buttons that can be programmed with features or as line keys with corresponding status indicators

2.6.4.3.3 Shall support manager/secretary features

2.6.5 **ATTENDANT CONSOLE** وحدة التحكم

2.6.5.1 The telephony system shall be supplied with both Auto Attendant and human operator attendant console(s)

2.6.5.2 The Auto Attendant shall be customizable and shall be able, as a minimum to provide automated call handling and routing by name or extension number or transfer to human attendant operators.

2.6.5.3 The attendant console shall support the following features by default and as a minimum:

2.6.5.3.1 Answering and holding of multiple calls at the console

2.6.5.3.2 Dialing by name or extension numbers

2.6.5.3.3 Programmable hot keys for commonly accessed number, groups and features

2.6.5.3.4 Night service feature

2.6.5.3.5 System alarm notification

2.6.5.3.6 Operator override and interruption on busy lines (this feature shall include a warning tone of the interruption to the called user)

2.6.5.4 It shall be possible to use ordinary telephones on the system as back up or overflow consoles to provide the basic operator call handling features.

2.6.6 **(SYSTEM ADMINISTRATION)** إدارة النظام

2.6.6.1 The following shall be implemented by the CONTRACTOR.

2.6.6.1.1 A browser-based administration system that is accessible from any workstation on the LAN/WAN shall be provided as part of the system. It shall include all hardware and software for proper operation.

2.6.6.1.2 The systems shall support remote administration via a secure connection through an MOI-approved remote connection mechanism and with appropriate credentials.

2.6.6.1.3 File Exports: Allows the export of system data in an industry standard format for use with MOI provided programs. Please describe any formats that are available.

2.6.6.1.4 Designation Strip Printing: Permits the system administrator to automatically print designation strips for the administered system's telephones. Preference is for a solution where this is not necessary.

2.6.6.1.5 Global Change: Provides the ability for the system administrator to change all telephones within the system or within a specific group.

2.6.6.1.6 LAN Network Adaptability: Allows the system administration software to reside on a PC located anywhere on the LAN.

2.6.6.1.7 Multiple Administrators: Provides the ability to allow multiple system administrators on a single system.

2.6.6.1.8 On-Line Help: Provides the system administrator with context specific assistance.

2.6.6.1.9 Port Assignment: Automatically assigns spare ports when adding stations. It shall also be possible to turn off this ability if desired for security.

2.6.6.1.10 Scheduled Changes: Permits the system administrator to prepare changes in the present for future implementation.

2.6.6.1.11 Search: Permits the system administrator to locate station information based on multiple criteria (e.g. extension number, name etc.). Define search criteria available.

2.6.6.1.12 Telephone Set Graphics: Provides a visual representation of the telephone sets to be administered.

2.6.6.1.13 Templates: Allows the system administrator to program multiple telephones with similar features/functions at the same time.

2.6.6.1.14 Voice Mail Integration: Permits the system administrator to update/change the voice mail system to reflect station changes from the same terminal without being required to "log-out" of either system.

2.6.6.2 All system parameters and user information must be under the control of the system administrator.

2.6.6.3 System Administration Reports – The proposed system administration software must provide the following reports as standard:

2.6.6.3.1 Class of Service

2.6.6.3.2 Directories

2.6.6.3.3 Intercom Groups

2.6.6.3.4 Pickup Groups

2.6.6.3.5 Private Lines

2.6.6.3.6 Speed Call Groups

2.6.6.3.7 Traffic Summary Reports by Extension

2.6.7 **متطلبات التثبيت (INSTALLATION REQUIREMENTS)**

2.6.7.1 The CONTRACTOR shall implement all solutions based on the following:

2.6.7.1.1 Where undefined by codes and standards, CONTRACTOR shall apply a safety factor of at least 2 times the rated load to all fastenings and supports of system components.

2.6.7.1.2The CONTRACTOR shall install all hardware and all system components including furnished equipment in accordance with the manufacturer's instructions and shall furnish all cables, connectors, terminators, interconnections, services, and adjustments required for a complete and operable system.

2.6.7.1.3Grounding shall be installed as necessary to preclude ground loops, noise, and surges from adversely affecting system operation.

2.6.7.1.4For equipment mounted in drawers or on slides, provide the interconnecting cables with a service loop of not less than three feet and ensure that the cable is long enough to allow full extension of drawer or slide.

2.6.7.1.5The CONTRACTOR shall not apply power to any solution until after the system and components have been installed and inspected in accordance with the manufacturer's installation instructions.

2.6.7.1.6The CONTRACTOR shall test and verify system wiring and shall share the corresponding report with the MOI prior to applying power to any of the solutions.

2.6.7.1.7The CONTRACTOR shall test and verify that all system grounding and transient protection systems have been verified as properly installed and connected and shall share the corresponding report with the MOI prior to applying power to any of the solutions.

2.6.7.1.8Power supplies to be connected to the system and equipment have been verified as the correct voltage, phasing, and frequency as indicated.

2.6.7.2The BIDDER shall note that satisfaction of the above requirements shall not relieve the CONTRACTOR of responsibility for incorrect installations, defective equipment items, or collateral damage as a result of CONTRACTOR work/equipment.

2.6.7.3In addition to the testing requirements identified elsewhere in this document, the following shall be included in the testing for the telephony and recording solution:

2.6.7.3.1Failover and failback between paired centers.

2.6.7.3.2Failover of primary/secondary trunk lines.

2.6.7.3.3Power failure.

2.7 البنية التحتية التقنية الخاصة بالسحابة (PRIVATE CLOUD TECHNICAL INFRASTRUCTURE)

2.7.1 متطلبات التصميم (DESIGN REQUIREMENTS)

2.7.1.1The CONTRACTOR shall implement all requirements in this section with the understanding that:

2.7.1.1.1 Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.

2.7.1.1.2 Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.

2.7.1.1.3 Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.

2.7.1.2 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.

2.7.1.3 The CONTRACTOR shall deliver a Technical Infrastructure solution with AI-enabled capabilities for OSS911 use.

2.7.1.4 The BIDDER shall confirm that the proposed Technical Infrastructure includes all servers, all systems, all network, all security, all storage with the associated hardware, equipment, software, firmware and systems to deliver the operations of the OSS911 center.

2.7.1.5 Within their technical submittals, the BIDDER shall clearly articulate their space, HVAC, power, and other such requirements for hosting their equipment in any datacenters.

2.7.1.6 Within their technical submittals, the BIDDER shall indicate the amount of as-is scalability and expandability available in their technical infrastructure without adding additional hardware, software or firmware.

2.7.1.7 Prior to the completion of the acceptance testing and as per the schedules finalized by the MOI, the CONTRACTOR shall provide complete technical infrastructure documentation including the as-built design, configurations, solutions, access credentials, full inventory of all items with their barcodes, the licenses details with their unique numbers and validity periods, standard operating procedures for routine maintenance, location of the equipment/license, description, model, and serial numbers, the user guides and the system manuals.

2.7.1.8 The CONTRACTOR shall implement the technical Infrastructure to provide the basic authentication, LDAP, domain, storage, network, security, file and print services within the OSS911 Center.

2.7.1.9 At each OSS911 center, the CONTRACTOR shall utilize the MiFON connection that are terminated in a CE device and connect them to the CONTRACTOR provided network edge equipment for integration of a that OSS911 center with MiFON backbone and thus, with the other OSS911 centers.

2.7.1.10 The CONTRACTOR shall implement a centralized LDAP solution for the OSS911 center in scope of the project and shall configure authentication of all OSS911 solutions to the centralized domain services for the OSS911 center in scope.

- 2.7.1.11The CONTRACTOR shall supply and configure the required racks, rack management and any peripheral components
- 2.7.1.12The CONTRACTOR shall deliver all OSS911 applications user interface and system administration interface in Arabic language as default, and shall additionally support the English language interface.
- 2.7.1.13The CONTRACTOR shall provide computers that includes the installation of the latest version of desktop automation suite (capability of word processing, spreadsheet, presentation and collaboration including email) software at the time of installation to all supplied personal computers, together with the appropriate licenses.
- 2.7.1.14All items in the technical infrastructure delivered by the CONTRACTOR for production environment shall:
- 2.7.1.14.1Be configured by the CONTRACTOR to be fully resilient against any single point of failure.
- 2.7.1.14.2Be configured by the CONTRACTOR to deliver high availability disaster recovery – fully redundant at link, component and device levels, that complies with the availability and service level requirements within this RFP.
- 2.7.1.14.3Be connected by the CONTRACTOR to two independent power sources.
- 2.7.1.15The most current, full release server operating system shall be supplied for use as the operating system for domain, file, print and application servers. If the operating system proposed by the BIDDER is no longer supported by its manufacturer prior to the final acceptance, the server operating system shall be upgraded by the CONTRACTOR without any additional cost to the MOI.
- 2.7.1.16All technical Infrastructure and OSS911 applications shall support Network Time Protocol (NTP) and shall be synchronized by the CONTRACTOR to the same time source as identified by MOI.
- 2.7.1.17All times shall be set to local Saudi Arabian Time by the CONTRACTOR.
- 2.7.1.18The CONTRACTOR shall implement a centralized LDAP solution for all the OSS911 solutions under the scope of delivery of the contractor.
- 2.7.1.19The CONTRACTOR shall implement the LDAP service to assist in the management, monitoring and tracking of all aspects of the OSS911 Center Technical Infrastructure, including the assignment of policies, deployment of software and application of patches and critical updates.
- 2.7.1.20The CONTRACTOR shall implement the following for authorization and authentication with the LDAP:
- 2.7.1.20.1A mandatory single sign-On scheme based on a user name and a complex password for any user of OSS911 solutions.
- 2.7.1.20.2A multi-factor authentication scheme for administering, monitoring and managing any technical infrastructure and/or OSS911 solution and/or service.

2.7.1.21 The CONTRACTOR shall implement LDAP domain for production environment within OSS911 that is separate than the LDAP domain for testing and the LDAP domain for training systems.

2.7.1.22 Within each domain, users shall be arranged into functional units that represent a function within OSS911 Center.

2.7.1.23 All users shall be required to log on to the LDAP before they can access any PC features or LAN resources.

2.7.1.24 All PCs within the OSS911 shall be a member of the directory before they can access any services within the directory.

2.7.1.25 Policy shall be used to control the features of the PC and apply security settings. The CONTRACTOR shall develop, document and implement the policies as per the approval of the MOI.

2.7.2 أنظمة OSS911 وترخيصها VIRTUALIZATION SOLUTION:

2.7.2.1 Any and all technical infrastructure implemented by the CONTRACTOR shall be based on virtualized solutions in the OSS911 center in scope, sized for the concept of operations outlined in this document and for the OSS911 applications that shall be delivered by the CONTRACTOR based on this concept of operations.

2.7.2.2 The CONTRACTOR shall deliver the OSS911 solutions using technical infrastructure is based on 100% virtualized systems hosted in hyperscale private cloud based on Software Defined Data Centers (SDDC) where all infrastructure is virtualized and delivered as a service, using highly available virtualization platforms.

2.7.2.3 The BIDDER shall provide evidence that their proposed virtualization solution has been used for similar 911 center solutions.

2.7.2.4 Technical Infrastructure serving an OSS911 center telephony and recording solution shall be hosted at that OSS911 center supported by the required network and cybersecurity technical infrastructure.

2.7.2.5 Technical Infrastructure serving all other OSS911 solutions supported by the required network and cybersecurity technical infrastructure shall be hosted the OSS911 center in scope of the project.

2.7.2.6 Technical infrastructure shall be compliant with authorized hardware specifications as found on the MOI and NIC websites.

2.7.2.7 The BIDDER shall include the following in their proposal while describing the virtualization solution:

2.7.2.7.1 Proposed server capacity to host the solution: number of VMs per server/socket, amount of memory per VM

2.7.2.7.2 Proposed storage capacity

2.7.2.7.3 Total storage requirements per VM and for the solution

2.7.2.7.4 Total network capacity requirement per VM.

2.7.2.7.5 The overall architecture of the virtualized systems

- 2.7.2.7.6Diagrams of the core virtualized system components
 - 2.7.2.7.7Design including the appropriate products and licensing
 - 2.7.2.7.8All necessary documentation of the solution including product literature, and specification sheets
 - 2.7.2.7.9Provide equipment list with version/model numbers and design description
 - 2.7.2.7.10An itemized list of all proposed equipment
 - 2.7.2.7.11An itemized list of software and software licenses
 - 2.7.2.7.12An itemized list of services (implementation, training)
 - 2.7.2.7.13A full description of the high availability and failover capabilities of the solution
 - 2.7.2.7.14A summary of the proposed solution and specific supported features
- 2.7.2.8The BIDDER shall describe the capabilities of the proposed virtualized solution in terms of VM creation and templating, physical to virtual transitioning, and provisioning of VMs across multiple physical servers.
- 2.7.2.9The BIDDER shall clearly articulate their space, HVAC, power, and other such requirements for hosting of their equipment.
- 2.7.2.10The BIDDER shall confirm that their solution supports network load balancing.
- 2.7.2.11The CONTRACTOR shall deliver the OSS911 solutions using virtualization that:
- 2.7.2.11.1Works on latest x86 server hardware available from all the leading vendors in the industry, is not proprietary and shall not be restricted to a particular vendor/make/model.
 - 2.7.2.11.2Supports NVMe, NVMe-oF, S3/RDMA, FC, ISCSI, FCoE, NFS, & SMB protocol to enable various types of storage traffic with the CONTRACTOR implementing the choice of the protocol based on the design finalization with the MOI.
 - 2.7.2.11.3Provides support for heterogeneous guest Operating systems.
- 2.7.2.12The CONTRACTOR shall implement the virtualized solution such that it is configured for scaling-up (by adding SSDs/ultra-high-capacity HDDs) and scaling-out (by adding nodes) architecture with no disruption to the workloads already running on the platform.
- 2.7.2.13The BIDDER proposed solution shall be compatible with all virtualization industry standards for interoperability compliant to ISO/IEC 19099.
- 2.7.2.14The CONTRACTOR shall deliver a technical infrastructure solution based on a virtualized environment containing: virtualized platform,

compute, storage and operating systems, virtualized hardware and required virtualized network devices.

2.7.2.15The CONTRACTOR shall implement the virtualization solution such that there is a capability to move a VM from one host server to another while the VM is running (Live Migration) and without downtime.

2.7.2.16The CONTRACTOR shall implement the virtualization solution such that it allows all OSS911 applications to run 24x7 while complying with the availability and reliability requirements of this RFP.

2.7.2.17The CONTRACTOR shall implement the virtualization solution such that there is prioritization between the VMs and such that VMs are able to restarted without any downtime.

2.7.2.18The CONTRACTOR shall implement the virtualization solution such that it allows automation and scheduling of VM migration and activity and re-activating of host servers.

2.7.2.19The CONTRACTOR shall implement the virtualization solution such that it allows self-service, VM provisioning and management.

2.7.2.20The CONTRACTOR shall implement the virtualization solution using a single highly available virtualization platform such that:

2.7.2.20.1The host nodes are connected using 10Gbps IP connectivity.

2.7.2.20.2Per node, a minimum of 2x10Gbps Ethernet connections is implemented.

2.7.2.20.3There is no dependency on any proprietary or specialized interconnects.

2.7.2.20.4In-built software storage capability integrated within the hypervisor kernel itself is included and the system shall work without the need for any specialized dedicated controller virtual appliance or any separate VM.

2.7.2.20.5There is zero-downtime, zero-data loss continuous availability against any failure.

2.7.2.20.6There is a converter to migrate physical workloads to virtual workloads.

2.7.2.21The CONTRACTOR shall deliver a single unified management console for the management of the entire environment including virtualized environment as well as software defined storage environment, underlying hardware and associated components.

2.7.2.22The CONTRACTOR solution shall provide the ability to hot-add memory and hot-plug disks and network interface cards

2.7.2.23The CONTRACTOR solution shall provide configurations at VM level that shall be tuned to help reduce latency.

2.7.2.24The CONTRACTOR solution shall provide zero downtime during scheduled host patching (maintenance mode) by moving active

workloads to other hosts manually and automatically on the platform with a consistent audit trail of the patching process.

2.7.2.25The CONTRACTOR solution shall provide centralized disk-based data protection (backup/restore) capabilities for the virtual workloads with file-level restore and built-in de-duplication to reduce backup storage requirements

2.7.2.26The CONTRACTOR shall implement management, monitoring and alerting of the physical hosts, the virtual machines, the compute, networking, storage and security through the virtualization management software and shall integrate this with the OSS911 management and monitoring solution.

2.7.3 **OSS911 SYSTEMS DIMENSIONING AND LICENSING**

2.7.3.1All the OSS911 systems software (applications, databases, operating systems, firmware and any other software) provided shall be of open license thus providing MOI the flexibility to scale up the position numbers up to the limit of dimensioning above, without buying additional backend hardware for compute, storage, network and security.

2.7.3.2The CONTRACTOR solution shall not place any restriction on the location where the supplied equipment and software to be installed. The MOI shall have complete freedom to instruct the deployment of the procured quantities or any increments anywhere within the Kingdom of Saudi Arabia.

2.7.3.3The different components of infrastructure supporting the OSS911 systems shall be dimensioned for the following maximum capacities:

2.7.3.3.1IRMS – 300 positions each for call takers and supervisors, 300 positions of dispatchers and supervisors

2.7.3.3.2Joint Operation – 100 positions each for JOA operators

2.7.3.4Unless specified explicitly in the Bill of Quantity for this RFP by the MOI, all the software (applications, databases, operating systems, firmware and any other software) licenses provided by the CONTRACTOR shall include and implement licenses such that the each of the software:

2.7.3.4.1Shall be the advanced and enterprise version / flavor for the complete suite as provided by the related product vendor without exclusions and/or exceptions.

2.7.3.4.2Shall be capable of being used from anywhere in Kingdom of Saudi Arabia through a web browser.

2.7.3.4.3Shall not have per seat / per user license.

2.7.3.4.4Shall be perpetual and shall not be subscription-based that needs to be renewed periodically.

2.7.3.4.5Shall be unrestricted by the number of computer workstations or other dimension limiting factor, and shall allow the operating system or applications to be loaded, or reloaded onto a substitute machine (or machines) without restriction to a specific

hardware serial number or registration number or IP address or MAC address or any other number.

2.7.3.5The CONTRACTOR shall provide complete software licenses for all software including third party software and including all components required to achieve the requirements of this RFP.

2.7.3.6The CONTRACTOR shall include the solution to install, activate and manage all software licenses (applications, databases, operating systems, firmware and any other software) centrally using a licensing server hosted on premises within the OSS911 center agreed with the MOI.

2.7.3.7For the duration of the contract, the CONTRACTOR shall be responsible for maintaining all the delivered software and solutions updated, upgraded and patched with their most recent stable major release as confirmed by the OEM vendors.

2.7.4 الخوادم والتخزين (SERVERS AND STORAGE)

2.7.4.1All the servers and storage required for each of the OSS911 solutions shall be supplied, configured and implemented by the CONTRACTOR.

2.7.4.2All servers, databases and storage in production environment shall be configured by the CONTRACTOR for high availability disaster recovery as required by the individual applications to ensure service availability as required in this document.

2.7.4.3All servers, databases and storage in production environment shall be configured by the CONTRACTOR with multiple, redundant 10Gbps connections to server switches.

2.7.4.4The CONTRACTOR shall implement an active-active object storage cluster (software-defined and/or HDD-based) for long term data retention, training datasets, etc. to deliver the RFP data retention for all data and AI-related training requirements for storage.

2.7.4.5Storage Area Network (SAN) shall be implemented by the CONTRACTOR that consolidates all the online storage accessible to all OSS911 solutions unless otherwise approved by MOI.

2.7.4.6The BIDDERS shall provide a preliminary SAN design with their proposal.

2.7.4.7The BIDDERS shall clearly document their rationale supporting the SAN capacities they intend to provide and how that meets all the requirements of the OSS911 applications.

2.7.4.8The CONTRACTOR shall implement an active-active cluster of high-performance compute nodes for real-time, concurrent inference and AI pre/post processing to deliver the RFP AI-requirements and related load including GPU-based parallel AI accelerators with NVMe storage.

2.7.4.9The CONTRACTOR shall implement an active-active cluster of high-performance compute nodes for training AI models to deliver the RFP AI-requirements and related load including GPU-based parallel AI accelerators with NVMe storage.

- 2.7.4.10The CONTRACTOR shall implement an active-active all-flash storage cluster for real-time AI inference to deliver the RFP AI-requirements and related load.
- 2.7.4.11Within the production environment, the CONTRACTOR shall configure and implement appropriate RAID levels in the SAN based on the final design agreed with the MOI.
- 2.7.4.12Within the production environment, the CONTRACTOR shall implement the SAN with clustering, mirroring (local and remote replication or cloning) and snapshots.
- 2.7.4.13Within the production environment, the CONTRACTOR shall implement the SAN to include non-disruptive component replacement and online microcode upgrades.
- 2.7.4.14It is the responsibility of the BIDDER to calculate the storage requirements for the proposed OSS911 Center, their systems, and operations, and then propose a suitable storage size for meeting them.
- 2.7.4.15Separate file servers shall be supplied for each domain.
- 2.7.4.16The CONTRACTOR shall implement MOI approved policy restrictions for each user to determine what access rights that user has.
- 2.7.4.17All printers shall be capable of being accessed via a print server.
- 2.7.4.18All file and print servers shall have redundant power supplies and hot swappable components as appropriate.

2.7.5 الشبكات (NETWORKS)

- 2.7.5.1The CONTRACTOR shall implement solutions for the OSS911 center to be connected to MOI Fiber Optic Network backbone (MiFON) network.
- 2.7.5.2The CONTRACTOR shall supply and implement an OSS911 network that is totally and absolutely physically isolated from the Internet except for the use cases and requirements explained in this document.
- 2.7.5.3The BIDDER shall note that for the fallback to the nominated/paired OSS911 center, the connection to the MOI MiFON backbone shall be used.
- 2.7.5.4Within their proposal, the BIDDER shall clearly state their requirements of the bandwidth, latency, QoS and other parameters for the MOI MiFON backbone for their performance and availability to be achievable.
- 2.7.5.5The CONTRACTOR shall implement end-to-end Quality of Service across the OSS911 networks to prioritize between different types of traffic.
- 2.7.5.6The CONTRACTOR shall implement Demilitarized Zones (DMZs) as agreed with the MOI and as per standard best practice of dual firewall DMZ or better.
- 2.7.5.7The CONTRACTOR shall implement security operations, monitoring and control system that is integrated in to the overall OSS911

infrastructure and systems management and monitoring tool that they shall implement as part of this project.

2.7.5.8The CONTRACTOR shall configure the network solutions for OSS911 network to support IPv4 and IPv6 for all components and devices.

2.7.5.9The CONTRACTOR OSS911 solutions shall work on Software Defined Network (SDN), be SDN compatible and shall utilize spine-leaf architecture and orchestration.

2.7.5.10The CONTRACTOR shall configure all the active elements within all the networks to support SNMPv3, HTTP/3 and RESTCONF & NETCONF.

2.7.5.11The CONTRACTOR shall implement API-based management using RESTCONF, NETCONF protocols and shall implement integration with AI/ML for predictive analytics in the OSS911 networks and for self-healing networks.

2.7.5.12The CONTRACTOR shall implement all servers to be dual-homed and shall connect to the network by multiple 10 Gbit/sec dedicated connections.

2.7.5.13The CONTRACTOR shall implement all structured cabling, cable trays / pathways / conduits and accessories required for connecting any and all CONTRACTOR implemented active and passive equipment for connecting to the existing structured cabling infrastructure the OSS911 center.

2.7.5.14The CONTRACTOR shall implement all copper UTP cabling that meets or exceed CAT6A specifications end-user connections, CAT7 for 10Gbps connections and CAT8 for 40Gbps connections, as per the respective ANSI/TIA-568 standards.

2.7.5.15The CONTRACTOR shall implement all UTP (copper), and fiber-optic cabling (multi-mode and single-mode), and the labeling as per the most recent approved TIA-942 and TIA-606 standards.

2.7.5.16The CONTRACTOR is required to implement any and all network equipment that is required by their solutions in addition to the MOI supplied equipment.

2.7.5.17The CONTRACTOR shall configure the switching capability within OSS911 network based on multiple Virtual Local Area Networks (VLANs) and Virtual Private Networks (VPNs).

2.7.5.18The CONTRACTOR shall implement all switches within the OSS911 network to be non-blocking and to support all unified communications requirements – ability to exchange video, voice and data with end-to-end Quality of Service (QoS).

2.7.5.19The CONTRACTOR shall implement spine switches within OSS911 network with the following specifications:

2.7.5.19.1The spine layer switches shall have hardware level redundancy (1+1) in terms of data plane and control plane. Issues with any of the plane should not impact the functioning of the switch

- 2.7.5.19.2The switch shall have redundant CPUs working in active-active or active-standby mode. CPU fail over/change over shall not disrupt/impact/degrade the functioning the switch The Switch shall support non-blocking Layer 2 switching and Layer 3 routing
- 2.7.5.19.3The switch shall not have any single point of failure like CPU, supervisor, switching fabric power supplies and fans etc. should have 1:1/N+1 level of redundancy
- 2.7.5.19.4Switch shall support in line hot insertion and removal of different parts like modules/power supplies/fan tray etc. This shall not require rebooting of the switch or create disruption in the working/functionality of the switch
- 2.7.5.19.5Switch shall support the complete stack of IP V4 and IP V6 services
- 2.7.5.19.6Switch with different modules shall function line rate and should not have any port with oversubscription ratio applied
- 2.7.5.19.7Switch shall support in service software upgrade of the switch without disturbing the traffic flow. There shall not be any impact on the performance in the event of the software upgrade/downgrade. It shall support in service patching of selected process/processes only without impacting other running processes
- 2.7.5.19.8Switch shall support non-blocking, wire speed performance per line card
- 2.7.5.19.9Switch shall be compatible with SDN architecture of all the OEM in future without any additional cost / hardware
- 2.7.5.19.10Switch shall be compatible with all the required parameters of Spine & leaf architecture
- 2.7.5.19.11Switch shall support Network Virtualization
- 2.7.5.19.12Switch shall support VXLAN and EVPN for supporting Spine - Leaf architecture to optimize the east - west traffic flow inside the data center
- 2.7.5.19.13Switch shall support Open Flow/Open Day light/Open Stack controller
- 2.7.5.19.14Switch shall support Data Center Bridging
- 2.7.5.19.15Switch shall support multi-OEM hypervisor environment and shall be able to sense movement of VM and configure network automatically
- 2.7.5.20The CONTRACTOR shall deliver and implement leaf, server leaf and border leaf switches and any other switches they require, within OSS911 network with the following specifications:
- 2.7.5.20.1The Switch shall support non-blocking Layer 2 switching and Layer 3 routing.

- 2.7.5.20.2 There switch shall not have any single point of failure like power supplies and fans etc. shall have 1:1/N+1 level of redundancy
- 2.7.5.20.3 Switch support in-line hot insertion and removal of different parts like modules/power supplies/fan tray etc. shall not require switch reboot and disrupt the functionality of the system
- 2.7.5.20.4 Switch shall support the complete STACK of IP V4 and IP V6 services
- 2.7.5.20.5 The Switch and different modules used shall function in line rate and shall not have any port with oversubscription ratio applied
- 2.7.5.20.6 Switch shall be compatible with SDN architecture of all the OEM in future without any additional cost / hardware
- 2.7.5.20.7 Switch shall be compatible with all the required parameters of Spine & leaf architecture
- 2.7.5.20.8 Switch shall support Network Virtualization using Virtual Over Lay Network using VXLAN /NVGRE.
- 2.7.5.20.9 Switch shall support VXLAN and EVPN or equivalent for supporting Spine - Leaf architecture to optimize the east - west traffic flow inside the data center.
- 2.7.5.20.10 Switch shall support Open Flow/Open Day light/Open Stack controller
- 2.7.5.20.11 Switch shall support Data Center Bridging
- 2.7.5.20.12 Switch shall support multi-OEM hypervisor environment and shall be able to sense movement of VM and configure network automatically.
- 2.7.5.21 The CONTRACTOR will supply, install and connect patch panels, and patch cables for their networking needs in the datacenter.
- 2.7.5.22 The CONTRACTOR shall implement hardware-based highly available load balancers for each of the zones within the OSS911 center under the scope of this contract for:
- 2.7.5.22.1 One pair for access to all datacenter solutions and services.
- 2.7.5.22.2 One pair for all access to the wide area network via MiFON.
- 2.7.5.22.3 One pair for all access to the DMZ.
- 2.7.5.23 The CONTRACTOR shall provide a DHCP capability for the dynamic assignment of IP addresses within the OSS911 network.
- 2.7.5.24 The CONTRACTOR shall provide a DNS capability for the resolution of names.
- 2.7.5.25 The CONTRACTOR shall configure DNS and DHCP in highly available mode each with primary and secondary servers and with load balancing between the two.

2.7.5.26 The CONTRACTOR shall supply 48-port access switches, aggregation switches and routers for use for NIC services local distribution within the OSS911 center as per the bill of quantity – these shall match the NIC specifications as mentioned on the MOI/NIC web site, and shall include all the required SFPs / GBICs and the necessary cables and accessories, and shall be based on products from vendors who are rated as leaders in the last released Gartner Magic Quadrant for the respective network function.

2.7.5.27 Based on the design discussions with the MOI, the CONTRACTOR shall finalize the required models for the different access, aggregation, leaf and spine switches, and the different routers that will be deployed by the CONTRACTOR.

2.7.6 الأمان (SECURITY)

2.7.6.1 The CONTRACTOR shall implement security solutions that ensure a hardened Technical Infrastructure for all OSS911 center systems and infrastructure.

2.7.6.2 The BIDDER shall note that the various security solutions' items in the BoQ are placeholders for all components of a solution. Other components as required by BIDDER's security solution will be included in the line item for firewalls in the BoQ, and the detailed breakdown, and functions of the components will be explained in technical and commercial proposals.

2.7.6.3 The requirements outlined in this section are not exhaustive and MOI expects the BIDDERS to utilize their knowhow and experience with critical infrastructure protection to propose highly secure and hardened security solutions over and beyond the stated requirements here.

2.7.6.4 The CONTRACTOR shall provide any and all security certificates, hardware, software, tokens and any other equipment that is required to achieve a secure and hardened technical infrastructure.

2.7.6.5 The CONTRACTOR shall implement and demonstrate compliance to the cybersecurity controls from the most recent National Cybersecurity Authority of Kingdom of Saudi Arabia for all domains, subdomains, main controls and subcontrols therein unless explicitly waived for implementation by MOI, and in accordance with ISO 27001 for information security.

2.7.6.6 The CONTRACTOR shall implement security controls that are required for securing the OSS911 center systems, data and infrastructure and for providing information assurance.

2.7.6.7 The CONTRACTOR shall implement all of the following Center for Internet Security (CIS) Critical Security Controls:

2.7.6.7.1 Inventory of Authorized and Unauthorized Hardware Devices

2.7.6.7.2 Inventory of Authorized and Unauthorized Software

2.7.6.7.3 Secure Configurations for Hardware and Software on Mobile Devices, Laptops, Workstations and Servers

2.7.6.7.4 Continuous Vulnerability Assessment and Remediation

2.7.6.7.5 Malware Defenses

2.7.6.7.6Application Software Security

2.7.6.7.7Wireless Access Control

2.7.6.7.8Data Recovery Capability

2.7.6.7.9Security Skills Assessment and Appropriate Training to Fill Gaps

2.7.6.7.10Secure Configurations for All Network Devices including Firewalls, Routers and Switches

2.7.6.7.11Limitation and Control of Network Ports, Protocols and Services

2.7.6.7.12Controlled Use of Administrative Privileges

2.7.6.7.13Boundary Defense

2.7.6.7.14Maintenance, Monitoring and Analysis of Audit Logs

2.7.6.7.15Controlled access based on the Need to Know

2.7.6.7.16Account Monitoring and Control

2.7.6.7.17Data Protection

2.7.6.7.18Incident Response and Management

2.7.6.7.19Secure Network Engineering

2.7.6.7.20Penetration Tests and Red Team Exercises

2.7.6.8 The BIDDERS shall note that the CONTRACTOR shall be provided with the official MOI security policy including the security controls compliance required. The requirements within the MOI security policy are reflected in the requirements of this RFP to a large extent. However, if by the award of the contract, newer MOI policy requirements surface, then the CONTRACTOR shall be required to implement these. Where there is a conflict between any stated requirement below and the MOI policy, the requirement from MOI policy shall take precedence.

2.7.6.9 All the unused input / output ports within all the devices (servers, networks, PCs, SAN) shall be disabled by default.

2.7.6.10 Workstations shall be capable of being screen locked and shall default to screen lock when left unattended for a pre-set period. Screen locks shall require a password to gain access.

2.7.6.11 The CONTRACTOR shall implement the following AI security controls unless it is agreed with the MOI that this control is not relevant for implementation in the OSS911 solutions as part of this contract:

2.7.6.11.1 Model integrity verification

2.7.6.11.2 Adversarial input detection

2.7.6.11.3 Training data poisoning prevention

2.7.6.11.4 Secure deployment pipeline

- 2.7.6.12 The CONTRACTOR shall design the security protection such that the protection is both extendable/customizable and prevents users from avoiding the protection (or shutting it off).
- 2.7.6.13 The security solution shall be configured by the CONTRACTOR to provide alerts to the system management tool based on different parameters exceeding predefined thresholds.
- 2.7.6.14 The CONTRACTOR shall remove all unnecessary protocols from all devices.
- 2.7.6.15 The CONTRACTOR shall configure the network equipment to close inactive sessions.
- 2.7.6.16 The CONTRACTOR shall configure the security solution with policy-based auditing of all OSS911 applications and systems to produce logs of all activities such that these logs can be analyzed with CONTRACTOR supplied tools for monitoring, alerting and reporting.
- 2.7.6.17 An antivirus solution shall be provided by the CONTRACTOR that shall be capable of monitoring, alerting and reporting on the status including the status of virus definitions being up-to-date
- 2.7.6.18 An anti-virus client license shall be delivered for each computer and server delivered under this RFP, using volume licensing method.
- 2.7.6.19 The anti-virus system solution shall provide real-time protection against any virus attacks without compromising the performance.
- 2.7.6.20 Any item which is not a core component of the OSS911 project servers shall be physically separated by firewalls and proxies that are configured such that full protection is provided to critical services.
- 2.7.6.21 The CONTRACTOR shall implement next generation, FIPS or equivalent certified firewalls that are of two types – midrange firewall and entry range firewalls.
- 2.7.6.22 Mid-range firewalls shall have at least 16*GE RJ45 ports, 6*GE SFP ports, 6*10GE SFP ports with 16G Memory, 30Mpps throughput at least.
- 2.7.6.23 All firewalls shall be modular, hardware appliance-based and shall be configured in load sharing and load balancing mode providing stateful failover.
- 2.7.6.23.1 Firewalls within a DMZ shall be from different vendors.
- 2.7.6.23.2 Firewalls shall support MD-5 and SHA-1 authentication.
- 2.7.6.24 The CONTRACTOR shall deliver a VPN solution with multifactor authentication as part of the firewall solution for secure remote access to OSS911 solutions by authorized users.
- 2.7.6.25 The CONTRACTOR shall implement Web Application Firewall “WAF” with HA cluster and a sandbox that are compatible with the installed firewalls.
- 2.7.6.26 The CONTRACTOR shall install a vulnerability scanner that is compatible with the installed firewalls, WAF and Sandbox.
- 2.7.6.27 The CONTRACTOR shall integrate the firewalls, the WAFs, the Sandbox and the vulnerability scanner for a hardened security posture.

2.7.6.28

The CONTRACTOR shall implement security solutions delivering the following features unless a feature is agreed with the MOI to be not applicable to the OSS911 solutions:

2.7.6.28.1Web Security

- a. Automatic profiling (white list)
- b. Web server and application signatures (black list)
- c. IP Reputation
- d. IP Geolocation
- e. HTTP RFC compliance
- f. Native support for HTTP/2
- g. OpenAPI 3.0 verification
- h. WebSocket protection and signature enforcement
- i. Man in the Browser (MiTB) protection
- j. AI-based machine learning

2.7.6.28.2Application Attack Protection

- a. OWASP Top 10
- b. Cross Site Scripting
- c. SQL Injection
- d. Cross Site Request Forgery
- e. Session Hijacking
- f. Built-in Vulnerability Scanner
- g. Third-party scanner integration (virtual patching)
- h. File upload scanning with AV and sandbox

2.7.6.28.3Security Services

- a. Web services signatures
- b. XML and JSON protocol conformance
- c. Malware detection
- d. Virtual patching
- e. Protocol validation
- f. Brute force protection
- g. Cookie signing and encryption

- h.** Threat scoring and weighting
- i.** Syntax-based SQLi detection
- j.** HTTP Header Security
- k.** Custom error message and error code handling
- l.** Operating system intrusion signatures
- m.** Known threat and zero-day attack protection
- n.** L4 Stateful Network Firewall
- o.** DoS prevention
- p.** Advanced correlation protection using multiple security elements
- q.** Data leak prevention
- r.** Web Defacement Protection

2.7.6.28.4 Application Delivery

- a.** Layer 7 server load balancing
- b.** URL Rewriting
- c.** Content Routing
- d.** HTTPS/SSL Offloading
- e.** HTTP Compression
- f.** Caching

2.7.6.28.5 Authentication

- a.** Active and passive authentication
- b.** Site Publishing and SSO
- c.** RSA Access for 2-factor authentication
- d.** LDAP, RADIUS, and SAML support
- e.** SSL client certificate support
- f.** CAPTCHA and Real Browser Enforcement (RBE)

2.7.6.29 The proposed vulnerability scanner shall provide the following:

2.7.6.29.1 Unlimited Web Scanning

2.7.6.29.2 DeepScan Crawler

2.7.6.29.3 Gray-box Vulnerability Testing

2.7.6.29.4 Out-of-band Vulnerability Testing

2.7.6.29.5Login Sequence Recorder

2.7.6.29.6Manual Intervention during Scan

2.7.6.29.7Malware URL Detection

2.7.6.29.8Manual Pen-testing Tool Suite

2.7.6.29.9Scanning of Online Web Application Assets

2.7.6.29.10Scanning of Internal Web Application assets

2.7.6.29.11Key Reports (Affected Items, Quick, Developer, Executive)

2.7.6.29.12OWASP TOP 10 Report

2.7.6.29.13CVSS (Common Vulnerability Scoring System) for Severity

2.7.6.29.14Remediation Advice

- 2.7.6.30 The CONTRACTOR shall implement centralized solutions that detects and prevents intrusions in each OSS911 center.
- 2.7.6.31 The CONTRACTOR shall configure the intrusion detection system to be capable of proactively deciphering if the OSS911 network is getting compromised based on traffic patterns.
- 2.7.6.32 The CONTRACTOR shall configure the intrusion prevention system to be capable of detecting and isolating any malicious traffic within OSS911 networks.
- 2.7.6.33 The CONTRACTOR shall configure the intrusion solutions to be able to generate alarms if required.
- 2.7.6.34 The CONTRACTOR implemented security solutions, monitoring and reporting shall comply with National Institute of Standards and Technology (NIST) 800-137 standard or equivalent.
- 2.7.6.35 A network access control solution shall be implemented by the CONTRACTOR.
- 2.7.6.36 The CONTRACTOR shall implement security measures for encrypting OSS911 data in rest and transit, with the latest standards as agreed with the MOI, while delivering the availability and performance requirements.
- 2.7.6.37 The CONTRACTOR shall develop and implement security policies and technical procedures for handling all security related matters.
- 2.7.6.38 The management and monitoring system shall be implemented and configured by the CONTRACTOR to provide analytics on the security related aspects.
- 2.7.6.39 The CONTRACTOR shall implement a privileged access management (PAM) solution for the OSS911 center with multi-factor authentication with following capabilities:
- 2.7.6.39.1The CONTRACTOR shall through the PAM solution provide secure, centralized control over all privileged access to business systems and underlying infrastructure (including servers, networks, databases, security devices, and endpoints).

2.7.6.39.2The CONTRACTOR shall through the PAM solution, enforce multi-factor authentication, granular role-based access controls, and least-privilege principles.

2.7.6.39.3The CONTRACTOR shall through the PAM solution broker and isolate all remote sessions without exposing credentials; and support credential vaulting with automatic rotation.

2.7.6.39.4The CONTRACTOR shall through the PAM solution provide full session monitoring, recording, and audit logging for accountability and compliance, with real-time alerting on suspicious activities.

2.7.6.39.5The CONTRACTOR shall integrate the PAM solution with existing identity and security tools, support approval workflows for access, and ensure all access is time-bound, policy-driven, and fully traceable.

2.7.6.40 The CONTRACTOR shall implement a security information and event management solution for the OSS911 center that is integrated with various OSS911 systems and solutions and that.

2.7.6.40.1Ingests, normalizes, correlates, and securely stores logs and telemetry from all critical business systems and infrastructure components—including servers, networks, databases, endpoints, and security devices—in near real time.

2.7.6.40.2Provides advanced analytics and correlation rules to detect anomalies, threats, and policy violations, and provides alerting.

2.7.6.40.3Delivers customizable alerting with prioritization and automated response workflows.

2.7.6.40.4Delivers comprehensive search, reporting, and dashboards for operational visibility and compliance.

2.7.6.40.5Ensures scalability, high availability, and data integrity.

2.7.6.40.6Integrates with other cybersecurity tools and identity systems.

2.7.6.40.7Enforces role-based access control, audit trails, and retention policies aligned with NCA and MOI cybersecurity requirements.

2.7.6.41 The CONTRACTOR shall implement an asset management solution that is not limited to asset discovery and classification - it shall be able to actively discover, monitor, control and manage all active assets, and to have a barcode associated with each.

2.7.6.42 The CONTRACTOR shall implement an update and patch management solution for managing and distributing updates and patches for all software/firmware/operating systems delivered by the CONTRACTOR.

2.7.6.43 The CONTRACTOR shall provide complete, independent vulnerability solution for internal and external vulnerability assessment/testing and shall provide a comprehensive assessment report of their infrastructure, security and network systems that includes the risk and threat analysis, and for suitability of the

CONTRACTOR proposed security architecture and mitigations for vulnerabilities and threats.

2.7.6.44 The CONTRACTOR shall deliver four rounds of penetration testing (from datacenter network and from external networks) during the first year of the contract with the exact schedule for each test to be decided by the MOI, while adhering to the below requirements:

2.7.6.44.1 The penetration testing will be conducted by a third-party that is approved by the MOI.

2.7.6.44.2 After each round of penetration testing, the CONTRACTOR shall submit within 3 business days a report identifying observations classified as high, medium and low according to their probable impact along with a clear mitigation plan for each.

2.7.6.44.3 For observations rated as high and medium impact, the CONTRACTOR shall mitigate the findings /vulnerabilities / observations within 5 calendar days which shall then lead to the conduct of a repeat penetration test that should confirm the fixing of the findings / vulnerabilities / observations. However, this repeat penetration test will be considered as part of the preceding penetration test round, and will not be considered a new round of penetration testing.

2.7.6.44.4 For observations rated as low impact, the CONTRACTOR shall mitigate the findings /vulnerabilities / observations within 10 calendar days and prior to the next round of penetration testing.

2.7.6.45 Prior to going-live of any solution that its exposed to the Internet, the CONTRACTOR shall conduct vulnerability assessment and penetration testing for the OSS911 solutions, and shall fix any identified vulnerabilities and observations prior to the launch of the solution for public use.

2.7.6.46 The CONTRACTOR shall provide detailed reports to authorized MOI personnel for any vulnerability assessment and penetration testing activity along with detailed mitigations.

2.7.7 DATA MIGRATION

2.7.7.1 As part of their CONTRACT implementation, the CONTRACTOR shall migrate the historical data in Makkah 911 center into the new OSS911 solution.

2.7.7.2 The CONTRACTOR shall ensure that during data migration is seamless and that there is no impact whatsoever to the operations, technology, systems and data in the existing Makkah 911 environment.

2.7.7.3 Further to their analysis of the existing Makkah 911 environment and within two months of the award of the contract, the CONTRACTOR shall submit to Mol a step-by-step data migration approach and plan and shall not commence the migration unless this plan is approved by the Mol.

2.7.7.4 The CONTRACTOR shall migrate all the datasets, records, metadata, system logs and access logs relevant to CAD, telephony, recording, reporting and management and the different databases therein.

2.7.7.5 The CONTRACTOR shall ensure that the migrated data is preserved with 100% fidelity including timestamps with original time zones, calls and events sequencing

chronologically, calls and call detail records (CDRs), call recordings, ANI/ALI data, events (single and multiagency), units' statuses and others, as determined by the Mol.

- 2.7.7.6 The CONTRACTOR shall ensure that during the migration there is no loss of data due to any rounding / truncation / alteration / format change.
- 2.7.7.7 The CONTRACTOR shall analyze the existing Makkah 911 environment datasets, datastore and data, and shall submit to the Mol documentation that explains the mapping of the different data elements and shall conduct the related workshops with the Mol team before arriving at a final and complete mapping of the data to the CONTRACTOR delivered systems and solutions.
- 2.7.7.8 Should data transformation be required, the CONTRACTOR shall submit to the Mol the transformation rules that preserve the business and operational meaning and relevance of the data.
- 2.7.7.9 The CONTRACTOR shall ensure that their solutions support the data structure (schema) evolution across historical datasets, including the ability to ingest, map, and normalize data originating from multiple schema versions over time for all the existing systems whose data shall be migrated into the CONTRACTOR delivered systems and solutions.
- 2.7.7.10 Further to the data migration activity, the CONTRACTOR shall undertake a complete data reconciliation and validation exercise at record-level without relying only on statistical sampling.
- 2.7.7.11 Further to the data migration activity, the CONTRACTOR shall undertake a complete functional validation exercise to verify calls and incidents information including but not limited to call distribution, handling and recording, events dispatch and closure, events chronology etc.
- 2.7.7.12 The CONTRACTOR shall submit to the Mol a complete audit record of the migration activity documenting clearly failures and/or exceptions, if any, and the measures planned to correct the failures.
- 2.7.7.13 Following the data migration and reconciliation activity, the CONTRACTOR shall ensure that the CONTRACTOR delivered systems and solutions deliver the capability for authorized users to query historical and new data seamlessly without manual intervention, and to produce reports consistently whether the data is historical or the current data.
- 2.7.8 **TRANSITION TO CONTRACTOR DELIVERED SYSTEMS AND SOLUTIONS**
- 2.7.8.1 Further to their analysis of the existing Makkah 911 environment and within two months of the award of the contract, the CONTRACTOR shall submit to the Mol their approach and plan to transition the Makkah 911 operations, to run from the contractor delivered systems and solutions.
- 2.7.8.2 The CONTRACTOR shall ensure that the cutover to the contractor-delivered solutions is seamless and without loss of any live calls and/or live incidents.
- 2.7.8.3 The CONTRACTOR shall ensure that the cutover to the contractor-delivered solutions does not result in any dropped calls.

2.7.9 DATA MANAGEMENT

- 2.7.9.1 As part of their CONTRACT implementation, the CONTRACTOR shall implement data governance by defining the different roles and responsibilities involved in data management and the ownership for the data vis-à-vis the defined roles.
- 2.7.9.2 The CONTRACTOR shall define and implement data governance policies, accesses and rules in accordance with the MOI and SDAIA policies for critical systems data, considering all OSS911 solutions data as critical data.
- 2.7.9.3 As part of the centralized data repository created and maintained by the CONTRACTOR for the duration of the contract, the CONTRACTOR shall deliver data quality management, data consistency and data cleansing, and provides metrics and reporting on data quality.
- 2.7.9.4 As part of their CONTRACT implementation, the CONTRACTOR shall implement data discovery capabilities that includes periodic automated scanning of all OSS911 databases and files to be able to extract meaningful metadata and to identify critical/sensitive data that aids in data classification.
- 2.7.9.5 The CONTRACTOR shall develop a data classification policy in conjunction with the MOI and shall have this policy approved by the MOI prior to implementing any database in production and training environments.
- 2.7.9.6 The CONTRACTOR shall classify all data as per the approved Data Classification Policy, irrespective of the data formats, and on all systems, and shall label according to the relevant KSA legal and regulatory frameworks as per the approved data classifications policy.
- 2.7.9.7 The CONTRACTOR shall implement a Data Loss Protection (DLP) solution that is centrally managed, and that secures, monitors and alerts on compromises on data and information security.
- 2.7.9.8 The CONTRACTOR shall configure the DLP solution for safeguarding all data exchanges including through REST APIs and SMS interfaces for data exposure, leakage and/or unauthorized transfer.
- 2.7.9.9 The CONTRACTOR shall configure the DLP solution for inspecting API and SMS payloads in real time and to redact, mask, encrypt or block fields, or deny entire transactions.
- 2.7.9.10 The CONTRACTOR shall create, monitor and maintain a DLP register for the different types of data and information to be exchanged internally and externally.
- 2.7.9.11 The CONTRACTOR shall submit this DLP register along with the data classification to the MOI for approval prior to implementing any data storage in production and training environments.
- 2.7.9.12 The CONTRACTOR shall create and maintain a list of vectors (networks, interfaces, applications etc.,) by which data and information contained in the DLP register can exit/transfer.
- 2.7.9.13 The CONTRACTOR shall configure the DLP solution to monitor and control the movement of data listed in the DLP register using a predefined approved set of rules.
- 2.7.9.14 The CONTRACTOR shall integrate the DLP solution with the CONTRACTOR deployed SIEM solution.

2.7.9.15 The CONTRACTOR shall implement context-aware controls for DLP for APIs including user role, consuming applications, end point location, transaction intent etc.

2.7.10 النسخ الاحتياطي (BACKUP)

2.7.10.1 The CONTRACTOR shall supply a centralized backup solution that performs automated / unattended backup of all data using:

2.7.10.1.1 Network-based backup solution using removable rewritable media such as DVD / tapes that can be stored offline.

2.7.10.1.2 Online network backup solution to high-speed secondary storage such as storage arrays.

2.7.10.2 All backed up data shall be indexed, retrievable and restorable.

2.7.10.3 The BIDDER shall include as part of their proposal a suggested backup policy covering frequency of backups, backup windows, retention period, and rotation schemes.

2.7.10.4 At a minimum, the CONTRACTOR shall implement the following minimum retention periods as part of the backup solution:

2.7.10.4.1 Online for all critical data including all incidents voice and data for up to one year. Video to be included if video is related to an event.

2.7.10.4.2 Offline for all critical data including all incidents voice and data for up to five years.

2.7.10.4.3 Extended Offline for identified incidents voice and data for up to eight years.

2.7.10.5 Within the backup solution, the CONTRACTOR shall implement capability to tag data as 'Not to be deleted'.

2.7.10.6 The CONTRACTOR shall implement the backup solution on open architecture and formats, and without any proprietary data formats – allowing data on any format on any operating system(s) to be restored to any other operating system.

2.7.10.7 At the end of each month of live operations starting with the live operation of the first agency, the CONTRACTOR shall deliver tapes containing backup of all operational systems and data that maybe available for each OSS911 center to be offsite by the MOI, and shall implement a tape labeling and rotation scheme that allows monthly tapes offsite for disaster recovery for the duration of the contract.

2.7.11 أداة إدارة البنية التحتية والنظام والتطبيقات (INFRASTRUCTURE, SYSTEM AND APPLICATIONS MANAGEMENT TOOL)

2.7.11.1 The CONTRACTOR shall implement a GUI-based centralized infrastructure management tool and a system and applications management tool to manage the PCs, servers, networks, security, databases, camera feeds, interfaces, integrations and applications in production and training environments across all OSS911 center.

2.7.11.2 These tools shall be configured by the CONTRACTOR to provide performance data and threshold-based alerts for real time performance

monitoring, service level monitoring, reporting, version control and historical trending for all manageable elements with documented deployment and rollback procedures.

2.7.11.3The CONTRACTOR shall implement centralized monitoring, management, logging and alerting for all technical infrastructure, applications, and services and integrations.

2.7.11.4The CONTRACTOR shall configure alarms and events to be sent to the centralized management and monitoring servers and to be integrated with the SIEM solution provided by the CONTRACTOR.

2.7.11.5Within each OSS911 center datacenter, the CONTRACTOR shall furnish an alarm management screen to provide a continuously updating display of the alerts and alarms.

2.7.11.6The CONTRACTOR shall configure the management and monitoring solutions to be capable of auditing each of the Technical Infrastructure to identify the following:

2.7.11.6.1Any changes that have been made to certain parameters (e.g., a system administrator password changed, a new user added)

2.7.11.6.2To identify all configuration changes made within a time interval

2.7.11.6.3To identify the person responsible for making changes

2.7.11.7The CONTRACTOR shall configure the management and monitoring system to collect, analyze and summarize management data from various OSS911 systems, applications and services for performance management.

2.7.11.8The CONTRACTOR shall configure the management and monitoring solution to provide at a minimum the following performance report:

2.7.11.8.1An Executive Summary report that gives an overall view of a group of elements, showing volume and other important metrics for the Technical Infrastructure being viewed.

2.7.11.8.2Capacity Planning report which provides a view of under-and-over-utilized elements.

2.7.11.8.3Service Level report that shows the Technical Infrastructure and applications with metrics used to monitor SLAs such as availability, response times, etc.

2.7.11.8.4Applications performance report including average and peak response times, error rates, concurrent users, requests per second, queries per second etc.

2.7.11.9The system and application management solution shall be integrated by the CONTRACTOR with the solution for service level agreement monitoring and threshold alert reporting.

2.7.11.10As part of the monitoring, alerting and reporting, the CONTRACTOR shall be responsible to provide their own solutions for

dispersing the alerts to authorized personnel using Email and SMS notifications. Such solutions shall be submitted to the MOI for their approval prior to their usage.

2.7.12 **حل مراكز البيانات ، DCIM والرفوف (DATACENTER SOLUTION, DCIM AND RACKS)**

2.7.12.1The BIDDER shall conduct the site surveys to understand the details of the datacenters and telecom rooms in the OSS911 center.

2.7.12.2The CONTRACTOR shall supply and install full height 42U-48U 19" racks with sufficient height, load capacity and airflow to support high-density AI compute infrastructure, in the MOI identified datacenters, to house all equipment meeting the following criteria and shall integrate them with the DCIM solution by including all the intelligent PDUs within and all the sensors:

2.7.12.2.1Shall provide standard 19" rack mounting options for equipment.

2.7.12.2.2Mounting rails shall be adjustable.

2.7.12.2.3Equipment cabinet layouts shall be provided for this design.

2.7.12.2.4Cabinets shall be designed to accommodate server rail kits from various equipment manufactures

2.7.12.2.5Include removable front & rear locking doors

2.7.12.2.6Lift-off locking side panels

2.7.12.2.7Perforated top panel for ventilation and fan mounting

2.7.12.2.8Open base

2.7.12.2.9All equipment cabinet assemblies shall include an exhaust fan, power distribution units, grounding bus, and any mounting hardware required

2.7.12.2.10All equipment cabinets shall be floor mountable by design and permanently fixed to the floor with bolt-down kits. Manufacturer's procedures for floor mounting shall be followed.

2.7.12.2.11A common make and type of cabinet shall be used for all cabinets.

2.7.12.2.12High-speed structured cabling (copper and fiber) with redundant paths capable of supporting $\geq 25/40/100$ Gb Ethernet.

2.7.12.2.13Each rack shall include capability for intake of dual power distribution (A/B feeds) to at least four metered/intelligent PDUs with each PDU rated for at least 5 KVA.

2.7.12.3The BIDDER shall note that cabling exists in the room connected to the power source and that the CONTRACTOR is required to supply and install the cabling required from the power units in the datacenter to each of the CONTRACTOR racks.

2.7.12.4The CONTRACTOR shall deliver perforated doors on the front and rear of all cabinets to improve air flow and equipment cooling.

- 2.7.12.5 The CONTRACTOR shall install the racks arranged in rows side by side with the side panels between adjoining cabinets removed allowing access between the cabinets, noting that the side panels shall not be removed in cases where they are essential to maintain the correct operating temperature within the cabinet.
- 2.7.12.6 The CONTRACTOR shall install the racks such that sufficient space shall be left between the rows of equipment to allow all doors to be opened without obstruction and for all equipment within the cabinets to be removed.
- 2.7.12.7 Where equipment has internally serviceable parts which require top access, that The CONTRACTOR shall install the racks with sliders and cable management devices to allow the unit to be safely pulled out for servicing without the need to disconnect any of the cables and without the need for any additional support.
- 2.7.12.8 The CONTRACTOR shall install the racks such that the side panels shall be left in place on the outside edges of all end-of-row racks or stand-alone racks.
- 2.7.12.9 Racks containing powered equipment shall use thermostatically controlled forced air cooling to maintain the temperature within the rack at the midpoint (or optimum) operating temperature.
- 2.7.12.10 For the OSS911 datacenter, the CONTRACTOR shall implement locally, a scalable and vendor-agnostic centralized datacenter infrastructure management (DCIM) solution that is accessible from anywhere on the OSS911 network and that is dimensioned for 500 devices and 15 cameras.
- 2.7.12.11 The CONTRACTOR delivered DCIM solution shall be based on COTS products from vendors who are rated as leaders in the last released Gartner Magic Quadrant for DCIM solution.
- 2.7.12.12 The CONTRACTOR delivered DCIM solution shall monitor, organize, and distribute critical alerts and information, providing a unified view of the datacenter infrastructure environments.
- 2.7.12.13 The CONTRACTOR delivered DCIM solution shall be integrated with the already existing OSS911 datacenter systems including the CCTV system, the security and access control system, the fire alarm system and fire protection system, cooling system, building management system.
- 2.7.12.14 The CONTRACTOR shall include all necessary solutions and sensors for centrally monitoring the racks temperature, humidity and cooling and for creating necessary alarms for each rack.
- 2.7.12.15 The CONTRACTOR shall install the equipment in the racks to be connected to two independent breakers within the rack such that the failure of one power feed to the rack shall have no impact on the operation of that equipment.
- 2.7.12.16 Where equipment in a rack has only a single power supply, the CONTRACTOR shall evenly distribute the connections for such equipment across the PDUs within the racks.

- 2.7.12.17The CONTRACTOR shall install horizontal and vertical cable management in the front of all racks where front routing of cables is required (e.g., patch panels and active equipment such as switches).
- 2.7.12.18The CONTRACTOR shall install cable management accessories for all cabling in the rear of the racks.
- 2.7.12.19The CONTRACTOR shall ensure appropriate separation of power and signal cables for all their installations.
- 2.7.12.20The CONTRACTOR shall install all cables such that the cables that are required to exit the rack shall pass through the side openings to adjacent and adjoining racks and such that the cables shall not exit the cabinet through the front or the rear, and the front and rear of the racks shall remain free of obstacles.
- 2.7.12.21The CONTRACTOR shall provide a cable schedule for all installed cables that includes all labels.
- 2.7.12.22All cables supplied by the CONTRACTOR shall be tested and certified for compliance to their relevant specification.
- 2.7.12.23The CONTRACTOR shall maintain a 3-foot working clearance in the front and in the back of each equipment cabinets as a minimum, and a 3 foot working clearance at both ends of the equipment cabinet or assemblies.
- 2.7.12.24The CONTRACTOR shall install cabinets with cable and jumper management and include the accessories to organize and support cabling and patch cords in the vertical and horizontal planes.
- 2.7.12.25The CONTRACTOR shall implement all installations in conformance with industry installation guidelines, and manufacturer requirements. Failure to follow the appropriate guidelines will require the CONTRACTOR to provide, in a timely fashion, the additional material and labor necessary to properly rectify the situation, at no additional cost to the MOI.
- 2.7.12.26The CONTRACTOR shall install all racks and cabinets to be floor mountable by design and permanently fixed to the floor with bolt-down kits, unless otherwise specified.
- 2.7.12.27The CONTRACTOR shall not use plastic cable ties; cables shall be bundled using Velcro wraps of the appropriate length. Velcro ties shall be installed in such a manner that cables may be easily removed or added to the cable bundle.

2.7.13 متطلبات التشييت (INSTALLATION REQUIREMENTS)

- 2.7.13.1Design plans for hardware selection, installation location, and installation details shall be provided by the CONTRACTOR.
- 2.7.13.2The CONTRACTOR shall ensure the ventilation requirements for the all hardware components are met.
- 2.7.13.3Installation of the equipment by the CONTRACTOR shall be in accordance with the manufacturer's installation instructions.

2.7.13.4 For the CONTRACTOR supplied equipment, the CONTRACTOR shall produce the final as-built design for the OSS911 equipment at each location. These designs shall include the following:

2.7.13.4.1 Floor layouts showing the position of all racks and other equipment that shall be installed in each location

2.7.13.4.2 Rack front layouts for each rack

2.7.13.4.3 Cable designs

2.7.13.4.4 Power requirements analysis showing the loading per rack and per circuit

2.7.13.4.5 Heat output analysis

2.7.13.4.6 Floor loadings

2.7.13.5 The CONTRACTOR shall implement:

2.7.13.5.1 Power distribution strips that shall conform to the local KSA standards from SASO and standards of the NIC and MOI as prescribed on their website.

2.7.13.5.2 Power cabling and termination blocks, if any, that shall be installed between the UPS panel/switchboard designated for the equipment room and the rack.

2.7.13.5.3 All power plugs on installed equipment shall conform to the KSA standards.

2.7.13.5.4 Each rack shall be designed to be fed from a minimum of two independent circuit breakers of an appropriate rating for the load they supply.

2.7.13.5.5 The rows of racks shall be configured in hot aisle/cold aisle layout to maximize cooling

2.7.14 أجهزة الكمبيوتر والشاشات والطابعات والفاكس (**COMPUTERS, MONITORS, PRINTERS AND FAX**)

2.7.14.1 Computers and peripherals shall be installed by the CONTRACTOR in each of the OSS911 Centers.

2.7.14.2 Within their proposal, the BIDDER shall include a detailed breakdown of the quantities of the different equipment under supply including breakdown by different areas within OSS911.

2.7.14.3 The computing solution from the CONTRACTOR shall use the OSS911 Center structured cabling system of the OSS911 center. Proprietary cabling systems are not acceptable.

2.7.14.4 The PCs implemented by the CONTRACTOR within all the OSS911 Centers shall be of the same basic manufacturer, and model.

2.7.14.5 The CONTRACTOR shall supply PCs in three different configurations, as per the related appendix#8.

2.7.14.6 Each PC deployed by the CONTRACTOR at the OSS911 centers shall be capable of driving the required number of high-resolution

displays simultaneously, without any noticeable degradation in performance of the applications.

2.7.14.7The CONTRACTOR shall supply two 24" LED monitors, each monitor with built-in speakers, for each operational position with each of the call taking PCs to be equipped with built-in webcam that will be integrated with OSS911 solutions for video calls handling.

2.7.14.8Each configuration of the PC shall be certified for use by the various COTS vendors and the CONTRACTOR for their solutions to work as per the performance and service level requirements of this RFP.

2.7.14.9The CONTRACTOR shall supply each of the PC with the most current, full release desktop operating system which is compatible with the OSS911 systems and architecture, stable as certified by the OSS911 systems' manufacturer, and supports the required security profile.

2.7.14.10The operating system version to be installed shall be approved by MOI during detailed design following the award of the contract.

2.7.14.11The CONTRACTOR shall provide the latest version of desktop automation suite (capability of word processing, spreadsheet, presentation and collaboration including email) on each supplied personal computer, together with the appropriate licenses.

2.7.14.12The CONTRACTOR shall produce a standard workstation base build with the operating systems and other common applications, and then add the appropriate specific application software related to the intended usage.

2.7.14.13The CONTRACTOR shall deliver the images of all builds to the MOI.

2.7.14.14The CONTRACTOR shall supply the PCs with a wired optical mouse with a navigation wheel and a minimum of 2 navigation buttons.

2.7.14.15The CONTRACTOR shall supply the PCs with full sized wired keyboards that include a numeric keypad. All relevant keys shall be printed with both Latin and Arabic letters.

2.7.14.16The CONTRACTOR shall supply monitors which shall be flat panel and support native full HD as a minimum.

2.7.14.17The monitor that the CONTRACTOR shall install shall have a Video Electronics Standards Association (VESA) universal mounting that allows the stand to be removed and replaced by an arm or other type of mounting device.

2.7.14.18The CONTRACTOR shall implement a fax solution shall allow Fax over IP which allows any authorized user and/or computer from any of the OSS911 center to transmit and receive faxes via a centrally located fax server.

2.7.14.19The CONTRACTOR shall implement any gateways and cabling that are required for connecting their fax server to the incoming fax lines.

- 2.7.14.20The CONTRACTOR shall implement the fax solution to be capable of storing and forwarding all inbound fax communication into the OSS911 Center.
- 2.7.14.21The CONTRACTOR shall implement the fax solution to be capable of having multiple Direct Dial Inward (DDI) numbers allowing routing of the inbound fax based on the dialed number.
- 2.7.14.22The CONTRACTOR shall implement the fax solution such that all inbound faxes are stored and indexed on the fax server and details such as the CLI of the originating call shall be recorded when available.
- 2.7.14.23The CONTRACTOR shall implement the fax solution such that access to data stored on the fax server is only available to authorized users.
- 2.7.14.24The CONTRACTOR shall implement the fax solution such that it is possible to archive to removable media all fax data that is older than a certain period of time.
- 2.7.14.25The CONTRACTOR shall implement the fax solution such that it is configured to restore to the fax server all or selected archived data.
- 2.7.14.26The CONTRACTOR shall implement the fax solution such that all inbound and outbound faxes are given a unique reference in the system.
- 2.7.14.27The CONTRACTOR shall implement the fax solution such that it is configured to produce reports on all inbound and outbound fax traffic.

2.8 INTEGRATION WITH EXISTING VIDEO DISPLAY SYSTEMS (VDS)

2.8.1 نطاق العمل (SCOPE OF WORK)

- 2.8.1.1The CONTRACTOR shall implement all requirements in this section with the understanding that:
- 2.8.1.1.1Unless otherwise explicitly stated, the phrase “OSS911 center” refers to OSS911 centers in each of the provinces that are included in the scope of this contract.
- 2.8.1.1.2Unless otherwise explicitly stated, the phrases “OSS911 solution” or “OSS911 systems” or “OSS911 subsystems” refer to all the OSS911 systems being delivered by the CONTRACTOR as part of this contract.
- 2.8.1.1.3Unless otherwise explicitly stated, the word “data” in this document includes all types of data including voice, text, video and SMS.
- 2.8.1.2 The CONTRACTOR shall deliver all the requirements of this section that are fully compliant to the concept of operations requirements stated in this RFP.
- 2.8.1.3The CONTRACTOR shall integrate the OSS911 solutions with MOI-provided VDS which comprises the existing Video walls, Video Management System (VMS) and Video Analytics (VA) solutions in the Makkah 911 center.

- 2.8.1.4 The CONTRACTOR integration with MOI-provided VDS shall be such that there is no limitation in the combinations of visualized sources.
- 2.8.1.5 The CONTRACTOR shall configure the integration with MOI-provided VDS such that a single image shall be able to be displayed on the whole of the video walls whilst maintaining the correct aspect ratio of the input signal.
- 2.8.1.6 The CONTRACTOR's integration with MOI-provided VDS shall allow authorized users in to display any of the incoming video feeds into the center to be displayed on any of video walls in these areas.